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1.0.1 GATE ECE 2018 | Question: 55


Let $X[k] = k + 1, 0 \leq k \leq 7$ be 8-point DFT of a sequence $x[n]$.

where $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/N}$.

The value (correct to two decimal places) of $\sum_{n=0}^3 x[2n]$ is _____.

gate2018-ec numerical-answers calculus

1.0.2 GATE ME 2012 | Question: 54


For a particular project, eight activities are to be carried out. Their relationships with other activities and expected durations are mentioned in the table below.

Activity	Predecessors	Duration (days)
<i>a</i>	–	3
<i>b</i>	<i>a</i>	4
<i>c</i>	<i>a</i>	5
<i>d</i>	<i>a</i>	4
<i>e</i>	<i>b</i>	2
<i>f</i>	<i>d</i>	9
<i>g</i>	<i>c, e</i>	6
<i>h</i>	<i>f, g</i>	2

The critical path for the project is

- A. $a-b-e-g-h$ B. $a-c-g-h$ C. $a-d-f-h$ D. $a-b-c-f-h$

gateme-2012 calculus

1.0.3 GATE ME 2012 | Question: 55


For a particular project, eight activities are to be carried out. Their relationships with other activities and expected durations are mentioned in the table below.

Activity	Predecessors	Duration (days)
<i>a</i>	–	3
<i>b</i>	<i>a</i>	4
<i>c</i>	<i>a</i>	5
<i>d</i>	<i>a</i>	4
<i>e</i>	<i>b</i>	2
<i>f</i>	<i>d</i>	9
<i>g</i>	<i>c, e</i>	6
<i>h</i>	<i>f, g</i>	2

If the duration of the activity *f* alone is changed from 9 to 10, then the

- A. critical path remains the same and the total duration to complete the project changed to 19 days
 B. critical path and the total duration to complete the project remain the same
 C. critical path changes but the total duration to complete the project remain the same
 D. critical path changes but the total duration to complete the project changes to 17 days

1.0.4 GATE ECE 2012 | Question: 25



If $x = \sqrt{-1}$, then the value of x^x is

- A. $e^{\frac{-\pi}{2}}$ B. $e^{\frac{\pi}{2}}$ C. x D. 1

gate2012-ec calculus

1.1 Area Under Curve (1)

1.1.1 Area Under Curve: GATE ME 2013 | Question: 26



The following surface integral is to be evaluated over a sphere for the given steady velocity vector field $F = xi + yj + zk$ defined with respect to a Cartesian coordinate system having i, j and k as unit base vectors.

$$\iint_S \frac{1}{4}(F \cdot n) dA$$

where S is the sphere, $x^2 + y^2 + z^2 = 1$ and n is the outward unit normal vector to the sphere. The value of the surface integral is

- A. π B. 2π C. $3\pi/4$ D. 4π

gateme-2013 calculus integrals area-under-curve

1.2 Cartesian Coordinates (2)

1.2.1 Cartesian Coordinates: GATE 2016 - 1



A straight line of the form $y = mx + c$ passes through the origin and the point $(x, y) = (2, 6)$. The value of m is _____.

gate2016-in numerical-answers calculus cartesian-coordinates

1.2.2 Cartesian Coordinates: GATE BT 2021 | Question: 5



The Cartesian coordinates (x, y) of a point A with polar coordinates $(4, \pi/4)$ is

- A. $(\sqrt{3}, 2\sqrt{2})$ B. $(2, 2\sqrt{3})$
C. $(2\sqrt{2}, \sqrt{3})$ D. $(2\sqrt{2}, 2\sqrt{2})$

gatebt-2021 calculus cartesian-coordinates

1.3 Complex Number (1)

1.3.1 Complex Number: GATE Electrical 2017 Set 1 | Question: 2



For a complex number z , $\lim_{z \rightarrow i} \frac{z^2 + 1}{z^3 + 2z - i(z^2 + 2)}$ is

- A. $-2i$ B. $-i$ C. i D. $2i$

gate2017-ee-1 calculus limits complex-number

1.4 Complex Variables (5)

1.4.1 Complex Variables: GATE Mechanical 2014 Set 1 | Question: 3



The argument of the complex number $\frac{1+i}{1-i}$ where $i = \sqrt{-1}$, is

- A. $-\pi$ B. $-\frac{\pi}{2}$ C. $\frac{\pi}{2}$ D. π

gateme-2014-set1 calculus complex-variables

1.4.2 Complex Variables: GATE Mechanical 2014 Set 2 | Question: 26



An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$, where $i = \sqrt{-1}$. If $u(x, y) = 2xy$, then $v(x, y)$ must be

- A. $x^2 + y^2 + \text{constant}$ B. $x^2 - y^2 + \text{constant}$
C. $-x^2 + y^2 + \text{constant}$ D. $-x^2 - y^2 + \text{constant}$

gateme-2014-set2 calculus complex-variables

1.4.3 Complex Variables: GATE Mechanical 2014 Set 3 | Question: 26



An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$, where $i = \sqrt{-1}$. If $u(x, y) = x^2 - y^2$, then expression for $v(x, y)$ in terms of x, y and a general constant c would be

- A. $xy + c$ B. $\frac{x^2 + y^2}{2} + c$ C. $2xy + c$ D. $\frac{(x - y)^2}{2} + c$

gateme-2014-set3 calculus complex-variables

1.4.4 Complex Variables: GATE Mechanical 2021 Set 1 | Question: 27



Let C represent the unit circle centered at origin in the complex plane, and complex variable, $z = x + iy$. The value of the contour integral $\oint_C \frac{\cosh 3z}{2z} dz$ (where integration is taken counter clockwise) is

- A. 0 B. 2 C. πi D. $2\pi i$

gateme-2021-set1 calculus complex-variables

1.4.5 Complex Variables: GATE Mechanical 2021 Set 2 | Question: 19



Value of $(1 + i)^8$, where $i = \sqrt{-1}$, is equal to

- A. 4 B. 16 C. $4i$ D. $16i$

gateme-2021-set2 calculus complex-variables

1.5 Continuity (9)

1.5.1 Continuity: GATE CSE 1996 | Question: 3



Let f be a function defined by

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 1 \\ ax^2 + bx + c & \text{for } 1 < x \leq 2 \\ x + d & \text{for } x > 2 \end{cases}$$

Find the values for the constants a, b, c and d so that f is continuous and differentiable everywhere on the real line.

Answer key 

1.5.2 Continuity: GATE CSE 1998 | Question: 1.4



Consider the function $y = |x|$ in the interval $[-1, 1]$. In this interval, the function is

- A. continuous and differentiable
 B. continuous but not differentiable
 C. differentiable but not continuous
 D. neither continuous nor differentiable

gate1998 calculus continuity differentiation easy

Answer key 

1.5.3 Continuity: GATE CSE 2007 | Question: 1



Consider the following two statements about the function $f(x) = |x|$:

- P. $f(x)$ is continuous for all real values of x .
- Q. $f(x)$ is differentiable for all real values of x .

Which of the following is **TRUE**?

- A. P is true and Q is false.
 B. P is false and Q is true.
 C. Both P and Q are true.
 D. Both P and Q are false.

gatecse-2007 calculus continuity differentiation easy

Answer key 

1.5.4 Continuity: GATE CSE 2013 | Question: 22



Which one of the following functions is continuous at $x = 3$?

- A. $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \\ \frac{x+3}{3} & \text{if } x < 3 \end{cases}$
 B. $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8 - x & \text{if } x \neq 3 \end{cases}$
 C. $f(x) = \begin{cases} x + 3, & \text{if } x \leq 3 \\ x - 4 & \text{if } x > 3 \end{cases}$
 D. $f(x) = \begin{cases} \frac{1}{x^3 - 27} & \text{if } x \neq 3 \end{cases}$

gatecse-2013 calculus continuity normal

Answer key 

1.5.5 Continuity: GATE CSE 2014 Set 1 | Question: 47



A function $f(x)$ is continuous in the interval $[0, 2]$. It is known that $f(0) = f(2) = -1$ and $f(1) = 1$. Which one of the following statements must be true?

- A. There exists a y in the interval $(0, 1)$ such that $f(y) = f(y + 1)$
 B. For every y in the interval $(0, 1)$, $f(y) = f(2 - y)$
 C. The maximum value of the function in the interval $(0, 2)$ is 1
 D. There exists a y in the interval $(0, 1)$ such that $f(y) = -f(2 - y)$

gatecse-2014-set1 calculus continuity normal

Answer key 

1.5.6 Continuity: GATE CSE 2015 Set 2 | Question: 26



Let $f(x) = x^{-\left(\frac{1}{3}\right)}$ and A denote the area of region bounded by $f(x)$ and the X-axis, when x varies from -1 to 1 . Which of the following statements is/are TRUE?

- I. f is continuous in $[-1, 1]$
- II. f is not bounded in $[-1, 1]$
- III. A is nonzero and finite

A. II only B. III only C. II and III only D. I, II and III

gatecse-2015-set2 continuity functions normal

Answer key

1.5.7 Continuity: GATE CSE 2021 Set 2 | Question: 25



Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \leq 2$. If $f(-3) = 7$, then $f(3)$ is at most

gatecse-2021-set2 numerical-answers calculus continuity 1-mark

Answer key

1.5.8 Continuity: GATE Electrical 2015 Set 1 | Question: 2



If a continuous function $f(x)$ does not have a root in the interval $[a, b]$, then which one of the following statements is TRUE?

- A. $f(a) \cdot f(b) = 0$
- B. $f(a) \cdot f(b) < 0$
- C. $f(a) \cdot f(b) > 0$
- D. $f(a)/f(b) \leq 0$

gate2015-ee-1 calculus continuity

1.5.9 Continuity: GATE Electrical 2017 Set 2 | Question: 26



Let $g(x) = \begin{cases} -x & x \leq 1 \\ x+1 & x \geq 1 \end{cases}$ and $f(x) = \begin{cases} 1-x & x \leq 0 \\ x^2 & x > 0 \end{cases}$.

Consider the composition of f and g , i.e., $(f \circ g)(x) = f(g(x))$. The number of discontinuities in $(f \circ g)(x)$ present in the interval $(-\infty, 0)$ is:

- A. 0 B. 1 C. 2 D. 4

gate2017-ee-2 calculus continuity

1.6 Continuity and Differentiability (5)

1.6.1 Continuity and Differentiability: GATE CH 2021 | Question: 16



For the function $f(x) = \begin{cases} -x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ the CORRECT statement(s) is/are

- A. $f(x)$ is continuous at $x = 1$
- B. $f(x)$ is differentiable at $x = 1$
- C. $f(x)$ is continuous at $x = 0$
- D. $f(x)$ is differentiable at $x = 0$

gatech-2021 calculus continuity-and-differentiability multiple-selects

1.6.2 Continuity and Differentiability: GATE ECE 2016 Set 1 | Question: 3



Given the following statements about a function $f: \mathbb{R} \rightarrow \mathbb{R}$, select the right option:

P: If $f(x)$ is continuous at $x = x_0$ then it is also differentiable at $x = x_0$.

Q: If $f(x)$ is continuous at $x = x_0$ then it may not be differentiable at $x = x_0$.

R: If $f(x)$ is differentiable at $x = x_0$ then it is also continuous at $x = x_0$.

- A. P is true, Q is false, R is false
C. P is false, Q is true, R is false

- B. P is false, Q is true, R is true
D. P is true, Q is false, R is true

gate2016-ec-1 calculus continuity-and-differentiability

1.6.3 Continuity and Differentiability: GATE Electrical 2017 Set 1 | Question: 26



A function $f(x)$ is defined as

$$f(x) = \begin{cases} e^x, & x < 1 \\ \ln x + ax^2 + bx, & x \geq 1 \end{cases}, \text{ where } x \in \mathbb{R}$$

Which one of the following statement is TRUE?

- A. $f(x)$ is NOT differentiable at $x = 1$ for any values of a and b .
B. $f(x)$ is differentiable at $x = 1$ for the unique values of a and b .
C. $f(x)$ is differentiable at $x = 1$ for all values of a and b such that $a + b = e$.
D. $f(x)$ is differentiable at $x = 1$ for all values of a and b .

gate2017-ee-1 calculus continuity-and-differentiability

1.6.4 Continuity and Differentiability: GATE Electrical 2018 | Question: 11



Let f be a real-valued function of a real variable defined as $f(x) = x^2$ for $x \geq 0$, and $f(x) = -x^2$ for $x < 0$. Which one of the following statements is true?

- A. $f(x)$ is discontinuous at $x = 0$
B. $f(x)$ is continuous but not differentiable at $x = 0$
C. $f(x)$ is differentiable but its first derivative is not continuous at $x = 0$
D. $f(x)$ is differentiable but its first derivative is not differentiable at $x = 0$

gate2018-ee calculus continuity-and-differentiability

1.6.5 Continuity and Differentiability: GATE ME 2012 | Question: 12



Consider the function $f(x) = |x|$ in the interval $-1 \leq x \leq 1$. At the point $x = 0$, $f(x)$ is

- A. continuous and differentiable
C. continuous and non-differentiable
B. non-continuous and differentiable
D. neither continuous nor differentiable

gateme-2012 calculus continuity-and-differentiability

1.7 Contour Integral (1)

1.7.1 Contour Integral: GATE Electrical 2017 Set 2 | Question: 27



The value of the contour integral in the complex plane $\oint \frac{z^3 - 2z + 3}{z - 2} dz$ along the contour $|z| = 3$, taken counter-clockwise is

- A. $-18\pi i$
B. 0
C. $14\pi i$
D. $48\pi i$

gate2017-ee-2 calculus contour-integral

1.8 Contour Plots (1)

1.8.1 Contour Plots: GATE Electrical 2021 | Question: 28



Let $(-1 - j)$, $(3 - j)$, $(3 + j)$ and $(-1 + j)$ be the vertices of a rectangle C in the complex plane. Assuming that C is traversed in counter-clockwise direction, the value of the contour integral

$$\oint_C \frac{dz}{z^2(z-4)}$$
 is

- A. $j\pi/2$ B. 0 C. $-j\pi/8$ D. $j\pi/16$

gateee-2021 calculus contour-plots

1.9 Convergence (3)

1.9.1 Convergence: GATE CH 2021 | Question: 34



To solve an algebraic equation $f(x) = 0$, an iterative scheme of the type $x_{n+1} = g(x_n)$ is proposed, where

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

At the solution $x = s$, $g'(s) = 0$ and $g''(s) \neq 0$. The order of convergence for this iterative scheme near the solution is _____

gatech-2021 numerical-answers calculus convergence

1.9.2 Convergence: GATE CSE 1993 | Question: 01.6



Which of the following improper integrals is (are) convergent?

- A. $\int_0^1 \frac{\sin x}{1-\cos x} dx$ B. $\int_0^\infty \frac{\cos x}{1+x} dx$
 C. $\int_0^\infty \frac{x}{1+x^2} dx$ D. $\int_0^1 \frac{1-\cos x}{x^5} dx$

gate1993 calculus integration convergence out-of-gatecse-syllabus multiple-selects

Answer key

1.9.3 Convergence: GATE CSE 1993 | Question: 02.2



The radius of convergence of the power series

$$\sum_{m=0}^{\infty} \frac{(3m)!}{(m!)^3} x^{3m}$$

is: _____

gate1993 calculus convergence normal out-of-gatecse-syllabus fill-in-the-blanks

1.10 Convergence Criteria (1)

1.10.1 Convergence Criteria: GATE ECE 2014 Set 4 | Question: 1



The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges to

- A. $2 \ln 2$ B. $\sqrt{2}$ C. 2 D. e

gate2014-ec-4 calculus taylor-series convergence-criteria

1.11 Curves (3)

1.11.1 Curves: GATE Electrical 2017 Set 2 | Question: 19



Let x and y be integers satisfying the following equations

$$2x^2 + y^2 = 34$$

$$x + 2y = 11$$

The value of $(x + y)$ is _____.

gate2017-ee-2 numerical-answers calculus curves

1.11.2 Curves: GATE Electrical 2017 Set 2 | Question: 20



Let $y^2 - 2y + 1 = x$ and $\sqrt{x} + y = 5$. The value of $x + \sqrt{y}$ equals _____. (Give the answer up to three decimal places).

gate2017-ee-2 numerical-answers calculus curves

1.11.3 Curves: GATE Electrical 2021 | Question: 13



Suppose the circles $x^2 + y^2 = 1$ and $(x - 1)^2 + (y - 1)^2 = r^2$ intersect each other orthogonally at the point (u, v) . Then $u + v =$ _____.

gateee-2021 numerical-answers calculus curves

1.12

Definite Integral (10)

1.12.1 Definite Integral: GATE CSE 2023 | Question: 21



The value of the definite integral

$$\int_{-3}^3 \int_{-2}^2 \int_{-1}^1 (4x^2 y - z^3) dz dy dx$$

is _____. (Rounded off to the nearest integer)

gatecse-2023 calculus definite-integral numerical-answers 1-mark

Answer key

1.12.2 Definite Integral: GATE CSE 2024 | Set 2 | Question: 6



Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x) dx$?

- A. 0 B. 1 C. 2 D. -1

gatecse2024-set2 calculus definite-integral

Answer key

1.12.3 Definite Integral: GATE Civil 2013 | Question: 28



The solution for $\int_0^{\pi/6} \cos^4 3\theta \sin^3 6\theta d\theta$ is

- A. 0 B. $\frac{1}{15}$ C. 1 D. $\frac{8}{3}$

gate2013-ce calculus definite-integral

1.12.4 Definite Integral: GATE Civil 2021 Set 2 | Question: 18

The value (round off to one decimal place) of $\int_{-1}^1 x e^{|x|} dx$ is _____

gatecivil-2021-set2 numerical-answers calculus definite-integral

1.12.5 Definite Integral: GATE Electrical 2016 Set 1 | Question: 5

The value of the integral

$$\oint_c \frac{2z + 5}{\left(z - \frac{1}{2}\right)(z^2 - 4z + 5)} dz$$

over the contour $|z| = 1$, taken in the anti-clockwise direction, would be

- A. $\frac{24\pi i}{13}$ B. $\frac{48\pi i}{13}$ C. $\frac{24}{13}$ D. $\frac{12}{13}$

gate2016-ee-1 calculus definite-integral

1.12.6 Definite Integral: GATE Electrical 2016 Set 1 | Question: 9

The value of $\int_{-\infty}^{+\infty} e^{-t} \delta(2t - 2) dt$, where $\delta(t)$ is the Dirac delta function, is

- A. $\frac{1}{2e}$ B. $\frac{2}{e}$ C. $\frac{1}{e^2}$ D. $\frac{1}{2e^2}$

gate2016-ee-1 calculus definite-integral

1.12.7 Definite Integral: GATE Electrical 2016 Set 2 | Question: 29

The value of the integral $2 \int_{-\infty}^{\infty} \left(\frac{\sin 2\pi t}{\pi t}\right) dt$ is equal to

- A. 0 B. 0.5 C. 1 D. 2

gate2016-ee-2 calculus definite-integral

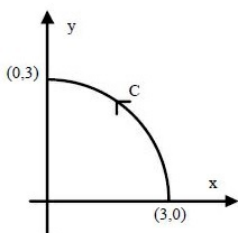
1.12.8 Definite Integral: GATE Electrical 2018 | Question: 18

Let f be a real-valued function of a real variable defined as $f(x) = x - [x]$, where $[x]$ denotes the largest integer less than or equal to x . The value of $\int_{0.25}^{1.25} f(x) dx$ is _____ (up to 2 decimal places).

gate2018-ee numerical-answers calculus definite-integral

1.12.9 Definite Integral: GATE Electrical 2018 | Question: 42

As shown in the figure, C is the arc from the point $(3, 0)$ to the point $(0, 3)$ on the circle $x^2 + y^2 = 9$. The value of the integral $\int_C (y^2 + 2yx) dx + (2xy + x^2) dy$ is _____ (up to 2 decimal places).



1.12.10 Definite Integral: GATE Electrical 2020 | Question: 2



Which of the following is true for all possible non-zero choices of integers $m, n; m \neq n$, or all possible non-zero choices of real numbers $p, q; p \neq q$, as applicable?

A.

$$\frac{1}{\pi} \int_0^\pi \sin m\theta \sin n\theta d\theta = 0$$

B.

$$\frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin p\theta \sin q\theta d\theta = 0$$

C.

$$\frac{1}{2\pi} \int_{-\pi}^\pi \sin p\theta \cos q\theta d\theta = 0$$

D.

$$\lim_{\alpha \rightarrow \infty} \frac{1}{2\alpha} \int_{-\alpha}^\alpha \sin p\theta \sin q\theta d\theta = 0$$

1.13

Definite Integrals (19)

1.13.1 Definite Integrals: GATE BT 2021 | Question: 49



Calculate the following integral $\int_0^{\pi^2/4} \sin \sqrt{x} dx =$ _____.

1.13.2 Definite Integrals: GATE Chemical 2012 | Question: 27



If a is a constant, then the value of the integral $a^2 \int_0^\infty x e^{-ax} dx$ is

A. $1/a$ B. a

C. 1

D. 0

1.13.3 Definite Integrals: GATE Chemical 2014 | Question: 3



If $f^*(x)$ is the complex conjugate of $f(x) = \cos(x) + i \sin(x)$, then for real a and b , $\int_a^b f^*(x) f(x) dx$ is *ALWAYS*

A. positive

B. negative

C. real

D. imaginary

1.13.4 Definite Integrals: GATE Chemical 2014 | Question: 9



Integral of the time-weighted absolute error (ITAE) is expressed as

A. $\int_0^\infty \frac{|\varepsilon(t)|}{t^2} dt$

B. $\int_0^\infty \frac{|\varepsilon(t)|}{t} dt$

C. $\int_0^\infty t |\varepsilon(t)| dt$

D. $\int_0^\infty t^2 |\varepsilon(t)| dt$

1.13.5 Definite Integrals: GATE ECE 2014 Set 2 | Question: 45

The value of the integral $\int_{-\infty}^{\infty} \operatorname{sinc}^2(5t) dt$ is _____.

gate2014-ec-2 numerical-answers calculus definite-integrals

1.13.6 Definite Integrals: GATE ECE 2015 Set 1 | Question: 29

The maximum area (in square units) of a rectangle whose vertices lie on the ellipse $x^2 + 4y^2 = 1$ is _____.

gate2015-ec-1 numerical-answers calculus definite-integrals

1.13.7 Definite Integrals: GATE ECE 2015 Set 2 | Question: 27

The value of the integral $\int_{-\infty}^{\infty} 12 \cos(2\pi t) \frac{\sin(4\pi t)}{4\pi t} dt$ is _____.

gate2015-ec-2 numerical-answers calculus definite-integrals

1.13.8 Definite Integrals: GATE ECE 2016 Set 1 | Question: 26

The integral $\frac{1}{2\pi} \iint_D (x + y + 10) dx dy$, where D denotes the disc: $x^2 + y^2 \leq 4$, evaluates to _____.

gate2016-ec-1 numerical-answers calculus definite-integrals

1.13.9 Definite Integrals: GATE ECE 2016 Set 3 | Question: 4

The integral $\int_0^1 \frac{dx}{\sqrt{1-x}}$ is equal to _____.

gate2016-ec-3 numerical-answers calculus definite-integrals

1.13.10 Definite Integrals: GATE ECE 2017 Set 2 | Question: 26

The values of the integrals

$$\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dy \right) dx$$

and

$$\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dx \right) dy$$

are

- A. same and equal to 0.5
C. 0.5 and -0.5 , respectively

- B. same and equal to -0.5
D. -0.5 and 0.5, respectively

gate2017-ec-2 calculus definite-integrals

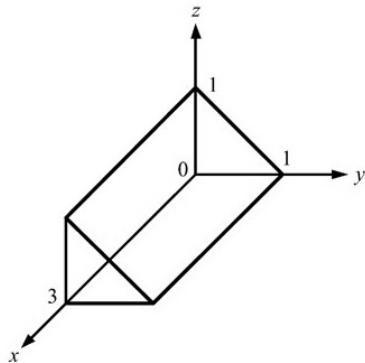
1.13.11 Definite Integrals: GATE ECE 2019 | Question: 19

The value of the integral $\int_0^\pi \int_y^\pi \frac{\sin x}{x} dx dy$, is equal to _____.

gate2019-ec numerical-answers calculus definite-integrals

1.13.12 Definite Integrals: GATE ECE 2020 | Question: 51

For the solid S shown below, the value of $\iiint_S x dx dy dz$ (rounded off to two decimal places) is _____.



gate2020-ec numerical-answers calculus definite-integrals

1.13.13 Definite Integrals: GATE ME 2012 | Question: 11

The area enclosed between the straight line $y = x$ and the parabola $y = x^2$ in the $x - y$ plane is

- A. $1/6$ B. $1/4$ C. $1/3$ D. $1/2$

gateme-2012 calculus definite-integrals area-under-curve

1.13.14 Definite Integrals: GATE ME 2013 | Question: 47

The value of the definite integral $\int_1^e \sqrt{x} \ln(x) dx$ is

- A. $\frac{4}{9}\sqrt{e^3} + \frac{2}{9}$ B. $\frac{2}{9}\sqrt{e^3} - \frac{4}{9}$
C. $\frac{2}{9}\sqrt{e^3} + \frac{4}{9}$ D. $\frac{4}{9}\sqrt{e^3} - \frac{2}{9}$

gateme-2013 calculus definite-integrals

1.13.15 Definite Integrals: GATE Mechanical 2014 Set 1 | Question: 26

The integral $\oint_c (y dx - x dy)$ is evaluated along the circle $x^2 + y^2 = \frac{1}{4}$ traversed in counter clockwise direction. The integral is equal to

- A. 0 B. $-\frac{\pi}{4}$ C. $\frac{-\pi}{2}$ D. $\frac{\pi}{4}$

gateme-2014-set1 calculus definite-integrals area-under-curve

1.13.16 Gradient: GATE Chemical 2014 | Question: 1

Gradient of a scalar variable is always

- A. a vector B. a scalar C. a dot product D. zero

gate2014 calculus vector-calculus gradient

1.13.17 Definite Integrals: GATE Mechanical 2014 Set 4 | Question: 26

If z is a complex variable, the value of $\int_5^{3i} \frac{dz}{z}$ is

- A. $-0.511 - 1.57i$
C. $0.511 - 1.57i$

- B. $-0.511 + 1.57i$
D. $0.511 + 1.57i$

gateme-2014-set4 calculus definite-integrals complex-variables

1.13.18 Definite Integrals: GATE Mechanical 2014 Set 4 | Question: 27



The value of the integral $\int_0^2 \int_0^x e^{x+y} dy dx$ is

A.

$$\frac{1}{2}(e-1)$$

B.

$$\frac{1}{2}(e^2-1)^2$$

C.

$$\frac{1}{2}(e^2-e)$$

D. $\frac{1}{2}(e - \frac{1}{e})^2$

gateme-2014-set4 calculus definite-integrals

1.13.19 Definite Integrals: GATE Mechanical 2021 Set 2 | Question: 26



The value of $\int_0^{\pi/2} \int_0^{\cos \theta} r \sin \theta dr d\theta$ is

A. 0

B. $\frac{1}{6}$

C. $\frac{4}{3}$

D. π

gateme-2021-set2 calculus definite-integrals double-integrals

1.14

Degree of Polynomial (1)

1.14.1 Degree of Polynomial: GATE Electrical 2016 Set 1 | Question: 33



Given the following polynomial equation $s^3 + 5.5s^2 + 8.5s + 3 = 0$ the number of roots of the polynomial, which have real parts strictly less than -1 , is _____.

gate2016-ee-1 calculus degree-of-polynomial numerical-answers

1.15

Derivatives (7)

1.15.1 Derivatives: GATE BT 2021 | Question: 8



$\frac{d}{dx} [\ln(2x)]$ is equal to

A.

$$1/2x$$

B.

$$1/x$$

C.

$$1/2$$

D. x

gatebt-2021 calculus derivatives

1.15.2 Derivatives: GATE ECE 2015 Set 3 | Question: 2



The contour on the $x-y$ plane, where the partial derivative of $x^2 + y^2$ with respect to y is equal to the partial derivative of $6y + 4x$ with respect to x , is

A. $y=2$

B. $x=2$

C. $x+y=4$

D. $x-y=0$

gate2015-ec-3 calculus derivatives partial-derivatives

1.15.3 Integrals: GATE Chemical 2013 | Question: 4



Evaluate

$$\int \frac{dx}{e^x - 1}$$

(Note: C is a constant of integration)

A. $\frac{e^x}{e^x-1} + C$

C. $\ln\left(\frac{e^x}{e^x-1}\right) + C$

B. $\frac{\ln(e^x-1)}{e^x} + C$

D. $\ln(1 - e^{-x}) + C$

gate2013 calculus integrals

1.15.4 Derivatives: GATE ECE 2020 | Question: 3



The partial derivative of the function

$$f(x, y, z) = e^{1-x \cos y} + xze^{-1/(1+y^2)}$$

with respect to x at the point $(1, 0, e)$ is

A. -1

B. 0

C. 1

D. $\frac{1}{e}$

gate2020-ec calculus derivatives partial-derivatives

1.15.5 Derivatives: GATE Electrical 2013 | Question: 46



A function $y = 5x^2 + 10x$ is defined over an open interval $x = (1, 2)$. At least at one point in this interval, $\frac{dy}{dx}$ is exactly

A. 20

B. 25

C. 30

D. 35

gate2013-ee calculus derivatives

1.15.6 Derivatives: GATE Electrical 2017 Set 2 | Question: 18



Consider a function $f(x, y, z)$ given by

$$f(x, y, z) = (x^2 + y^2 - 2z^2)(y^2 + z^2)$$

The partial derivative of this function with respect to x at the point, $x = 2, y = 1$ and $z = 3$ is _____.

gate2017-ee-2 numerical-answers calculus derivatives partial-derivatives

1.15.7 Derivatives: GATE Mechanical 2021 Set 2 | Question: 13



A two dimensional flow has velocities in x and y directions given by $u = 2xyt$ and $v = -y^2t$, where t denotes time. The equation for streamline passing through $x = 1, y = 1$ is

A. $x^2y = 1$

B. $xy^2 = 1$

C. $x^2y^2 = 1$

D. $x/y^2 = 1$

gateme-2021-set2 calculus derivatives

1.16 Differential Equation (1)

1.16.1 Differential Equation: GATE CSE 1993 | Question: 01.2



The differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + \sin y = 0$ is:

A. linear

B. non-linear

C. homogeneous

D. of degree two

gate1993 calculus differential-equation easy out-of-gatecse-syllabus multiple-selects

1.17 Differential Equations (1)

1.17.1 Differential Equations: GATE Electrical 2012 | Question: 38



The direction of vector $\mathbf{A}$ is radially outward from the origin, with $|\mathbf{A}| = kr^n$ where $r^2 = x^2 + y^2 + z^2$ and k is a constant. The value of n for which $\nabla \cdot \mathbf{A} = 0$ is

- A. -2 B. 2 C. 1 D. 0

gate2012-ee differential-equations

1.18 Differentiation (6)

1.18.1 Differentiation: GATE CSE 1996 | Question: 1.6



The formula used to compute an approximation for the second derivative of a function f at a point x_0 is

- A. $\frac{f(x_0+h) + f(x_0-h)}{2}$ B. $\frac{f(x_0+h) - f(x_0-h)}{2h}$
 C. $\frac{f(x_0+h) + 2f(x_0) + f(x_0-h)}{h^2}$ D. $\frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2}$

gate1996 calculus differentiation normal

1.18.2 Differentiation: GATE CSE 2014 Set 1 | Question: 46



The function $f(x) = x \sin x$ satisfies the following equation:

$$f''(x) + f(x) + t \cos x = 0$$

The value of t is _____.

gatecse-2014-set1 calculus easy numerical-answers differentiation

1.18.3 Differentiation: GATE CSE 2014 Set 1 | Question: 6



Let the function

$$f(\theta) = \begin{vmatrix} \sin \theta & \cos \theta & \tan \theta \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) & \tan(\frac{\pi}{6}) \\ \sin(\frac{\pi}{3}) & \cos(\frac{\pi}{3}) & \tan(\frac{\pi}{3}) \end{vmatrix}$$

where

$\theta \in [\frac{\pi}{6}, \frac{\pi}{3}]$ and $f'(\theta)$ denote the derivative of f with respect to θ . Which of the following statements is/are **TRUE**?

- I. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) = 0$
 II. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) \neq 0$

- A. I only B. II only C. Both I and II D. Neither I nor II

gatecse-2014-set1 calculus differentiation normal

1.18.4 Differentiation: GATE CSE 2016 Set 2 | Question: 02



Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____.

gatecse-2016-set2 calculus normal numerical-answers differentiation

Answer key

1.18.5 Differentiation: GATE CSE 2017 Set 2 | Question: 10



If $f(x) = R \sin(\frac{\pi x}{2}) + S$, $f'(\frac{1}{2}) = \sqrt{2}$ and $\int_0^1 f(x) dx = \frac{2R}{\pi}$, then the constants R and S are

- A. $\frac{2}{\pi}$ and $\frac{16}{\pi}$ B. $\frac{2}{\pi}$ and 0 C. $\frac{4}{\pi}$ and 0 D. $\frac{4}{\pi}$ and $\frac{16}{\pi}$

gatecse-2017-set2 engineering-mathematics calculus differentiation

Answer key

1.18.6 Differentiation: GATE CSE 2024 | Set 1 | Question: 1



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

- A. $\{-1, 1, 2\}$ B. $\{-2, -1, 1\}$ C. $\{0, 1\}$ D. $\{-1, 0, 1\}$

gatecse2024-set1 calculus differentiation

Answer key

1.19 Directional Derivatives (3)

1.19.1 Directional Derivatives: GATE Chemical 2019 | Question: 34



If x, y and z are directions in a Cartesian coordinate system and i, j and k are the respective unit vectors, the directional derivative of the function $u(x, y, z) = x^2 - 3yz$ at the point $(2, 0, -4)$ in the direction $(i + j - 2k)/\sqrt{6}$ _____ (rounded off to two decimal places).

gate2019 numerical-answers calculus vector-calculus directional-derivatives

1.19.2 Directional Derivatives: GATE Civil 2021 Set 2 | Question: 36



A function is defined in Cartesian coordinate system as $f(x, y) = xe^y$. The value of the directional derivative of the function (*in integer*) at the point $(2, 0)$ along the direction of the straight line segment from point $(2, 0)$ to point $(\frac{1}{2}, 2)$ is _____

gatecivil-2021-set2 numerical-answers calculus directional-derivatives

1.19.3 Directional Derivatives: GATE Electrical 2018 | Question: 12



The value of the directional derivative of the function $\Phi(x, y, z) = xy^2 + yz^2 + zx^2$ at the point $(2, -1, 1)$ in the direction of the vector $\mathbf{p} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ is

- A. 1 B. 0.95 C. 0.93 D. 0.9

gate2018-ee calculus directional-derivatives

1.20 Divergence (2)

1.20.1 Mean Value Theorems: GATE Chemical 2016 | Question: 29



The Lagrange mean-value theorem is satisfied for $f(x) = x^3 + 5$, in the interval $(1, 4)$ at a value (rounded off to the second decimal place) of x equal to _____

1.20.2 Divergence: GATE Electrical 2019 | Question: 39



If $\mathbf{A} = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}$ and $u = x^2 + y^2 + z^2$, then $\text{div}(u\mathbf{A})$ at $(1, 1, 1)$ is _____

gate2019-ee numerical-answers calculus divergence

1.21

Double Integral (2)

1.21.1 Double Integral: GATE Electrical 2014 Set 2 | Question: 26



To evaluate the double integral $\int_0^8 \left(\int_{(y/2)}^{y/2+1} \left(\frac{2x-y}{2} \right) dx \right) dy$, we make the substitution $u = \left(\frac{2x-y}{2} \right)$ and $v = \frac{y}{2}$. The integral will reduce

A.

$$\int_0^4 \left(\int_0^2 2u \, du \right) dv$$

B.

$$\int_0^4 \left(\int_0^1 2u \, du \right) dv$$

C.

$$\int_0^4 \left(\int_0^1 u \, du \right) dv$$

D.

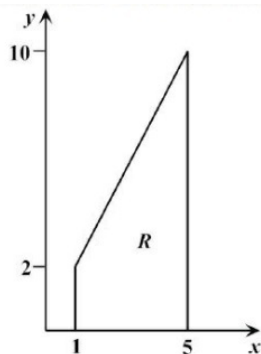
$$\int_0^4 \left(\int_0^2 u \, du \right) dv$$

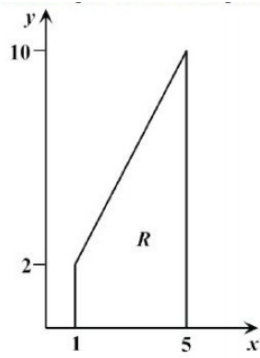
gate2014-ee-2 calculus definite-integral double-integral

1.21.2 Double Integral: GATE Electrical 2017 Set 1 | Question: 17



Let $I = c \int_R xy^2 dx dy$, where R is the region shown in the figure and $c = 6 \times 10^{-4}$. The value of I equals _____. (Give the answer up to two decimal places.)





gate2017-ee-1 numerical-answers calculus double-integral

1.22

Field Vector (1)

1.22.1 Field Vector: GATE Electrical 2013 | Question: 24



Given a vector field $\mathbf{F} = y^2 x \mathbf{a}_x - yz \mathbf{a}_y - x^2 \mathbf{a}_z$ the line integral $\int \mathbf{F} \cdot d\mathbf{l}$ evaluated along a segment on the x -axis from $x = 1$ to $x = 2$ is

- A. -2.33 B. 0 C. 2.33 D. 7

gate2013-ee calculus field-vector integral

1.23

Field Vectors (4)

1.23.1 Field Vectors: GATE Electrical 2014 Set 3 | Question: 3



Let $\nabla \cdot (fv) = x^2 y + y^2 z + z^2 x$, where f and v are scalar and vector fields respectively. If $v = yi + zj + xk$ then $v \cdot \Delta f$ is

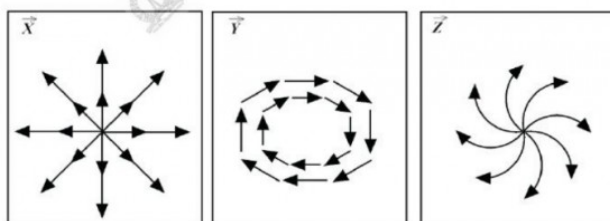
- A. $x^2 y + y^2 z + z^2 x$ B. $2xy + 2yz + 2zx$ C. $x + y + z$ D. 0

gate2014-ee-3 calculus field-vectors

1.23.2 Field Vectors: GATE Electrical 2017 Set 2 | Question: 3



The figures show diagrammatic representations of vector fields $\vec{X}$, $\vec{Y}$, and $\vec{Z}$ respectively. Which one of the following choices is true?



- A. $\nabla \cdot \vec{X} = 0, \nabla \times \vec{Y} \neq 0, \nabla \times \vec{Z} = 0$
 B. $\nabla \cdot \vec{X} \neq 0, \nabla \times \vec{Y} = 0, \nabla \times \vec{Z} \neq 0$
 C. $\nabla \cdot \vec{X} \neq 0, \nabla \times \vec{Y} \neq 0, \nabla \times \vec{Z} \neq 0$
 D. $\nabla \cdot \vec{X} = 0, \nabla \times \vec{Y} = 0, \nabla \times \vec{Z} = 0$

gate2017-ee-2 calculus field-vectors

1.23.3 Field Vectors: GATE Electrical 2020 | Question: 27

The vector function expressed by

$$\vec{F} = a_x (5y - k_1 z) + a_y (3z + k_2 x) + a_z (k_3 y - 4x)$$

represents a conservative field, where a_x, a_y, a_z are unit vectors along x, y and z directions, respectively. The values of constants k_1, k_2, k_3 are given by:

A. $k_1 = 3, k_2 = 3, k_3 = 7$

B. $k_1 = 3, k_2 = 8, k_3 = 5$

C. $k_1 = 4, k_2 = 5, k_3 = 3$

D. $k_1 = 0, k_2 = 0, k_3 = 0$

gate2020-ee calculus field-vectors

1.23.4 Field Vectors: GATE Electrical 2021 | Question: 5

Which one of the following vector functions represents a magnetic field $\vec{B}$?

($\hat{X}, \hat{Y}$ and $\hat{Z}$ are unit vectors along x-axis, y-axis, and z-axis, respectively)

A. $10x\hat{X} + 20y\hat{Y} - 30z\hat{Z}$

B. $10y\hat{X} + 20x\hat{Y} - 10z\hat{Z}$

C. $10z\hat{X} + 20y\hat{Y} - 30x\hat{Z}$

D. $10x\hat{X} - 30z\hat{Y} + 20y\hat{Z}$

gateee-2021 calculus field-vectors

1.24**Fourier Series (2)****1.24.1 Fourier Series: GATE Electrical 2016 Set 2 | Question: 10**

Let $f(x)$ be a real, periodic function satisfying $f(-x) = -f(x)$. The general form of its Fourier series representation would be

A. $f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx)$

B. $f(x) = \sum_{k=1}^{\infty} b_k \sin(kx)$

C. $f(x) = a_0 + \sum_{k=1}^{\infty} a_{2k} \cos(kx)$

D. $f(x) = \sum_{k=0}^{\infty} a_{2k+1} \sin(2k+1)x$

gate2016-ee-2 calculus fourier-series

1.24.2 Fourier Series: GATE Electrical 2019 | Question: 28

A periodic function $f(t)$, with a period of 2π , is represented as its Fourier series,

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt.$$

if

$$f(t) = \begin{cases} A \sin t, & 0 \leq t \leq \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

the Fourier series coefficients a_1 and b_1 of $f(t)$ are

A. $a_1 = \frac{A}{\pi}; b_1 = 0$

B. $a_1 = \frac{A}{2}; b_1 = 0$

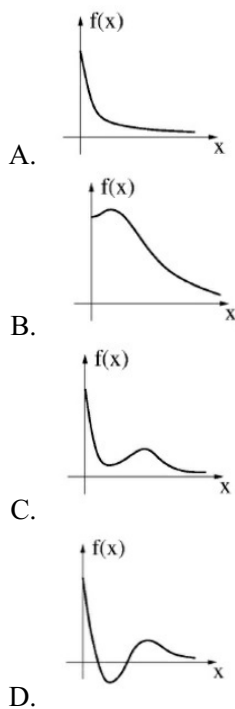
C. $a_1 = 0; b_1 = \frac{A}{\pi}$

D. $a_1 = 0; b_1 = \frac{A}{2}$

gate2019-ee calculus fourier-series

1.25**Functions (4)****1.25.1 Functions: GATE ECE 2015 Set 1 | Question: 28**

Which one of the following graphs describes the function $f(x) = e^{-x}(x^2 + x + 1)$?



gate2015-ec-1 calculus functions

1.25.2 Functions: GATE ECE 2016 Set 2 | Question: 4



How many distinct value of x satisfy the equation $\sin(x) = x/2$, where x is in radians?

- A. 1 B. 2 C. 3 D. 4 or more

gate2016-ec-2 calculus functions

1.25.3 Functions: GATE ME 2013 | Question: 25



Choose the CORRECT set of functions, which are linearly dependent.

- A. $\sin x, \sin^2 x$ and $\cos^2 x$ B. $\cos x, \sin x$ and $\tan x$
 C. $\cos 2x, \sin^2 x$ and $\cos^2 x$ D. $\cos 2x, \sin x$ and $\cos x$

gateme-2013 calculus functions

1.25.4 Functions: GATE Mechanical 2014 Set 3 | Question: 29



The real root of the equation $5x - 2\cos x - 1 = 0$ (up to two decimal accuracy) is _____

gateme-2014-set3 numerical-answers calculus functions

1.26 Functions of Single Variable (1)

1.26.1 Functions of Single Variable: GATE ME 2012 | Question: 24



At $x = 0$, the function $f(x) = x^3 + 1$ has

- A. a maximum value B. a minimum value
 C. a singularity D. a point of inflection

gateme-2012 calculus functions-of-single-variable maxima-minima

1.27 Gradient (3)

1.27.1 Gradient: GATE Civil 2013 | Question: 50

For a portion of national highway where a descending gradient of 1 in 25 meets with an ascending gradient of 1 in 20, a valley curve needs to be designed for a vehicle travelling at 90 kmph based on the following conditions.

- headlight sight distance equal to the stopping sight distance (SSD) of a level terrain considering length of valley curve $>$ SSD.
- comfort condition with allowable rate of change of centrifugal acceleration $\text{acceleration} = 0.5 \text{ m/sec}^3$.

Assume total reaction time = 2.5 seconds; coefficient of longitudinal friction of the pavement = 0.35; height of head light of the vehicle = 0.75 m; and beam angle 1° .

What is the length of valley curve (in m) based on the head light sight distance condition? _____

gate2013-ce numerical-answers calculus gradient

1.27.2 Gradient: GATE Civil 2013 | Question: 51

For a portion of national highway where a descending gradient of 1 in 25 meets with an ascending gradient of 1 in 20, a valley curve needs to be designed for a vehicle travelling at 90 kmph based on the following conditions.

- headlight sight distance equal to the stopping sight distance (SSD) of a level terrain considering length of valley curve $>$ SSD.
- comfort condition with allowable rate of change of centrifugal acceleration $\text{acceleration} = 0.5 \text{ m/sec}^3$.

Assume total reaction time = 2.5 seconds; coefficient of longitudinal friction of the pavement = 0.35; height of head light of the vehicle = 0.75 m; and beam angle 1° .

What is the length of valley curve (in m) based on the comfort condition? _____

gate2013-ce numerical-answers calculus gradient

1.27.3 Gradient: GATE Civil 2024 Set 1 | Question: 27

A vector field $\vec{p}$ and a scalar field r are given by

$$\vec{p} = (2x^2 - 3xy + z^2) \hat{i} + (2y^2 - 3yz + x^2) \hat{j} + (2z^2 - 3xz + x^2) \hat{k}$$
$$r = 6x^2 + 4y^2 - z^2 - 9xyz - 2xy + 3xz - yz$$

Consider the statements P and Q.

P: Curl of the gradient of the scalar field r is a null vector.

Q: Divergence of curl of the vector field $\vec{p}$ is zero.

Which one of the following options is CORRECT?

- | | |
|-----------------------------|-----------------------------|
| A. Both P and Q are FALSE | B. P is TRUE and Q is FALSE |
| C. P is FALSE and Q is TRUE | D. Both P and Q are TRUE |

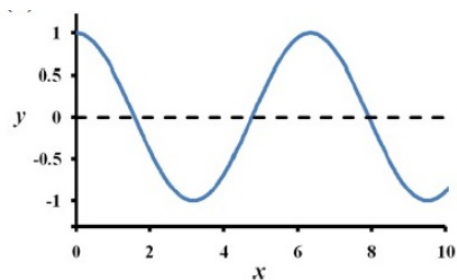
gatecivil-2024-set1 vector-calculus gradient divergence

Answer key

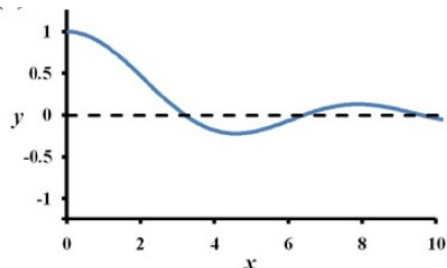
1.28.1 Graphs: GATE Chemical 2018 | Question: 4



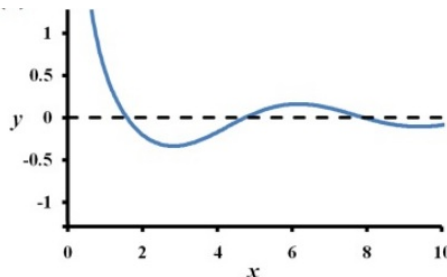
The figure which represents for $y = \frac{\sin x}{x}$ for $x > 0$ (x in radians) is



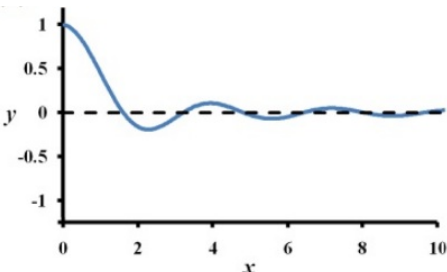
A.



B.



C.



D.

gate2018 calculus functions graphs

1.29

Initial and Boundary Value Problems (1)

1.29.1 Initial and Boundary Value Problems: GATE Mechanical 2014 Set 1 | Question: 27



If $y = f(x)$ is the solution of $\frac{d^2 y}{dx^2} = 0$ with the boundary conditions $y = 5$ at $x = 0$ and $\frac{dy}{dx} = 2$ at $x = 10$, $f(15) =$ _____.

gateme-2014-set1 numerical-answers calculus initial-and-boundary-value-problems

1.30

Integrals (2)

1.30.1 Integrals: GATE ECE 2019 | Question: 16

The value of the contour integral

$$\frac{1}{2\pi j} \oint \left(z + \frac{1}{z} \right)^2 dz$$

evaluated over the unit circle $|z| = 1$ is _____.

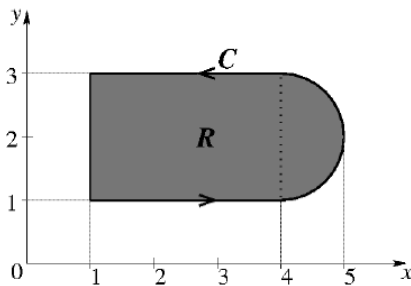
gate2019-ec numerical-answers calculus integrals

1.30.2 Integrals: GATE ECE 2019 | Question: 27

Consider the line integral

$$\int_C (x dy - y dx)$$

the integral being taken in a counterclockwise direction over the closed curve C that forms the boundary of the region R shown in the figure below. The region R is the area enclosed by the union of a 2×3 rectangle and a semi-circle of radius 1. The line integral evaluates to



- A. $6 + \frac{\pi}{2}$ B. $8 + \pi$ C. $12 + \pi$ D. $16 + 2\pi$

gate2019-ec integrals calculus

1.31 Integration (11)**1.31.1 Integration: GATE CSE 1993 | Question: 02.6**

The value of the double integral $\int_0^1 \int_0^{\frac{1}{x}} \frac{x}{1+y^2} dx dy$ is _____.

gate1993 calculus integration normal fill-in-the-blanks

Answer key

1.31.2 Integration: GATE CSE 1998 | Question: 8

- a. Find the points of local maxima and minima, if any, of the following function defined in $0 \leq x \leq 6$

$$x^3 - 6x^2 + 9x + 15$$

- b. Integrate

$$\int_{-\pi}^{\pi} x \cos x dx$$

$$\int_{\frac{1}{\pi}}^{\frac{2}{\pi}} \frac{\cos(1/x)}{x^2} dx$$

gatecse-2015-set1 calculus integration normal numerical-answers

Answer key 

1.31.9 Integration: GATE CSE 2015 Set 3 | Question: 45



If for non-zero x , $af(x) + bf(\frac{1}{x}) = \frac{1}{x} - 25$ where $a \neq b$ then $\int_1^2 f(x)dx$ is

- A. $\frac{1}{a^2-b^2} \left[a(\ln 2 - 25) + \frac{47b}{2} \right]$ B. $\frac{1}{a^2-b^2} \left[a(2\ln 2 - 25) - \frac{47b}{2} \right]$
 C. $\frac{1}{a^2-b^2} \left[a(2\ln 2 - 25) + \frac{47b}{2} \right]$ D. $\frac{1}{a^2-b^2} \left[a(\ln 2 - 25) - \frac{47b}{2} \right]$

gatecse-2015-set3 calculus integration normal

Answer key 

1.31.10 Integration: GATE CSE 2018 | Question: 16



The value of $\int_0^{\pi/4} x \cos(x^2) dx$ correct to three decimal places (assuming that $\pi = 3.14$) is ____

gatecse-2018 calculus integration normal numerical-answers 1-mark

Answer key 

1.31.11 Integration: GATE IT 2005 | Question: 35



What is the value of $\int_0^{2\pi} (x - \pi)^2 (\sin x) dx$

- A. -1 B. 0 C. 1 D. π

gateit-2005 calculus integration normal

Answer key 

1.32 Limits (26)

1.32.1 Limits: GATE BT 2021 | Question: 23



The value of $\lim_{x \rightarrow 0} \left[\frac{x - \sin 2x}{x - \sin 5x} \right]$ (rounded off to two decimal places) is _____

gatebt-2021 numerical-answers calculus limits

1.32.2 Limits: GATE CSE 1993 | Question: 02.1



$\lim_{x \rightarrow 0} \frac{x(e^x - 1) + 2(\cos x - 1)}{x(1 - \cos x)}$ is _____

gate1993 limits calculus normal fill-in-the-blanks

Answer key 

1.32.3 Limits: GATE CSE 1995 | Question: 7(B)



Compute without using power series expansion $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

gate1995 calculus limits numerical-answers

Answer key 

1.32.4 Limits: GATE CSE 2008 | Question: 1



$\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \cos x}$ equals

- A. 1 B. -1 C. ∞ D. $-\infty$

gatecse-2008 calculus limits easy

Answer key 

1.32.5 Limits: GATE CSE 2010 | Question: 5



What is the value of $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{2n}$?

- A. 0 B. e^{-2} C. $e^{-1/2}$ D. 1

gatecse-2010 calculus limits normal

Answer key 

1.32.6 Limits: GATE CSE 2015 Set 1 | Question: 4



$\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$ is

- A. ∞ B. 0 C. 1 D. Not defined

gatecse-2015-set1 calculus limits normal

Answer key 

1.32.7 Limits: GATE CSE 2015 Set 3 | Question: 9



The value of $\lim_{x \rightarrow \infty} (1 + x^2)^{e^{-x}}$ is

- A. 0 B. $\frac{1}{2}$ C. 1 D. ∞

gatecse-2015-set3 calculus limits normal

Answer key 

1.32.8 Limits: GATE CSE 2016 Set 1 | Question: 3



$$\lim_{x \rightarrow 4} \frac{\sin(x-4)}{x-4} = \underline{\hspace{2cm}}$$

gatecse-2016-set1 calculus limits easy numerical-answers

Answer key 

1.32.9 Limits: GATE CSE 2017 Set 1 | Question: 28



The value of $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

A. is 0

B. is -1

C. is 1

D. does not exist

gatecse-2017-set1 calculus limits normal

Answer key 

1.32.10 Limits: GATE CSE 2019 | Question: 13



Compute $\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

A. 1

B. 53/12

C. 108/7

D. Limit does not exist

gatecse-2019 engineering-mathematics calculus limits 1-mark

Answer key 

1.32.11 Limits: GATE CSE 2021 Set 1 | Question: 20



Consider the following expression.

$$\lim_{x \rightarrow -3} \frac{\sqrt{2x+22} - 4}{x+3}$$

The value of the above expression (rounded to 2 decimal places) is _____.

gatecse-2021-set1 calculus limits numerical-answers 1-mark

Answer key 

1.32.12 Limits: GATE CSE 2022 | Question: 24



The value of the following limit is _____.

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$

gatecse-2022 numerical-answers calculus limits 1-mark

Answer key 

1.32.13 Limits: GATE Chemical 2017 | Question: 1



The value of $\lim_{x \rightarrow 0} \frac{\tan(x)}{x}$ is _____.

gate2017 numerical-answers calculus limits

1.32.14 Limits: GATE Chemical 2018 | Question: 46



If $y = e^{-x^2}$ then the value of $\lim_{x \rightarrow \infty} \frac{1}{x} \frac{dy}{dx}$ is _____

gate2018 numerical-answers calculus limits

1.32.15 Limits: GATE Chemical 2019 | Question: 2



The value of the expression $\lim_{x \rightarrow \frac{\pi}{2}} \left| \frac{\tan x}{x} \right|$ is

A. ∞

B. 0

C. 1

D. -1

gate2019 calculus limits

1.32.16 Limits: GATE Civil 2021 Set 1 | Question: 18

Consider the limit:

$$\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$$

The limit (correct up to one decimal place) is _____

gatecivil-2021-set1 numerical-answers calculus limits

1.32.17 Limits: GATE Civil 2021 Set 2 | Question: 1

The value of $\lim_{x \rightarrow \infty} \frac{x \ln(x)}{1+x^2}$ is

- A. 0 B. 1.0 C. 0.5 D. ∞

gatecivil-2021-set2 calculus limits

1.32.18 Limits: GATE DS&AI 2024 | Question: 50

Evaluate the following limit:

$$\lim_{x \rightarrow 0} \frac{\ln((x^2 + 1) \cos x)}{x^2} =$$

gate-ds-ai-2024 numerical-answers limits engineering-mathematics

Answer key

1.32.19 Limits: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 5

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x-2}$$

- A. 0 B. $\sqrt{2}$ C. $\frac{1}{2\sqrt{2}}$ D. $\frac{1}{\sqrt{2}}$

gateda-sample-paper-2024 limits

Answer key

1.32.20 Limits: GATE ECE 2014 Set 2 | Question: 4

The value of

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

is

- A. $\ln 2$ B. 1.0 C. e D. ∞

gate2014-ec-2 calculus limits

1.32.21 Limits: GATE IN 2021 | Question: 2

Consider the sequence $x_n = 0.5x_{n-1} + 1, n = 1, 2, \dots$ with $x_0 = 0$. Then $\lim_{n \rightarrow \infty} x_n$ is

A. 0

B. 1

C. 2

D. ∞

gatein-2021 calculus limits

1.32.22 Limits: GATE ME 2012 | Question: 12

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \right) \text{ is}$$

A. $1/4$ B. $1/2$

C. 1

D. 2

gateme-2012 calculus limits

1.32.23 Limits: GATE Mechanical 2014 Set 1 | Question: 2

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{1 - \cos x} \text{ is}$$

A. 0

B. 1

C. 3

D. not defined

gateme-2014-set1 calculus limits

1.32.24 Limits: GATE Mechanical 2014 Set 2 | Question: 2

$$\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{\sin(4x)} \right) \text{ is equal to}$$

A. 0

B. 0.5

C. 1

D. 2

gateme-2014-set2 calculus limits

1.32.25 Limits: GATE Mechanical 2014 Set 3 | Question: 2

If a function is continuous at a point,

- A. the limit of the function may not exist at the point
- B. the function must be derivable at the point
- C. the limit of the function at the point tends to infinity
- D. the limit must exist at the point and the value of limit should be same as the value of the function at that point

gateme-2014-set3 calculus limits

1.32.26 Limits: GATE Mechanical 2021 Set 1 | Question: 2

$$\text{The value of } \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \right) \text{ is}$$

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$

D. 1

gateme-2021-set1 calculus limits

1.33**Line Integral (5)****1.33.1 Line Integral: GATE Chemical 2015 | Question: 37**

A vector $u = -2y\hat{i} + 2x\hat{j}$, where $\hat{i}$ and $\hat{j}$ are unit vectors in x and y directions, respectively. Evaluate the line integral

$$I = \oint_C u \cdot dr$$

where C is a closed loop formed by connecting points $(1, 1)$, $(3, 1)$, $(3, 2)$ and $(1, 2)$ in that order. The value of I is _____.

gate2015 numerical-answers calculus line-integral vector-identities

1.33.2 Line Integral: GATE Electrical 2014 Set 1 | Question: 28



The line integral of function $F = yzi$, in the counterclockwise direction, along the circle $x^2 + y^2 = 1$ at $z = 1$ is

- A. -2π B. $-\pi$ C. π D. 2π

gate2014-ee-1 calculus line-integral circle-equation

1.33.3 Line Integral: GATE Electrical 2016 Set 2 | Question: 9



The value of the line integral

$$\int_C (2xy^2 dx + 2x^2 y dy + dz)$$

along a path joining the origin $(0, 0, 0)$ and the point $(1, 1, 1)$ is

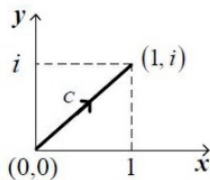
- A. 0 B. 2 C. 4 D. 6

gate2016-ee-2 calculus line-integral

1.33.4 Line Integral: GATE Electrical 2017 Set 1 | Question: 28



Consider the line integral $I = \int_C (x^2 + iy^2) dz$, where $z = x + iy$. The line c is shown in the figure below.



The value of I is

- A. $\frac{1}{2}i$ B. $\frac{2}{3}i$ C. $\frac{3}{4}i$ D. $\frac{4}{5}i$

gate2017-ee-1 calculus line-integral

1.33.5 Line Integral: GATE Electrical 2019 | Question: 18



If $f = 2x^3 + 3y^2 + 4z$, the value of line integral $\int_C \text{grad} f \cdot d\mathbf{r}$ evaluated over contour C formed by the segments $(-3, -3, 2) \rightarrow (2, -3, 2) \rightarrow (2, 6, 2) \rightarrow (2, 6, -1)$ is _____.

gate2019-ee numerical-answers calculus line-integral

1.34 Matrices (1)

1.34.1 Matrices: GATE Electrical 2015 Set 1 | Question: 3



If the sum of the diagonal elements of a 2×2 matrix is -6 , then the maximum possible value of determinant of the matrix is _____.

1.35

Maxima Minima (32)

1.35.1 Maxima Minima: GATE CSE 1987 | Question: 1-xxvi



If $f(x_i) \cdot f(x_{i+1}) < 0$ then

- A. There must be a root of $f(x)$ between x_i and x_{i+1}
- B. There need not be a root of $f(x)$ between x_i and x_{i+1}
- C. The fourth derivative of $f(x)$ with respect to x vanishes at x_i
- D. The fourth derivative of $f(x)$ with respect to x vanishes at x_{i+1}

gate1987 calculus maxima-minima

Answer key

1.35.2 Maxima Minima: GATE CSE 1995 | Question: 1.21



In the interval $[0, \pi]$ the equation $x = \cos x$ has

- A. No solution
- B. Exactly one solution
- C. Exactly two solutions
- D. An infinite number of solutions

gate1995 calculus normal maxima-minima

Answer key

1.35.3 Maxima Minima: GATE CSE 1995 | Question: 25a



Find the minimum value of $3 - 4x + 2x^2$.

gate1995 calculus maxima-minima easy descriptive

Answer key

1.35.4 Maxima Minima: GATE CSE 1997 | Question: 4.1



What is the maximum value of the function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$?

- A. 6
- B. 10
- C. 12
- D. 5.5

gate1997 calculus maxima-minima normal

Answer key

1.35.5 Maxima Minima: GATE CSE 2008 | Question: 25



A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is

- A. 0
- B. 1
- C. 2
- D. 3

gatecse-2008 calculus maxima-minima easy

Answer key

1.35.6 Maxima Minima: GATE CSE 2012 | Question: 9



Consider the function $f(x) = \sin(x)$ in the interval $x = \left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$. The number and location(s) of the local minima of this function are

- A. One, at $\frac{\pi}{2}$
 C. Two, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

gatecse-2012 calculus maxima-minima normal

Answer key 

- B. One, at $\frac{3\pi}{2}$
 D. Two, at $\frac{\pi}{4}$ and $\frac{3\pi}{2}$

1.35.7 Maxima Minima: GATE CSE 2015 Set 2 | Question: GA-3



Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

- A. 0, -1 B. -1, 0 C. 0, 1 D. -1, 2

gatecse-2015-set2 set-theory&algebra functions normal maxima-minima

Answer key 

1.35.8 Maxima Minima: GATE CSE 2020 | Question: 1



Consider the functions

- I. e^{-x}
 II. $x^2 - \sin x$
 III. $\sqrt{x^3 + 1}$

Which of the above functions is/are increasing everywhere in $[0, 1]$?

- A. III only B. II only
 C. II and III only D. I and III only

gatecse-2020 engineering-mathematics calculus maxima-minima 1-mark

Answer key 

1.35.9 Maxima Minima: GATE CSE 2023 | Question: 18



Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

- A. $f(x)$ does not have a local maximum. B. $f(x)$ has a local maximum.
 C. $f(x)$ does not have a local minimum. D. $f(x)$ has a local minimum.

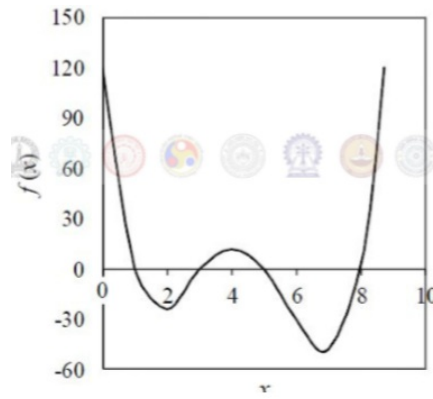
gatecse-2023 calculus maxima-minima multiple-selects 1-mark

Answer key 

1.35.10 Maxima Minima: GATE Chemical 2017 | Question: 3



The number of positive roots of the function $f(x)$ shown below in the range $0 < x < 6$ is _____.



gate2017 numerical-answers calculus maxima-minima

1.35.11 Maxima Minima: GATE Chemical 2020 | Question: 27



The maximum value of the function $f(x) = -\frac{5}{3}x^3 + 10x^2 - 15x + 16$ in the interval $(0.5, 3.5)$ is

- A. 0 B. 8 C. 16 D. 48

gate2020 calculus maxima-minima

1.35.12 Maxima Minima: GATE DS&AI 2024 | Question: 5



For any twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, if at some $x^* \in \mathbb{R}$, $f'(x^*) = 0$ and $f''(x^*) > 0$, then the function f necessarily has a _____ at $x = x^*$.

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. local minimum B. global minimum
C. local maximum D. global maximum

gate-ds-ai-2024 calculus maxima-minima

Answer key

1.35.13 Maxima Minima: GATE ECE 2010 | Question: 26



If $e^y = x^{\frac{1}{x}}$, then y has a

- A. maximum at $x = e$ B. minimum at $x = e$
C. maximum at $x = e^{-1}$ D. minimum at $x = e^{-1}$

gate2010-ec calculus maxima-minima

1.35.14 Maxima Minima: GATE ECE 2012 | Question: 46



The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

- A. 21 B. 25 C. 41 D. 46

gate2012-ec calculus maxima-minima

1.35.15 Maxima Minima: GATE ECE 2014 Set 2 | Question: 3



For $0 \leq t < \infty$, the maximum value of the function $f(t) = e^{-t} - 2e^{-2t}$ occurs at

- A. $t = \log_e 4$ B. $t = \log_e 2$ C. $t = 0$ D. $t = \log_e 8$

gate2014-ec-2 calculus maxima-minima

1.35.16 Maxima Minima: GATE ECE 2014 Set 3 | Question: 1

The maximum value of the function $f(x) = \ln(1+x) - x$ (where $x > -1$) occurs at $x =$ _____.

gate2014-ec-3 calculus maxima-minima numerical-answers

1.35.17 Maxima Minima: GATE ECE 2014 Set 3 | Question: 26

The maximum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the interval $0 \leq x \leq 3$ is _____.

gate2014-ec-3 calculus maxima-minima numerical-answers

1.35.18 Maxima Minima: GATE ECE 2014 Set 4 | Question: 29

For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the triangle, the angle between the hypotenuse and the side is

- A. 12° B. 36° C. 60° D. 45°

gate2014-ec-4 calculus maxima-minima

1.35.19 Maxima Minima: GATE ECE 2016 Set 2 | Question: 3

As x varies from -1 to $+3$, which one of the following describes the behaviour of the function $f(x) = x^3 - 3x^2 + 1$?

- A. $f(x)$ increases monotonically.
B. $f(x)$ increases, then decreases and increases again.
C. $f(x)$ decreases, then increases and decreases again.
D. $f(x)$ increases and then decreases.

gate2016-ec-2 calculus maxima-minima

1.35.20 Maxima Minima: GATE ECE 2017 Set 2 | Question: 30

The minimum value of the function $f(x) = \frac{1}{3}x(x^2 - 3)$ in the interval $-100 \leq x \leq 100$ occurs at $x =$ _____.

gate2017-ec-2 numerical-answers calculus maxima-minima

1.35.21 Maxima Minima: GATE ECE 2019 | Question: 26

Consider a differentiable function $f(x)$ on the set of real numbers, such that $f(-1) = 0$ and $|f'(x)| \leq 2$. Given these conditions, which one of the following inequalities is necessarily true for all $x \in [-2, 2]$?

- A. $f(x) \leq \frac{1}{2} |x+1|$ B. $f(x) \leq 2 |x+1|$
C. $f(x) \leq \frac{1}{2} |x|$ D. $f(x) \leq 2 |x|$

gate2019-ec calculus maxima-minima

1.35.22 Maxima Minima: GATE Electrical 2012 | Question: 27

The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

- A. 21 B. 25 C. 41 D. 46

gate2012-ee calculus maxima-minima

Answer key

1.35.23 Maxima Minima: GATE Electrical 2014 Set 1 | Question: 2

Let $f(x) = xe^{-x}$. The maximum value of the function in the interval $(0, \infty)$ is

A. e^{-1} B. e C. $1 - e^{-1}$ D. $1 + e^{-1}$

gate2014-ee-1 calculus maxima-minima

1.35.24 Maxima Minima: GATE Electrical 2014 Set 2 | Question: 28

The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is

A. 20

B. 28

C. 16

D. 32

gate2014-ee-2 calculus maxima-minima

1.35.25 Maxima Minima: GATE Electrical 2014 Set 2 | Question: 3

Minimum of the real valued function $f(x) = (x - 1)^{2/3}$ occurs at x equal to

A. $-\infty$

B. 0

C. 1

D. ∞

gate2014-ee-2 calculus maxima-minima

1.35.26 Maxima Minima: GATE Electrical 2016 Set 1 | Question: 1

The maximum value attained by the function. $f(x) = x(x - 1)(x - 2)$ in the interval $[1, 2]$ is _____.

gate2016-ee-1 calculus maxima-minima numerical-answers

1.35.27 Maxima Minima: GATE Electrical 2018 | Question: 43

Let $f(x) = 3x^3 - 7x^2 + 5x + 6$. The maximum value of $f(x)$ over the interval $[0, 2]$ is _____ (up to one decimal place).

gate2018-ee numerical-answers calculus maxima-minima

1.35.28 Maxima Minima: GATE Electrical 2020 | Question: 26

For real numbers, x and y , with $y = 3x^2 + 3x + 1$, the maximum and minimum value of y for $x \in [-2, 0]$ are respectively, _____.

A. 7 and $1/4$

B. 7 and 1

C. -2 and $-1/2$ D. 1 and $1/4$

gate2020-ee calculus maxima-minima

1.35.29 Maxima Minima: GATE Electrical 2021 | Question: 3

Let $f(x)$ be a real-valued function such that $f'(x_0) = 0$ for some $x_0 \in (0, 1)$, and $f''(x) > 0$ for all $x \in (0, 1)$. Then $f(x)$ has

A. no local minimum in $(0, 1)$ B. one local maximum in $(0, 1)$ C. exactly one local minimum in $(0, 1)$ D. two distinct local minima in $(0, 1)$

gateee-2021 calculus maxima-minima

1.35.30 Maxima Minima: GATE IN 2021 | Question: 23

Consider the function $f(x) = -x^2 + 10x + 100$. The minimum value of the function in interval $[5, 10]$ is _____.

gatein-2021 numerical-answers calculus maxima-minima

Answer key

1.35.31 Maxima Minima: GATE IT 2008 | Question: 31

If $f(x)$ is defined as follows, what is the minimum value of $f(x)$ for $x \in (0, 2]$?

$$f(x) = \begin{cases} \frac{25}{8x} & \text{when } x \leq \frac{3}{2} \\ x + \frac{1}{x} & \text{otherwise} \end{cases}$$

A. 2

B. $2\frac{1}{12}$

C. $2\frac{1}{6}$

D. $2\frac{1}{2}$

gateit-2008 calculus maxima-minima normal

Answer key 

1.35.32 Maxima Minima: GATE Mechanical 2021 Set 1 | Question: 34



Let $f(x) = x^2 - 2x + 2$ be a continuous function defined on $x \in [1, 3]$. The point x at which the tangent of $f(x)$ becomes parallel to the straight line joining $f(1)$ and $f(3)$ is

A. 0

B. 1

C. 2

D. 3

gateme-2021-set1 calculus maxima-minima

1.36

Mean Value Theorem (1)

1.36.1 Mean Value Theorem: GATE ECE 2015 Set 1 | Question: 2



A function $f(x) = 1 - x^2 + x^3$ is defined in the closed interval $[-1, 1]$. The value of x , in the open interval $(-1, 1)$ for which the mean value theorem is satisfied, is

A. $-1/2$

B. $-1/3$

C. $1/3$

D. $1/2$

gate2015-ec-1 calculus mean-value-theorem

1.37

Out of Gatecse Syllabus (4)

1.37.1 Out of Gatecse Syllabus: GATE CSE 1993 | Question: 01.5



Fourier series of the periodic function (period 2π) defined by

$$f(x) = \begin{cases} 0, & -p < x < 0 \\ x, & 0 < x < p \end{cases} \text{ is}$$

$$\frac{\pi}{4} + \sum \left[\frac{1}{\pi n^2} (\cos n\pi - 1) \cos nx - \frac{1}{n} \cos n\pi \sin nx \right]$$

But putting $x = \pi$, we get the sum of the series

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \text{ is}$$

A. $\frac{\pi^2}{4}$

B. $\frac{\pi^2}{6}$

C. $\frac{\pi^2}{8}$

D. $\frac{\pi^2}{12}$

gate1993 calculus normal out-of-gatecse-syllabus multiple-selects

Answer key 

1.37.2 Out of Gatecse Syllabus: GATE CSE 1993 | Question: 01.7



The function $f(x, y) = x^2y - 3xy + 2y + x$ has

A. no local extremum

B. one local minimum but no local maximum

C. one local maximum but no local minimum

D. one local minimum and one local maximum

gate1993 calculus maxima-minima normal out-of-gatecse-syllabus multiple-selects

Answer key 

1.37.3 Out of Gatecse Syllabus: GATE CSE 1993 | Question: 02.8

Given $\vec{v} = x \cos^2 y \hat{i} + x^2 e^z \hat{j} + z \sin^2 y \hat{k}$ and S the surface of a unit cube with one corner at the origin and edges parallel to the coordinate axes, the value of integral $\int_0^1 \int_S \vec{V} \cdot \hat{n} dS$ is _____.

gate1993 calculus normal out-of-gatecse-syllabus fill-in-the-blanks

1.37.4 Out of Gatecse Syllabus: GATE CSE 1995 | Question: 2.18

The solution of differential equation $y'' + 3y' + 2y = 0$ is of the form

- A. $C_1 e^x + C_2 e^{2x}$ B. $C_1 e^{-x} + C_2 e^{3x}$
C. $C_1 e^{-x} + C_2 e^{-2x}$ D. $C_1 e^{-2x} + C_2 e^{-x}$

gate1995 calculus out-of-gatecse-syllabus

Answer key

1.38**Partial Derivatives (3)****1.38.1 Partial Derivatives: GATE Chemical 2013 | Question: 7**

An equation of state is explicit in pressure p and cubic in the specific volume v . At the critical point ' c ', the isotherm passing through ' c ' satisfies

- A. $\frac{\partial p}{\partial v} < 0, \frac{\partial^2 p}{\partial v^2} = 0$
B. $\frac{\partial p}{\partial v} > 0, \frac{\partial^2 p}{\partial v^2} > 0$
C. $\frac{\partial p}{\partial v} = 0, \frac{\partial^2 p}{\partial v^2} > 0$
D. $\frac{\partial p}{\partial v} = 0, \frac{\partial^2 p}{\partial v^2} = 0$

gate2013 calculus partial-derivatives

1.38.2 Partial Derivatives: GATE ECE 2018 | Question: 52

Let $r = x^2 + y - z$ and $z^3 - xy + yz + y^3 = 1$. Assume that x and y are independent variables. At $(x, y, z) = (2, -1, 1)$, the value (correct to two decimal places) of $\frac{\partial r}{\partial x}$ is _____.

gate2018-ec numerical-answers calculus partial-derivatives

1.38.3 Partial Derivatives: GATE Mechanical 2021 Set 2 | Question: 11

For a two-dimensional, incompressible flow having velocity components u and v in the x and y directions, respectively, the expression

$$\frac{\partial(u^2)}{\partial x} + \frac{\partial(uv)}{\partial y}$$

can be simplified to

- A. $u \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial y}$
B. $2u \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial y}$
C. $2u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}$
D. $u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}$

Answer key 

1.39

Polynomials (5)

1.39.1 Polynomials: GATE CSE 1987 | Question: 1-xxii



The equation $7x^7 + 14x^6 + 12x^5 + 3x^4 + 12x^3 + 10x^2 + 5x + 7 = 0$ has

- A. All complex roots
 B. At least one real root
 C. Four pairs of imaginary roots
 D. None of the above

gate1987 calculus polynomials

Answer key 

1.39.2 Polynomials: GATE CSE 1995 | Question: 2.8



If the cube roots of unity are $1, \omega$ and ω^2 , then the roots of the following equation are

$$(x - 1)^3 + 8 = 0$$

- A. $-1, 1 + 2\omega, 1 + 2\omega^2$
 B. $1, 1 - 2\omega, 1 - 2\omega^2$
 C. $-1, 1 - 2\omega, 1 - 2\omega^2$
 D. $-1, 1 + 2\omega, -1 + 2\omega^2$

gate1995 calculus normal polynomials

Answer key 

1.39.3 Polynomials: GATE ECE 2013 | Question: 20



A polynomial $f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$ with all coefficients positive has

- A. no real roots
 B. no negative real root
 C. odd number of real roots
 D. at least one positive and one negative real root

gate2013-ec calculus polynomials

1.39.4 Polynomials: GATE Electrical 2020 | Question: 1



$ax^3 + bx^2 + cx + d$ is a polynomial on real x over real coefficients a, b, c, d wherein $a \neq 0$. Which of the following statements is true?

- A. d can be chosen to ensure that $x = 0$ is a root for any given set a, b, c .
 B. No choice of coefficient can make all roots identical.
 C. a, b, c, d can be chosen to ensure that all roots are complex.
 D. c alone cannot ensure that all roots are real.

gate2020-ee calculus polynomials

Answer key 

1.39.5 Polynomials: GATE Electrical 2021 | Question: 26



In the open interval $(0, 1)$, the polynomial $p(x) = x^4 - 4x^3 + 2$ has

- A. two real roots
 B. one real root
 C. three real roots
 D. no real roots

gateee-2021 calculus polynomials

1.40

Routh Hurwitz Array (1)

1.40.1 Routh Hurwitz Array: GATE Electrical 2014 Set 1 | Question: 17

In the formation of Routh-Hurwitz array for a polynomial, all the elements of a row have zero values. This premature termination of the array indicates the presence of

- A. only one root at the origin
- B. Imaginary roots
- C. only positive real roots
- D. only negative real roots

gate2014-ee-1 calculus polynomial routh-hurwitz-array

1.41**Sequences and Series (1)****1.41.1 Sequences and Series: GATE BT 2021 | Question: 20**

The sum of the infinite geometric series $1 + 1/3 + 1/3^2 + 1/3^3 + \dots$ (rounded off to one decimal place) is _____

gatebt-2021 numerical-answers calculus sequences-and-series infinite-geometric-series

1.42**Taylor Series (6)****1.42.1 Taylor Series: GATE CH 2021 | Question: 2**

The function $\cos(x)$ is approximated using Taylor series around $x = 0$ as $\cos(x) \approx 1 + ax + bx^2 + cx^3 + dx^4$. The values of a, b, c and d are

- A. $a = 1, b = -0.5, c = -1, d = -0.25$
- B. $a = 0, b = -0.5, c = 0, d = 0.042$
- C. $a = 0, b = 0.5, c = 0, d = 0.042$
- D. $a = -0.5, b = 0, c = 0.042, d = 0$

gatech-2021 calculus taylor-series

1.42.2 Taylor Series: GATE Chemical 2012 | Question: 3

For the function $f(t) = e^{-t}/\tau$, the Taylor series approximation for $t \ll \tau$ is

- A. $1 + \frac{t}{\tau}$
- B. $1 - \frac{t}{\tau}$
- C. $1 - \frac{t^2}{2\tau^2}$
- D. $1 + t$

gate2012 calculus taylor-series

1.42.3 Taylor Series: GATE ECE 2014 Set 1 | Question: 26

The Taylor series expansion of $3\sin x + 2\cos x$ is

- A. $2 + 3x - x^2 - \frac{x^3}{2} + \dots$
- B. $2 - 3x + x^2 - \frac{x^3}{2} + \dots$
- C. $2 + 3x + x^2 + \frac{x^3}{2} + \dots$
- D. $2 - 3x - x^2 + \frac{x^3}{2} + \dots$

gate2014-ec-1 calculus taylor-series

1.42.4 Taylor Series: GATE ECE 2015 Set 3 | Question: 5

The value of $\sum_{n=0}^{\infty} n \left(\frac{1}{2}\right)^n$ is _____.

gate2015-ec-3 numerical-answers calculus taylor-series

1.42.5 Taylor Series: GATE ECE 2017 Set 1 | Question: 26

Let $f(x) = e^{x+x^2}$ for real x . From among the following, choose the Taylor series approximation of $f(x)$ around $x = 0$, which includes all powers of x less than or equal to 3.

- A. $1 + x + x^2 + x^3$ B. $1 + x + \frac{3}{2}x^2 + x^3$
 C. $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$ D. $1 + x + 3x^2 + 7x^3$

gate2017-ec-1 calculus taylor-series

1.42.6 Taylor Series: GATE ECE 2018 | Question: 24

Taylor series expansion of $f(x) = \int_0^x e^{-\left(\frac{t^2}{2}\right)} dt$ around $x = 0$ has the form

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

The coefficient a_2 (correct to two decimal places) is equal to _____.

gate2018-ec numerical-answers calculus taylor-series

1.43 Triangles (1)**1.43.1 Triangles: GATE BT 2021 | Question: 41**

If the area of a triangle with the vertices $(k, 0)$, $(2, 0)$ and $(0, -2)$ is 2 square units, the value of k is _____.

gatebt-2021 numerical-answers calculus geometry triangles area

1.44 Tripple Integrals (1)**1.44.1 Tripple Integrals: GATE Civil 2021 Set 1 | Question: 19**

The volume determined from $\int \int \int_V 8xyz \, dV$ for $V = [2, 3] \times [1, 2] \times [0, 1]$ will be (in integer) _____.

gatecivil-2021-set1 numerical-answers calculus tripple-integrals volume

1.45 Unit Vectors (2)**1.45.1 Unit Vectors: GATE Chemical 2015 | Question: 15**

A scalar function in the xy -plane is given by $\phi(x, y) = x^2 + y^2$. If $\hat{i}$ and $\hat{j}$ are unit vectors in the x and y directions, the direction of maximum increase in the value of ϕ at $(1, 1)$ is along:

- A. $-2\hat{i} + 2\hat{j}$ B. $2\hat{i} + 2\hat{j}$
 C. $-2\hat{i} - 2\hat{j}$ D. $2\hat{i} - 2\hat{j}$

gate2015 calculus vector-calculus unit-vectors

1.45.2 Unit Vectors: GATE Chemical 2017 | Question: 4

Let $\hat{i}$ and $\hat{j}$ be the unit vectors in the x and y directions, respectively. For the function $F(x, y) = x^3 + y^2$ the gradient of the function, i.e., ∇F is given by

- A. $3x^2\hat{i} - 2y\hat{j}$ B. $6x^2y$ C. $3x^2\hat{i} + 2y\hat{j}$ D. $2y\hat{i} - 3x^2\hat{j}$

gate2017 calculus vector-calculus unit-vectors gradient

1.46 Vector Identities (10)

1.46.1 Vector Identities: GATE CH 2021 | Question: 5

A three-dimensional velocity field is given by $V = 5x^2y\mathbf{i} + Cy\mathbf{j} - 10xyz\mathbf{k}$, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the unit vectors in x, y, z directions, respectively, describing a cartesian coordinate system. The coefficient C is a constant. If V describes an incompressible fluid flow, the value of C is

- A. -1 B. 0 C. 1 D. 5

gatech-2021 calculus vector-calculus vector-identities

1.46.2 Vector Identities: GATE Chemical 2015 | Question: 11

The following set of the three vectors

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} x \\ 6 \\ x \end{pmatrix} \text{ and } \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix},$$

is linearly dependent when x is equal to

- A. 0 B. 1 C. 2 D. 3

gate2015 calculus vector-calculus vector-identities

1.46.3 Vector Identities: GATE Chemical 2020 | Question: 3

Consider the following continuously differentiable function

$$\mathbf{v}(x, y, z) = 3x^2y\mathbf{i} + 8y^2z\mathbf{j} + 5xyz\mathbf{k}$$

where $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ represent the respective unit vectors along the x, y , and z directions in the Cartesian coordinate system. The curl of this function is

- A. $-3x^2\mathbf{i} - 8y^2\mathbf{j} + 5z(x+y)\mathbf{k}$ B. $6xy\mathbf{i} - 16yz\mathbf{j} + 5xy\mathbf{k}$
C. $(5xz - 8y^2)\mathbf{i} - 5yz\mathbf{j} - 3x^2\mathbf{k}$ D. $y(11x + 16z)$

gate2020 calculus vector-calculus vector-identities

1.46.4 Vector Identities: GATE Civil 2012 | Question: 29

For the parallelogram OPQR shown in the sketch, $\overrightarrow{OP} = a\hat{i} + b\hat{j}$ and $\overrightarrow{OR} = c\hat{i} + d\hat{j}$. The area of the parallelogram is

- A. $ad - bc$ B. $ac + bd$ C. $ad + bc$ D. $ab - cd$

gate2012-ce calculus vector-identities

1.46.5 Vector Identities: GATE Civil 2021 Set 2 | Question: 3

The unit normal vector to the surface $X^2 + Y^2 + Z^2 - 48 = 0$ at the point $(4, 4, 4)$ is

- A. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$
B. $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$
C. $\frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}$
D. $\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}$

gatecivil-2021-set2 calculus vector-calculus vector-identities unit-normal-vector

1.46.6 Vector Identities: GATE ME 2012 | Question: 25

For the spherical surface $x^2 + y^2 + z^2 = 1$, the unit outward normal vector at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ is given by

- A. $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$
 B. $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$
 C. $\hat{k}$
 D. $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}$

gateme-2012 calculus vector-identities

1.46.7 Vector Identities: GATE Mechanical 2014 Set 1 | Question: 5

Which one of the following describes the relationship among the three vectors, $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $5\hat{i} + 6\hat{j} + 4\hat{k}$?

- A. The vectors are mutually perpendicular
 B. The vectors are linearly dependent
 C. The vectors are linearly independent
 D. The vectors are unit vectors

gateme-2014-set1 calculus vector-identities

1.46.8 Vector Identities: GATE Mechanical 2014 Set 2 | Question: 3

Curl of vector $\vec{F} = x^2 z^2 \hat{i} - 2xy^2 z \hat{j} + 2y^2 z^3 \hat{k}$ is

- A. $(4yz^3 + 2xy^2)\hat{i} + 2x^2 z \hat{j} - 2y^2 z \hat{k}$
 B. $(4yz^3 + 2xy^2)\hat{i} - 2x^2 z \hat{j} - 2y^2 z \hat{k}$
 C. $2xz^2 \hat{i} - 4xyz \hat{j} + 6y^2 z^2 \hat{k}$
 D. $2xz^2 \hat{i} + 4xyz \hat{j} + 6y^2 z^2 \hat{k}$

gateme-2014-set2 calculus vector-identities

1.46.9 Vector Identities: GATE Mechanical 2014 Set 3 | Question: 3

Divergence of the vector field $x_2 z \hat{i} + xy \hat{j} - yz \hat{k}$ at $(1, -1, 1)$ is

- A. 0
 B. 3
 C. 5
 D. 6

gateme-2014-set3 calculus vector-identities divergence-and-curl

1.46.10 Taylor Series: GATE Civil 2012 | Question: 3

The infinite series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ corresponds to

- A. $\sec x$
 B. e^x
 C. $\cos x$
 D. $1 + \sin^2 x$

gate2012-ce calculus taylor-series

1.47**Volume Integral (1)****1.47.1 Volume Integral: GATE Electrical 2015 Set 2 | Question: 26**

The volume enclosed by the surface $f(x, y) = e^x$ over the triangle bounded by the lines $x = y$; $x = 0$; $y = 1$ in the xy plane is _____.

gate2015-ee-2 calculus volume-integral numerical-answers

Answer Keys

1.0.1	2.90 : 3.10	1.0.2	C	1.0.3	A	1.0.4	A	1.1.1	A
1.2.1	3:3	1.2.2	D	1.3.1	D	1.4.1	C	1.4.2	C
1.4.3	C	1.4.4	C	1.4.5	B	1.5.1	N/A	1.5.2	B
1.5.3	A	1.5.4	A	1.5.5	A	1.5.6	C	1.5.7	19 : 19
1.5.8	C	1.5.9	A	1.6.1	A;B;C	1.6.2	B	1.6.3	B
1.6.4	D	1.6.5	C	1.7.1	C	1.8.1	C	1.9.1	2 : 2
1.9.2	Q-Q	1.9.3	N/A	1.10.1	D	1.11.1	7 : 7	1.11.2	5.7 : 5.8
1.11.3	1 : 1	1.12.1	0	1.12.2	B	1.12.3	B	1.12.4	0.0 : 0.0
1.12.5	B	1.12.6	A	1.12.7	D	1.12.8	0.49 : 0.51	1.12.9	0.0 : 0.0
1.12.10	D	1.13.1	2 : 2	1.13.2	C	1.13.3	C	1.13.4	C
1.13.5	0.19 : 0.21	1.13.6	0.95 : 1.05	1.13.7	3 : 3	1.13.8	18 : 22	1.13.9	2.0 : 2.0
1.13.10	C	1.13.11	1.99 : 2.01	1.13.12	2.25 : 2.25	1.13.13	A	1.13.14	C
1.13.15	C	1.13.16	A	1.13.17	B	1.13.18	B	1.13.19	B
1.14.1	2.0 : 2.0	1.15.1	B	1.15.2	A	1.15.3	D	1.15.4	B
1.15.5	B	1.15.6	40 : 40	1.15.7	B	1.16.1	A	1.17.1	X
1.18.1	D	1.18.2	-2	1.18.3	C	1.18.4	9	1.18.5	C
1.18.6	D	1.19.1	6.52:6.54	1.19.2	1 : 1	1.19.3	A	1.20.1	2.6:2.7
1.20.2	45 : 45	1.21.1	B	1.21.2	0.99 : 1.01	1.22.1	B	1.23.1	A
1.23.2	C	1.23.3	C	1.23.4	A	1.24.1	B	1.24.2	D
1.25.1	B	1.25.2	C	1.25.3	C	1.25.4	0.53 : 0.56	1.26.1	D
1.27.1	308 : 311	1.27.2	106 : 107	1.27.3	D	1.28.1	B	1.29.1	34 : 36
1.30.1	-0.0001 : 0.0001	1.30.2	C	1.31.1	N/A	1.31.2	N/A	1.31.3	A
1.31.4	D	1.31.5	D	1.31.6	A	1.31.7	4	1.31.8	-1
1.31.9	A	1.31.10	0.288 : 0.289	1.31.11	B	1.32.1	0.25 : 0.25	1.32.2	1
1.32.3	1	1.32.4	A	1.32.5	B	1.32.6	C	1.32.7	C
1.32.8	1	1.32.9	C	1.32.10	C	1.32.11	0.25 : 0.25	1.32.12	-0.5
1.32.13	1:1	1.32.14	0:0	1.32.15	A	1.32.16	0.5 : 0.5	1.32.17	A
1.32.18	0.5	1.32.19	C	1.32.20	C	1.32.21	C	1.32.22	B
1.32.23	A	1.32.24	B	1.32.25	D	1.32.26	C	1.33.1	8
1.33.2	B	1.33.3	B	1.33.4	B	1.33.5	139	1.34.1	9

1.35.1	A
1.35.6	D
1.35.11	C
1.35.16	-0.01 : 0.01
1.35.21	B
1.35.26	0.0 : 0.0
1.35.31	B
1.37.3	N/A
1.39.1	B
1.40.1	B
1.42.4	2 : 2
1.45.1	B
1.46.4	A
1.46.9	C

1.35.2	B
1.35.7	C
1.35.12	A
1.35.17	5.9 : 6.1
1.35.22	C
1.35.27	11.5 : 12.5
1.35.32	C
1.37.4	C
1.39.2	C
1.41.1	1.5 : 1.5
1.42.5	C
1.45.2	C
1.46.5	B
1.46.10	B

1.35.3	1
1.35.8	A
1.35.13	Q-Q
1.35.18	C
1.35.23	A
1.35.28	A
1.36.1	B
1.38.1	D
1.39.3	D
1.42.1	B
1.42.6	-0.01 : 0.01
1.46.1	B
1.46.6	A
1.47.1	0.70 : 0.76

1.35.4	B
1.35.9	B;D
1.35.14	C
1.35.19	B
1.35.24	B
1.35.29	C
1.37.1	C
1.38.2	4.4 : 4.6
1.39.4	A
1.42.2	B
1.43.1	0 : 0 ; 4 : 4
1.46.2	D
1.46.7	B

1.35.5	B
1.35.10	3:3
1.35.15	A
1.35.20	- 100.01 : - 99.99
1.35.25	C
1.35.30	100 : 100
1.37.2	A
1.38.3	D
1.39.5	B
1.42.3	A
1.44.1	15 : 15
1.46.3	C
1.46.8	A

2.1

Cartesian Coordinates (1)

2.1.1 Cartesian Coordinates: GATE IT 2007 | Question: 80



Let $P_1, P_2, \dots, P_n$ be n points in the xy -plane such that no three of them are collinear. For every pair of points P_i and P_j , let L_{ij} be the line passing through them. Let L_{ab} be the line with the steepest gradient amongst all $\frac{n(n-1)}{2}$ lines.

Which one of the following properties should necessarily be satisfied ?

- A. P_a and P_b are adjacent to each other with respect to their x -coordinate
- B. Either P_a or P_b has the largest or the smallest y -coordinate among all the points
- C. The difference between x -coordinates P_a and P_b is minimum
- D. None of the above

gateit-2007 cartesian-coordinates

Answer key

2.2

Determinant (10)

2.2.1 Determinant: GATE CSE 1997 | Question: 1.3



The determinant of the matrix $\begin{bmatrix} 6 & -8 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

- A. 11
- B. -48
- C. 0
- D. -24

gate1997 linear-algebra normal determinant

Answer key

2.2.2 Determinant: GATE CSE 2000 | Question: 1.3



The determinant of the matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 8 & 1 & 7 & 2 \\ 2 & 0 & 2 & 0 \\ 9 & 0 & 6 & 1 \end{bmatrix}$$

- A. 4
- B. 0
- C. 15
- D. 20

gatecse-2000 linear-algebra easy determinant

Answer key

2.2.3 Determinant: GATE CSE 2013 | Question: 3



Which one of the following does NOT equal

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} ?$$

A. $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \\ 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

gatecse-2013 linear-algebra normal determinant

Answer key 

B. $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \\ 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

D. $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \\ 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

2.2.4 Determinant: GATE CSE 2014 Set 2 | Question: 4



If the matrix A is such that

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \ 9 \ 5]$$

then the determinant of A is equal to _____.

gatecse-2014-set2 linear-algebra numerical-answers easy determinant

Answer key 

2.2.5 Determinant: GATE CSE 2019 | Question: 9



Let X be a square matrix. Consider the following two statements on X .

- I. X is invertible
- II. Determinant of X is non-zero

Which one of the following is TRUE?

- A. I implies II; II does not imply I
- B. II implies I; I does not imply II
- C. I does not imply II; II does not imply I
- D. I and II are equivalent statements

gatecse-2019 engineering-mathematics linear-algebra determinant 1-mark

Answer key 

2.2.6 Determinant: GATE CSE 2023 | Question: 8



Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B , respectively.

Which one of the options given below is TRUE?

- A. $\det(A) = \det(B)$
- B. $\det(B) = -\det(A)$
- C. $\det(A) = 0$
- D. $\det(AB) = \det(A) + \det(B)$

gatecse-2023 linear-algebra determinant 1-mark easy

Answer key 

2.2.7 Determinant: GATE CSE 2024 | Set 2 | Question: 37



Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- A. The determinant of B is the negative of the determinant of A
- B. If A is invertible, then B is also invertible
- C. If A is symmetric, then B is also symmetric
- D. If the trace of A is zero, then the trace of B is also zero

gatecse2024-set2 linear-algebra multiple-selects matrix determinant

Answer key 

2.2.8 Determinant: GATE DS&AI 2024 | Question: 25



Consider the 3×3 matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$.

The determinant of $(M^2 + 12M)$ is _____.

gate-ds-ai-2024 numerical-answers matrix determinant linear-algebra

Answer key 

2.2.9 Determinant: GATE IT 2004 | Question: 32



Let A be an $n \times n$ matrix of the following form.

$$A = \begin{bmatrix} 3 & 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & \dots & 0 & 0 & 0 \\ \dots & & & & & & & & \\ \dots & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & 1 & 3 \end{bmatrix}_{n \times n}$$

What is the value of the determinant of A ?

- A. $\left(\frac{5+\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}+7}{2\sqrt{3}}\right) + \left(\frac{5-\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}-7}{2\sqrt{3}}\right)$
- B. $\left(\frac{7+\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}+3}{2\sqrt{5}}\right) + \left(\frac{7-\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}-3}{2\sqrt{5}}\right)$
- C. $\left(\frac{3+\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}+5}{2\sqrt{7}}\right) + \left(\frac{3-\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}-5}{2\sqrt{7}}\right)$
- D. $\left(\frac{3+\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}+7}{2\sqrt{5}}\right) + \left(\frac{3-\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}-7}{2\sqrt{5}}\right)$

Answer key **2.2.10 Determinant: GATE IT 2005 | Question: 3**

The determinant of the matrix given below is

$$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

- A. -1 B. 0 C. 1 D. 2

Answer key **2.3 Eigen Value (30)****2.3.1 Eigen Value: GATE CSE 1993 | Question: 01.1**The eigen vector (s) of the matrix

$$\begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \alpha \neq 0$$

is (are)

- A. $(0,0,\alpha)$ B. $(\alpha,0,0)$ C. $(0,0,1)$ D. $(0,\alpha,0)$

Answer key **2.3.2 Eigen Value: GATE CSE 2002 | Question: 5a**

Obtain the eigen values of the matrix

$$A = \begin{bmatrix} 1 & 2 & 34 & 49 \\ 0 & 2 & 43 & 94 \\ 0 & 0 & -2 & 104 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Answer key **2.3.3 Eigen Value: GATE CSE 2005 | Question: 49**What are the eigenvalues of the following 2×2 matrix?

$$\begin{pmatrix} 2 & -1 \\ -4 & 5 \end{pmatrix}$$

- A. -1 and 1 B. 1 and 6 C. 2 and 5 D. 4 and -1

Answer key **2.3.4 Eigen Value: GATE CSE 2007 | Question: 25**

Let A be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is an eigen value of the matrix

$\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix?

- A. -5 B. -7 C. 2 D. 1

gatecse-2007 eigen-value linear-algebra difficult

Answer key **2.3.5 Eigen Value: GATE CSE 2008 | Question: 28**

How many of the following matrices have an eigenvalue 1?

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$

- A. one B. two C. three D. four

gatecse-2008 eigen-value linear-algebra

Answer key **2.3.6 Eigen Value: GATE CSE 2010 | Question: 29**

Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

- A. $x = 4, y = 10$ B. $x = 5, y = 8$ C. $x = 3, y = 9$ D. $x = -4, y = 10$

gatecse-2010 linear-algebra eigen-value easy

Answer key **2.3.7 Eigen Value: GATE CSE 2011 | Question: 40**

Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following options provides the **CORRECT** values of the eigenvalues of the matrix?

- A. 1, 4, 3 B. 3, 7, 3 C. 7, 3, 2 D. 1, 2, 3

gatecse-2011 linear-algebra eigen-value easy

Answer key **2.3.8 Eigen Value: GATE CSE 2012 | Question: 11**

Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

- A. 1024 and -1024
C. $4\sqrt{2}$ and $-4\sqrt{2}$

- B. $1024\sqrt{2}$ and $-1024\sqrt{2}$
D. $512\sqrt{2}$ and $-512\sqrt{2}$

gatecse-2012 linear-algebra eigen-value

Answer key 

2.3.9 Eigen Value: GATE CSE 2014 Set 1 | Question: 5



The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4×4 symmetric positive definite matrix is _____

gatecse-2014-set1 linear-algebra eigen-value numerical-answers normal

Answer key 

2.3.10 Eigen Value: GATE CSE 2014 Set 2 | Question: 47



The product of the non-zero eigenvalues of the matrix is _____

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

gatecse-2014-set2 linear-algebra eigen-value normal numerical-answers

Answer key 

2.3.11 Eigen Value: GATE CSE 2014 Set 3 | Question: 4



Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

- A. If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.
B. If the trace of the matrix is positive, all its eigenvalues are positive.
C. If the determinant of the matrix is positive, all its eigenvalues are positive.
D. If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

gatecse-2014-set3 linear-algebra eigen-value normal

Answer key 

2.3.12 Eigen Value: GATE CSE 2015 Set 1 | Question: 36



Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b . The eigenvalues of this matrix are -1 and 7 . What are the values of a and b ?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$

- A. $a=6, b=4$ B. $a=4, b=6$ C. $a=3, b=5$ D. $a=5, b=3$

gatecse-2015-set1 linear-algebra eigen-value easy

Answer key 

2.3.13 Eigen Value: GATE CSE 2015 Set 2 | Question: 5



The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is _____.

gatecse-2015-set2 linear-algebra eigen-value easy numerical-answers

Answer key 

2.3.14 Eigen Value: GATE CSE 2015 Set 3 | Question: 15



In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1. The eigenvectors corresponding to the eigenvalue 1 are

- A. $\{a(4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$
- B. $\{a(-4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$
- C. $\{a(\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$
- D. $\{a(-\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$

gatecse-2015-set3 linear-algebra eigen-value normal

Answer key 

2.3.15 Eigen Value: GATE CSE 2016 Set 1 | Question: 05



Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____

gatecse-2016-set1 linear-algebra eigen-value numerical-answers normal

Answer key 

2.3.16 Eigen Value: GATE CSE 2016 Set 2 | Question: 06



Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____.

gatecse-2016-set2 linear-algebra eigen-value normal numerical-answers

Answer key 

2.3.17 Eigen Value: GATE CSE 2017 Set 1 | Question: 31



Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2 = 50$. Consider the following statements.

- I. One eigenvalue must be in $[-5, 5]$
- II. The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

- A. Both I and II
- B. I only
- C. II only
- D. Neither I nor II

gatecse-2017-set1 linear-algebra eigen-value normal

Answer key 

2.3.18 Eigen Value: GATE CSE 2017 Set 2 | Question: 22



Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices.

Then the rank of $P + Q$ is _____.

gatecse-2017-set2 linear-algebra eigen-value numerical-answers

Answer key 

2.3.19 Eigen Value: GATE CSE 2017 Set 2 | Question: 52

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____

gatecse-2017-set2 engineering-mathematics linear-algebra numerical-answers eigen-value

Answer key

2.3.20 Eigen Value: GATE CSE 2018 | Question: 17

Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v . The largest eigenvalue of A is _____

gatecse-2018 linear-algebra eigen-value normal numerical-answers 1-mark

Answer key

2.3.21 Eigen Value: GATE CSE 2018 | Question: 26

Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

- I. P does not have an inverse
- II. P has a repeated eigenvalue
- III. P cannot be diagonalized

Which one of the following options is correct?

- A. Only I and III are necessarily true
- B. Only II is necessarily true
- C. Only I and II are necessarily true
- D. Only II and III are necessarily true

gatecse-2018 linear-algebra matrix eigen-value normal 2-marks

Answer key

2.3.22 Eigen Value: GATE CSE 2019 | Question: 44

Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____

gatecse-2019 numerical-answers engineering-mathematics linear-algebra eigen-value 2-marks

Answer key

2.3.23 Eigen Value: GATE CSE 2021 Set 1 | Question: 52

Consider the following matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

gatecse-2021-set1 linear-algebra matrix eigen-value numerical-answers 2-marks

Answer key 

2.3.24 Eigen Value: GATE CSE 2022 | Question: 43



Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

A. $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$

D. $\begin{pmatrix} 0 \\ 0 \\ 1 \\ -3 \end{pmatrix}$

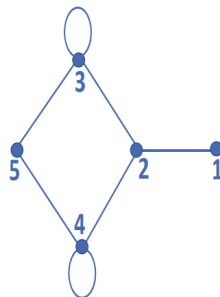
gatecse-2022 linear-algebra eigen-value multiple-selects 2-marks

Answer key 

2.3.25 Eigen Value: GATE CSE 2023 | Question: 20



Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.



Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, and λ_5 be the five eigenvalues of A . Note that these eigenvalues need not be distinct.

The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$ _____

gatecse-2023 linear-algebra eigen-value numerical-answers 1-mark

Answer key 

2.3.26 Eigen Value: GATE CSE 2024 | Set 1 | Question: 2



The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

A. -1

B. 0

C. 1

D. 2

gatecse2024-set1 linear-algebra eigen-value

Answer key 

2.3.27 Eigen Value: GATE DS&AI 2024 | Question: 3

Consider the matrix $M = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$.

Which ONE of the following statements is TRUE?

- A. The eigenvalues of M are non-negative and real.
- B. The eigenvalues of M are complex conjugate pairs.
- C. One eigenvalue of M is positive and real, and another eigenvalue of M is zero.
- D. One eigenvalue of M is non-negative and real, and another eigenvalue of M is negative and real.

gate-ds-ai-2024 eigen-value linear-algebra

Answer key

2.3.28 Eigen Value: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 17

For matrix $H = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$, one of the eigenvalues is 5. Then, the other eigenvalue is

- A. 12
- B. 10
- C. 8
- D. 6

gateda-sample-paper-2024 linear-algebra eigen-value

Answer key

2.3.29 Eigen Value: GATE IT 2006 | Question: 26

What are the eigenvalues of the matrix P given below

$$P = \begin{pmatrix} a & 1 & 0 \\ 1 & a & 1 \\ 0 & 1 & a \end{pmatrix}$$

- A. $a, a - \sqrt{2}, a + \sqrt{2}$
- B. a, a, a
- C. $0, a, 2a$
- D. $-a, 2a, 2a$

gateit-2006 linear-algebra eigen-value normal

Answer key

2.3.30 Eigen Value: GATE IT 2007 | Question: 2

Let A be the matrix $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. What is the maximum value of $x^T A x$ where the maximum is taken over all x that are the unit eigenvectors of A ?

- A. 5
- B. $\frac{(5+\sqrt{5})}{2}$
- C. 3
- D. $\frac{(5-\sqrt{5})}{2}$

gateit-2007 linear-algebra eigen-value normal

Answer key

2.4 Eigen Values (40)**2.4.1 Eigen Values: GATE Chemical 2012 | Question: 5**

Consider the following (2×2) matrix

$$\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

Which one of the following vectors is NOT a valid eigenvector of the above matrix ?

- A. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ B. $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$
 C. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$ D. $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

gate2012 linear-algebra matrices eigen-values eigen-vectors

2.4.2 Eigen Values: GATE Chemical 2013 | Question: 3



Which of the following statements are TRUE?

P. The eigenvalues of a symmetric matrix are real

Q. The value of the determinant of an orthogonal matrix can only be +1

R. The transpose of a square matrix **A** has the same eigenvalues as those of **A**

S. The inverse of an 'n x n' matrix exists if and only if the rank less than 'n'

- A. P and Q only B. P and R only C. Q and R only D. P and S only

gate2013 linear-algebra matrices eigen-values

2.4.3 Eigen Values: GATE Chemical 2015 | Question: 12



For the matrix $\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$, if $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector, the corresponding eigenvalue is _____.

gate2015 numerical-answers linear-algebra matrices eigen-values eigen-vectors

2.4.4 Eigen Values: GATE Chemical 2019 | Question: 22



The product of the eigenvalues of the matrix $\begin{pmatrix} 2 & 3 \\ 0 & 7 \end{pmatrix}$ is _____ (rounded off to one decimal place).

gate2019 numerical-answers linear-algebra matrices eigen-values

2.4.5 Eigen Values: GATE Chemical 2020 | Question: 25



Sum of the eigenvalues of the matrix $\begin{bmatrix} 2 & 4 & 6 \\ 3 & 5 & 9 \\ 12 & 1 & 7 \end{bmatrix}$ is _____ (round off to nearest integer).

gate2020 numerical-answers linear-algebra matrices eigen-values

Answer key

2.4.6 Eigen Values: GATE Civil 2012 | Question: 28



The eigenvalues of matrix $\begin{bmatrix} 9 & 5 \\ 5 & 8 \end{bmatrix}$ are

- A. -2.42 and 6.86 B. 3.48 and 13.53 C. 4.70 and 6.86 D. 6.86 and 9.50

gate2012-ce linear-algebra matrices eigen-values

2.4.7 Eigen Values: GATE Civil 2021 Set 2 | Question: 27



The smallest eigenvalue and the corresponding eigenvector of the matrix $\begin{bmatrix} 2 & -2 \\ -1 & 6 \end{bmatrix}$, respectively, are

- A. 1.55 and $\begin{Bmatrix} 2.00 \\ 0.45 \end{Bmatrix}$
B. 2.00 and $\begin{Bmatrix} 1.00 \\ 1.00 \end{Bmatrix}$
C. 1.55 and $\begin{Bmatrix} -2.55 \\ -0.45 \end{Bmatrix}$
D. 1.55 and $\begin{Bmatrix} 2.00 \\ -0.45 \end{Bmatrix}$

gatecivil-2021-set2 linear-algebra matrices eigen-values eigen-vectors

2.4.8 Eigen Values: GATE Civil 2024 Set 1 | Question: 26



What are the eigenvalues of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 2 \end{bmatrix}$?

- A. 1, 2, 5 B. 1, 3, 4 C. -5, 1, 2 D. -5, -1, 2

gatecivil-2024-set1 linear-algebra eigen-values

2.4.9 Eigen Values: GATE ECE 2010 | Question: 1



The eigenvalues of a skew-symmetric matrix are

- A. always zero B. always pure imaginary
C. either zero or pure imaginary D. always real

gate2010-ec linear-algebra eigen-values

2.4.10 Eigen Values: GATE ECE 2013 | Question: 19



The minimum eigenvalue of the following matrix is

$$\begin{bmatrix} 3 & 5 & 2 \\ 5 & 12 & 7 \\ 2 & 7 & 5 \end{bmatrix}$$

- A. 0 B. 1 C. 2 D. 3

gate2013-ec linear-algebra matrices eigen-values

Answer key

2.4.11 Eigen Values: GATE ECE 2014 Set 1 | Question: 4



A real (4×4) matrix A satisfies the equation $A^2 = I$, where I is the (4×4) identity matrix. The positive eigen value of A is _____.

gate2014-ec-1 linear-algebra matrices eigen-values numerical-answers

2.4.12 Eigen Values: GATE ECE 2014 Set 3 | Question: 27



Which one of the following statements is NOT true for a square matrix A ?

- A. If A is upper triangular, the eigenvalues of A are the diagonal elements of it
- B. If A is real symmetric, the eigenvalues of A are always real and positive
- C. If A is real, the eigenvalues of A and A^T are always the same
- D. If all the principal minors of A are positive, all the eigenvalues of A are also positive

gate2014-ec-3 linear-algebra matrices eigen-values

2.4.13 Eigen Values: GATE ECE 2015 Set 1 | Question: 5



The value of p such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$ is _____.

gate2015-ec-1 numerical-answers linear-algebra matrices eigen-values

2.4.14 Eigen Values: GATE ECE 2016 Set 2 | Question: 1



The value of x for which the matrix $A = \begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9+x \end{bmatrix}$ has zero as an eigenvalue is _____

gate2016-ec-2 numerical-answers linear-algebra matrices eigen-values

2.4.15 Eigen Values: GATE ECE 2017 Set 1 | Question: 1



Consider the 5×5 matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}$$

It is given that A has only one real eigenvalue. Then the real eigenvalue of A is

- A. -2.5
- B. 0
- C. 15
- D. 25

gate2017-ec-1 linear-algebra matrices eigen-values

Answer key

2.4.16 Eigen Values: GATE ECE 2018 | Question: 11



Let M be a real 4×4 matrix. Consider the following statements:

$S1 : M$ has 4 linearly independent eigenvectors.

$S2 : M$ has 4 distinct eigenvalues.

$S3 : M$ is non-singular (invertible).

Which one among the following is TRUE?

- A. $S1$ implies $S2$
- B. $S1$ implies $S3$
- C. $S2$ implies $S1$
- D. $S3$ implies $S2$

gate2018-ec linear-algebra matrices eigen-values

2.4.17 Eigen Values: GATE ECE 2019 | Question: 17



The number of distinct eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

is equal to _____.

gate2019-ec numerical-answers linear-algebra matrices eigen-values

Answer key 

2.4.18 Eigen Values: GATE ECE 2021 | Question: 36



A real 2×2 non-singular matrix A with repeated eigenvalue is given as

$$A = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$$

where x is a real positive number. The value of x (rounded off to one decimal place) is _____

gateec-2021 numerical-answers linear-algebra eigen-values

Answer key 

2.4.19 Eigen Values: GATE Electrical 2012 | Question: 26



Given that $\mathbf{A} = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix}$ and $\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the value of A^3 is

- A. $15\mathbf{A} + 12\mathbf{I}$ B. $19\mathbf{A} + 30\mathbf{I}$
C. $17\mathbf{A} + 15\mathbf{I}$ D. $17\mathbf{A} + 21\mathbf{I}$

gate2012-ee linear-algebra matrices eigen-values

Answer key 

2.4.20 Eigen Values: GATE Electrical 2013 | Question: 35



A matrix has eigenvalues -1 and -2 . The corresponding eigenvectors are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ respectively.

The matrix is

- A. $\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 2 \\ -2 & 4 \end{bmatrix}$
C. $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ D. $\begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}$

gate2013-ee linear-algebra matrices eigen-values eigen-vectors

2.4.21 Eigen Values: GATE Electrical 2014 Set 1 | Question: 46



A system matrix is given as follows.

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -6 & -11 & 6 \\ -6 & -11 & 5 \end{bmatrix}$$

The absolute value of the ratio of the maximum eigenvalue to the minimum eigenvalue is _____

gate2014-ee-1 linear-algebra matrices eigen-values numerical-answers

2.4.22 Eigen Values: GATE Electrical 2014 Set 2 | Question: 1



Which one of the following statements is true for all real symmetric matrices?

- A. All the eigenvalues are real.
- B. All the eigenvalues are positive.
- C. All the eigenvalues are distinct.
- D. Sum of all the eigenvalues is zero.

gate2014-ee-2 linear-algebra eigen-values

Answer key

2.4.23 Eigen Values: GATE Electrical 2015 Set 1 | Question: 26



The maximum value of "a" such that the matrix $\begin{pmatrix} -3 & 0 & -2 \\ 1 & -1 & 0 \\ 0 & a & -2 \end{pmatrix}$ has three linearly independent real eigenvectors is

- A. $\frac{2}{3\sqrt{3}}$
- B. $\frac{1}{3\sqrt{3}}$
- C. $\frac{1+2\sqrt{3}}{3\sqrt{3}}$
- D. $\frac{1+\sqrt{3}}{3\sqrt{3}}$

gate2015-ee-1 linear-algebra matrices eigen-values eigen-vectors

2.4.24 Eigen Values: GATE Electrical 2015 Set 2 | Question: 2



We have a set of 3 linear equations in 3 unknowns. ' $X \equiv Y$ ' means X and Y are equivalent statements and ' $X \not\equiv Y$ ' means X and Y are not equivalent statements.

P: There is a unique solution.

Q: The equations are linearly independent.

R: All eigenvalues of the coefficient matrix are nonzero.

S: The determinant of the coefficient matrix is nonzero.

Which one of the following is TRUE?

- A. $P \equiv Q \equiv R \equiv S$
- B. $P \equiv R \not\equiv Q \equiv S$
- C. $P \equiv Q \not\equiv R \equiv S$
- D. $P \not\equiv Q \not\equiv R \not\equiv S$

gate2015-ee-2 linear-algebra system-of-linear-equations eigen-values

2.4.25 Eigen Values: GATE Electrical 2016 Set 1 | Question: 2



Consider a 3×3 matrix with every element being equal to 1. Its only non-zero eigenvalue is _____.

gate2016-ee-1 linear-algebra matrices eigen-values

2.4.26 Eigen Values: GATE Electrical 2016 Set 1 | Question: 28



Let the eigenvalues of a 2×2 matrix A be 1, -2 with eigenvectors x_1 and x_2 respectively. Then the eigenvalues and eigenvectors of the matrix $A^2 - 3A + 4I$ would, respectively, be

- A. 2, 14; x_1, x_2
- B. 2, 14; $x_1 + x_2, x_1 - x_2$
- C. 2, 0; x_1, x_2
- D. 2, 0; $x_1 + x_2, x_1 - x_2$

gate2016-ee-1 linear-algebra matrices eigen-values eigen-vectors

2.4.27 Eigen Values: GATE Electrical 2016 Set 2 | Question: 32

Let $P = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ Consider the set S of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that $a^2 + b^2 = 1$ where $\begin{pmatrix} a \\ b \end{pmatrix} = P \begin{pmatrix} x \\ y \end{pmatrix}$.
Then S is

- A. A circle of radius $\sqrt{10}$.
- B. A circle of radius $\frac{1}{\sqrt{10}}$
- C. An ellipse with major axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- D. An ellipse with minor axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

gate2016-ee-2 linear-algebra matrices eigen-values eigen-vectors

2.4.28 Eigen Values: GATE Electrical 2016 Set 2 | Question: 49

Consider a linear time invariant system $\dot{x} = Ax$ with initial condition $x(0)$ at $t = 0$. Suppose α and β are eigenvectors of (2×2) matrix A corresponding to distinct eigenvalues λ_1 and λ_2 respectively. Then the response $x(t)$ of the system due to initial condition $x(0) = \alpha$ is

- A. $e^{\lambda_1 t} \alpha$
- B. $e^{\lambda_2 t} \beta$
- C. $e^{\lambda_2 t} \alpha$
- D. $e^{\lambda_1 t} \alpha + e^{\lambda_2 t} \beta$

gate2016-ee-2 linear-algebra eigen-values eigen-vectors

2.4.29 Eigen Values: GATE Electrical 2016 Set 2 | Question: 7

A 3×3 matrix P is such that, $P^3 = P$. Then the eigenvalues of P are

- A. $1, 1, -1$
- B. $1, 0.5 + j0.866, 0.5 - j0.866$
- C. $1, -0.5 + j0.866, -0.5 - j0.866$
- D. $0, 1, -1$

gate2016-ee-2 linear-algebra matrices eigen-values

2.4.30 Eigen Values: GATE ECE 2015 Set 2 | Question: 2

The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

- A. $5+j$
- B. $5-j$
- C. $1-5j$
- D. $1+5j$

gate2015-ec-2 linear-algebra matrices eigen-values

2.4.31 Eigen Values: GATE Electrical 2017 Set 2 | Question: 28

The eigenvalues of the matrix given below are

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix}$$

- A. $(0, -1, -3)$
- B. $(0, -2, -3)$
- C. $(0, 2, 3)$
- D. $(0, 1, 3)$

gate2017-ee-2 linear-algebra matrices eigen-values

2.4.32 Eigen Values: GATE Electrical 2019 | Question: 2

M is 2×2 matrix with eigenvalues 4 and 9. The eigenvalues of M^2 are

- A. 4 and 9 B. 2 and 3 C. -2 and -3 D. 16 and 81

gate2019-ee linear-algebra matrices eigen-values

Answer key

2.4.33 Eigen Values: GATE Electrical 2019 | Question: 26

Consider a 2×2 matrix $M = [v_1 \ v_2]$, where v_1 and v_2 are the column vectors. Suppose $M^{-1} = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$, where u_1^T and u_2^T are the row vecotrs. Consider the Following statements:

Statement 1: $u_1^T v_1 = 1$ and $u_2^T v_2 = 1$

Statement 2: $u_1^T v_2 = 0$ and $u_2^T v_1 = 0$

- A. Statement 1 is True and Statement 2 is false B. Statement 2 is true and Statement 1 is false
C. Both the Statements are true D. Both the statements are false

gate2019-ee linear-algebra matrices eigen-values eigen-vectors

Answer key

2.4.34 Eigen Values: GATE Electrical 2021 | Question: 1

Let p and q be real numbers such that $p^2 + q^2 = 1$. The eigenvalues of the matrix $\begin{bmatrix} p & q \\ q & -p \end{bmatrix}$ are

- A. 1 and 1 B. 1 and -1 C. j and $-j$ D. pq and $-pq$

gateeee-2021 linear-algebra matrices eigen-values

Answer key

2.4.35 Eigen Values: GATE ME 2012 | Question: 36

For the matrix $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, ONE of the normalized eigen vectors is given as

- A. $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$ B. $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$
C. $\begin{pmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{pmatrix}$ D. $\begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$

gateme-2012 linear-algebra eigen-values eigen-vectors

2.4.36 Eigen Values: GATE ME 2013 | Question: 2

The eigen values of a symmetric matrix are all

- A. complex with non-zero positive imaginary part. B. complex with non-zero negative imaginary part.
C. real. D. pure imaginary.

2.4.37 Eigen Values: GATE Mechanical 2014 Set 2 | Question: 1

One of the eigen vectors of the matrix $\begin{bmatrix} -5 & 2 \\ -9 & 6 \end{bmatrix}$ is

- A. $\begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$ B. $\begin{Bmatrix} -2 \\ 9 \end{Bmatrix}$
 C. $\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$ D. $\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

gateme-2014-set2 linear-algebra eigen-values eigen-vectors

2.4.38 Eigen Values: GATE Mechanical 2014 Set 3 | Question: 1

Consider a 3×3 real symmetric matrix S such that two of its eigenvalues are $a \neq 0, b \neq 0$ with respective eigenvectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. If $a \neq b$ then $x_1 y_1 + x_2 y_2 + x_3 y_3$ equals

- A. a B. b C. ab D. 0

gateme-2014-set3 linear-algebra matrices eigen-values eigen-vectors

2.4.39 Eigen Values: GATE Mechanical 2021 Set 1 | Question: 26

Consider a vector p in 2-dimensional space. Let its direction (counter-clockwise angle with the positive x -axis) be θ . Let p be an eigenvector of a 2×2 matrix A with corresponding eigenvalue $\lambda, \lambda > 0$. If we denote the magnitude of a vector v by $\|v\|$, identify the VALID statement regarding, p' , where $p' = Ap$.

- A. Direction of $p' = \lambda\theta, \|p'\| = \|p\|$
 B. Direction of $p' = \theta, \|p'\| = \lambda\|p\|$
 C. Direction of $p' = \lambda\theta, \|p'\| = \lambda\|p\|$
 D. Direction of $p' = \theta, \|p'\| = \|p\|/\lambda$

gateme-2021-set1 linear-algebra eigen-values eigen-vectors

Answer key

2.4.40 Eigen Values: GATE Mechanical 2021 Set 2 | Question: 1

Consider an $n \times n$ matrix A and a non-zero $n \times 1$ vector p . Their product $Ap = \alpha^2 p$, where $\alpha \in \mathbb{R}$ and $\alpha \notin \{-1, 0, 1\}$. Based on the given information, the eigen value of A^2 is:

- A. α B. α^2 C. $\sqrt{\alpha}$ D. α^4

gateme-2021-set2 linear-algebra matrices eigen-values

Answer key

2.5**Fractions (1)****2.5.1 Fractions: GATE Civil 2024 Set 1 | Question: 3**

The probability that a student passes only in Mathematics is $\frac{1}{3}$. The probability that the student passes only in English is $\frac{4}{9}$. The probability that the student passes in both of these subjects is $\frac{1}{6}$. The probability

that the student will pass in at least one of these two subjects is

A. $\frac{17}{18}$

B. $\frac{11}{18}$

C. $\frac{14}{18}$

D. $\frac{1}{18}$

gatecivil-2024-set1 probability fractions

2.6 Inequality (1)

2.6.1 Inequality: GATE CSE 1987 | Question: 1-xxi



If a, b , and c are constants, which of the following is a linear inequality?

A. $ax + bcy = 0$

B. $ax^2 + cy^2 = 21$

C. $abx + a^2y \geq 15$

D. $xy + ax \geq 20$

gate1987 linear-algebra inequality out-of-gatecse-syllabus

Answer key

2.7 Linear Equation (1)

2.7.1 Linear Equation: GATE Electrical 2014 Set 1 | Question: 1



Given a system of equations:

$$x + 2y + 2z = b_1$$

$$5x + y + 3z = b_2$$

Which of the following is true regarding its solutions

- A. The system has a unique solution for any given b_1 and b_2
- B. The system will have infinitely many solutions for any given b_1 and b_2
- C. Whether or not a solution exists depends on the given b_1 and b_2
- D. The system would have no solution for any values of b_1 and b_2

gate2014-ee-1 linear-equation system-of-linear-equations

2.8 Matrices (36)

2.8.1 Matrices: GATE BT 2021 | Question: 44



The determinant of matrix $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}$ is _____.

gatebt-2021 numerical-answers linear-algebra matrices determinant

2.8.2 Matrices: GATE CH 2021 | Question: 20



A, **B**, **C** and **D** are vectors of length 4

$$\mathbf{A} = [a_1 \ a_2 \ a_3 \ a_4]$$

$$\mathbf{B} = [b_1 \ b_2 \ b_3 \ b_4]$$

$$\mathbf{C} = [c_1 \ c_2 \ c_3 \ c_4]$$

$$\mathbf{D} = [d_1 \ d_2 \ d_3 \ d_4]$$

It is known that **B** is not scalar multiple of **A**. Also, **C** is linearly independent of **A** and **B**. Further, $\mathbf{D} = 3\mathbf{A} + 2\mathbf{B} + \mathbf{C}$.

The rank of the matrix $\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{bmatrix}$ is _____

gatech-2021 numerical-answers linear-algebra matrices rank-of-matrix

2.8.3 Matrices: GATE Chemical 2012 | Question: 1



Consider the following set of linear algebraic equations

$$x_1 + 2x_2 + 3x_3 = 2$$

$$x_2 + x_3 = -1$$

$$2x_2 + 2x_3 = 0$$

The system has

- A. a unique solution
- B. no solution
- C. an infinite number of solutions
- D. only the trivial solution

gate2012 linear-algebra matrices system-of-equations

2.8.4 Matrices: GATE Chemical 2018 | Question: 47



For the matrix $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ if $\det$ stands for the determinant and A^T is the transpose of A then the value of $\det(A^T A)$ is _____

gate2018 numerical-answers linear-algebra matrices determinant

2.8.5 Matrices: GATE Chemical 2019 | Question: 1



A system of n homogeneous linear equations containing n unknowns will have non-trivial solutions if and only if the determinant of the coefficient matrix is

- A. 1
- B. -1
- C. 0
- D. ∞

gate2019 linear-algebra matrices system-of-equations

Answer key

2.8.6 Matrices: GATE Civil 2013 | Question: 2



What is the minimum number of multiplications involved in computing the matrix product PQR ? Matrix P has 4 rows and 2 columns, matrix Q has 2 rows and 4 columns, and matrix R has 4 rows and 1 column.

gate2013-ce numerical-answers linear-algebra matrices

2.8.7 Matrices: GATE Civil 2021 Set 1 | Question: 1



The rank of matrix $\begin{bmatrix} 1 & 2 & 2 & 3 \\ 3 & 4 & 2 & 5 \\ 5 & 6 & 2 & 7 \\ 7 & 8 & 2 & 9 \end{bmatrix}$ is

- A. 1
- B. 2
- C. 3
- D. 4

gatecivil-2021-set1 linear-algebra matrices rank-of-matrix

2.8.8 Matrices: GATE Civil 2021 Set 2 | Question: 2

The rank of the matrix $\begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 2 & 0 & 1 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ is

- A. 1 B. 2 C. 3 D. 4

gatecivil-2021-set2 linear-algebra matrices rank-of-matrix

2.8.9 Matrices: GATE ECE 2013 | Question: 27

Let A be an $m \times n$ matrix and B an $n \times m$ matrix. It is given that determinant $(I_m + AB) = \text{determinant}(I_n + BA)$, where I_k is the $k \times k$ identity matrix. Using the above property, the determinant of the matrix given below is

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

- A. 2 B. 5 C. 8 D. 16

gate2013-ec linear-algebra matrices determinant

2.8.10 Matrices: GATE ECE 2014 Set 1 | Question: 1

For matrices of same dimension M, N and scalar c , which one of these properties DOES NOT ALWAYS hold?

- A. $(M^T)^T = M$ B. $(cM)^T = c(M)^T$
C. $(M+N)^T = M^T + N^T$ D. $MN = NM$

gate2014-ec-1 linear-algebra matrices

2.8.11 Matrices: GATE ECE 2014 Set 1 | Question: 29

Consider the matrix

$$J_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which is obtained by reversing the order of the columns of the identity matrix I_6 . Let $P = I_6 + \alpha J_6$, where α is a non-negative real number. The value of α for which $\det(P) = 0$ is _____.

gate2014-ec-1 numerical-answers linear-algebra matrices determinant

2.8.12 Matrices: GATE ECE 2014 Set 2 | Question: 1

The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is _____.

gate2014-ec-2 numerical-answers linear-algebra matrices determinant

2.8.13 Matrices: GATE ECE 2014 Set 2 | Question: 26

The system of linear equations $\begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 14 \end{pmatrix}$ has

- A. a unique solution
 B. infinitely many solutions
 C. no solution
 D. exactly two solutions

gate2014-ec-2 linear-algebra matrices system-of-equations

2.8.14 Matrices: GATE ECE 2014 Set 2 | Question: 28

The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is _____.

gate2014-ec-2 numerical-answers linear-algebra matrices determinant

2.8.15 Matrices: GATE ECE 2014 Set 3 | Question: 47

The state equation of a second-order linear system is given by

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0$$

For $x_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$, and for $x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $x(t) = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$. When $x_0 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$, $x(t)$ is

- A. $\begin{bmatrix} -8e^{-t} + 11e^{-2t} \\ 8e^{-t} - 22e^{-2t} \end{bmatrix}$
 B. $\begin{bmatrix} 11e^{-t} - 8e^{-2t} \\ -11e^{-t} + 16e^{-2t} \end{bmatrix}$
 C. $\begin{bmatrix} 3e^{-t} - 5e^{-2t} \\ -3e^{-t} + 10e^{-2t} \end{bmatrix}$
 D. $\begin{bmatrix} 5e^{-t} - 3e^{-2t} \\ -5e^{-t} + 6e^{-2t} \end{bmatrix}$

gate2014-ec-3 linear-algebra matrices

2.8.16 Matrices: GATE ECE 2014 Set 4 | Question: 46

The state transition matrix $\phi(t)$ of a system $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is

- A. $\begin{bmatrix} t & 1 \\ 1 & 0 \end{bmatrix}$
 B. $\begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix}$
 C. $\begin{bmatrix} 0 & 1 \\ 1 & t \end{bmatrix}$
 D. $\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$

gate2014-ec-4 linear-algebra matrices

2.8.17 Matrices: GATE ECE 2015 Set 2 | Question: 46

The state variable representation of a system is given as

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x; x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$y = [0 \quad 1] x$$

The response $y(t)$ is

- A. $\sin(t)$
 B. $1 - e^t$
 C. $1 - \cos(t)$
 D. 0

2.8.18 Matrices: GATE ECE 2015 Set 3 | Question: 1

For $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, the determinant of $A^T A^{-1}$ is

- A. $\sec^2 x$ B. $\cos 4x$ C. 1 D. 0

gate2015-ec-3 linear-algebra matrices determinant

Answer key

2.8.19 Matrices: GATE ECE 2016 Set 1 | Question: 1

Let $M^4 = I$, (where I denotes the identity matrix) and $M \neq I$, $M^2 \neq I$ and $M^3 \neq I$. Then, for any natural number k , M^{-1} equals:

- A. M^{4k+1} B. M^{4k+2}
C. M^{4k+3} D. M^{4k}

gate2016-ec-1 linear-algebra matrices

2.8.20 Matrices: GATE ECE 2016 Set 1 | Question: 27

A sequence $x[n]$ is specified as $\begin{bmatrix} x[n] \\ x[n-1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, for $n \geq 2$.

The initial conditions are $x[0] = 1$, $x[1] = 1$, and $x[n] = 0$ for $n < 0$. The value of $x[12]$ is _____

gate2016-ec-1 numerical-answers linear-algebra matrices

2.8.21 Matrices: GATE ECE 2016 Set 2 | Question: 29

The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has $\det(A) = 100$ and $\text{trace}(A) = 14$. The value of $|a - b|$ is _____

gate2016-ec-2 numerical-answers linear-algebra matrices

2.8.22 Matrices: GATE ECE 2016 Set 3 | Question: 1

Consider a 2×2 square matrix

$$\mathbf{A} = \begin{bmatrix} \sigma & x \\ \omega & \sigma \end{bmatrix},$$

where x is unknown. If the eigen values of the matrix $\mathbf{A}$ are $(\sigma + j\omega)$ and $(\sigma - j\omega)$, then x is equal to

- A. $+j\omega$ B. $-j\omega$ C. $+\omega$ D. $-\omega$

gate2016-ec-3 linear-algebra matrices

2.8.23 Matrices: GATE ECE 2017 Set 1 | Question: 2

The rank of the matrix

$$\mathbf{M} = \begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$$

is

A. 0

B. 1

C. 2

D. 3

gate2017-ec-1 linear-algebra matrices rank-of-matrix

2.8.24 Matrices: GATE ECE 2017 Set 2 | Question: 1



The rank of the matrix

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

is _____.

gate2017-ec-2 linear-algebra matrices rank-of-matrix numerical-answers

2.8.25 Matrices: GATE Electrical 2012 | Question: 41



The state variable description of an LTI system is given by

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$
$$y = (1 \ 0 \ 0) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

where y is the output and u is the input. The system is controllable for

A. $a_1 \neq 0, a_2 = 0, a_3 \neq 0$

B. $a_1 = 0, a_2 \neq 0, a_3 \neq 0$

C. $a_1 = 0, a_2 \neq 0, a_3 = 0$

D. $a_1 \neq 0, a_2 \neq 0, a_3 = 0$

gate2012-ee linear-algebra matrices system-of-linear-equations

2.8.26 Matrices: GATE Electrical 2013 | Question: 25



The equation $\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ has

A. no solution

B. only one solution $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

C. non-zero unique solution

D. multiple solutions

gate2013-ee linear-algebra matrices system-of-linear-equations

2.8.27 Matrices: GATE Electrical 2014 Set 3 | Question: 1



Two matrices A and B are given below:

$$A = \begin{vmatrix} p & q \\ r & s \end{vmatrix}; B = \begin{vmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{vmatrix}$$

If the rank of matrix A is N , then the rank of matrix B is

- A. $N/2$ B. $N-1$ C. N D. $2N$

gate2014-ee-3 linear-algebra matrices rank-of-matrix

2.8.28 Matrices: GATE Electrical 2016 Set 1 | Question: 29



Let A be a 4×3 real matrix with rank 2. Which one of the following statement is TRUE?

- A. Rank of $A^T A$ is less than 2. B. Rank of $A^T A$ is equal to 2.
C. Rank of $A^T A$ is greater than 2. D. Rank of $A^T A$ can be any number between 1 and 3.

gate2016-ee-1 linear-algebra matrices rank-of-matrix

2.8.29 Matrices: GATE Electrical 2018 | Question: 17



Consider a non-singular 2×2 square matrix A . If $\text{trace}(A) = 4$ and $\text{trace}(A^2) = 5$, the determinant of the matrix A is _____ (up to 1 decimal place).

gate2018-ee numerical-answers linear-algebra matrices determinant

2.8.30 Matrices: GATE Electrical 2018 | Question: 44



Let $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ and $B = A^3 - A^2 - 4A + 5I$, where I is the 3×3 identity matrix. The determinant of B is _____ (up to 1 decimal place).

gate2018-ee numerical-answers linear-algebra matrices determinant

2.8.31 Matrices: GATE Electrical 2020 | Question: 42



The number of purely real elements in a lower triangular representation of the given 3×3 matrix, obtained through the given decomposition is _____.

$$\begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 1 \\ 3 & 1 & 7 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix}^T$$

- A. 5 B. 6 C. 8 D. 9

gate2020-ee linear-algebra matrices

Answer key

2.8.32 Matrices: GATE Electrical 2021 | Question: 38



Let A be a 10×10 matrix such that A^5 is a null matrix, and let I be the 10×10 identity matrix. The determinant of $A+I$ is _____.

gateee-2021 numerical-answers linear-algebra matrices determinant

Answer key

2.8.33 Matrices: GATE IN 2021 | Question: 1



Consider the row vectors $v = (1, 0)$ and $w = (2, 0)$. The rank of the matrix $M = 2v^T v + 3w^T w$, where the superscript T denotes the transpose, is

- A. 1 B. 2 C. 3 D. 4

gatein-2021 linear-algebra matrices rank-of-matrix vectors

2.8.34 Matrices: GATE IN 2021 | Question: 25

The determinant of the matrix M shown below is _____.

$$M = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

gatein-2021 numerical-answers linear-algebra matrices determinant

2.8.35 Matrices: GATE IN 2021 | Question: 38

Given $A = \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix}$. The value of the determinant $|A^4 - 5A^3 + 6A^2 + 2I| =$ _____.

gatein-2021 numerical-answers linear-algebra matrices determinant

2.8.36 Matrices: GATE Mechanical 2014 Set 1 | Question: 1

Given that the determinant of the matrix $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{bmatrix}$ is -12 , the determinant of the matrix

$$\begin{bmatrix} 2 & 6 & 0 \\ 4 & 12 & 18 \\ -2 & 0 & 4 \end{bmatrix} \text{ is}$$

- A. -96 B. -24 C. 24 D. 96

gateme-2014-set1 linear-algebra matrices determinant

2.9 Matrix (19)**2.9.1 Matrix: GATE CSE 1987 | Question: 1-xxiii**

A square matrix is singular whenever

- A. The rows are linearly independent B. The columns are linearly independent
C. The row are linearly dependent D. None of the above

gate1987 linear-algebra matrix

Answer key

2.9.2 Matrix: GATE CSE 1988 | Question: 16i

Assume that the matrix A given below, has factorization of the form $LU = PA$, where L is lower-triangular with all diagonal elements equal to 1, U is upper-triangular, and P is a permutation matrix. For

$$A = \begin{bmatrix} 2 & 5 & 9 \\ 4 & 6 & 5 \\ 8 & 2 & 3 \end{bmatrix}$$

Compute L , U , and P using Gaussian elimination with partial pivoting.

gate1988 normal descriptive linear-algebra matrix

Answer key 

2.9.3 Matrix: GATE CSE 1993 | Question: 02.7



If $A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & i & i \\ 0 & 0 & 0 & -i \end{pmatrix}$ the matrix A^4 , calculated by the use of Cayley-Hamilton theorem or otherwise, is _____

gate1993 linear-algebra normal matrix fill-in-the-blanks

Answer key 

2.9.4 Matrix: GATE CSE 1994 | Question: 1.2



Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

- A. $AA' = I$ B. $A = A^{-1}$ C. $AB = BA$ D. $(AB)' = BA$

gate1994 linear-algebra normal matrix

Answer key 

2.9.5 Matrix: GATE CSE 1994 | Question: 3.12



Find the inverse of the matrix $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

gate1994 linear-algebra matrix easy descriptive

Answer key 

2.9.6 Matrix: GATE CSE 1996 | Question: 10



Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B .

gate1996 linear-algebra matrix normal descriptive

Answer key 

2.9.7 Matrix: GATE CSE 1996 | Question: 2.6



The matrices $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication

- A. if $a = b$ or $\theta = n\pi, n$ an integer B. always
C. never D. if $a \cos \theta = b \sin \theta$

gate1996 linear-algebra normal matrix

Answer key 

2.9.8 Matrix: GATE CSE 1997 | Question: 4.2



Let $A = (a_{ij})$ be an n -rowed square matrix and I_{12} be the matrix obtained by interchanging the first and second rows of the n -rowed Identity matrix. Then AI_{12} is such that its first

- A. Row is the same as its second row B. Row is the same as the second row of A
C. Column is the same as the second column of A D. Row is all zero

Answer key 

2.9.9 Matrix: GATE CSE 1998 | Question: 2.2



Consider the following determinant $\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

Which of the following is a factor of Δ ?

- A. $a+b$ B. $a-b$ C. $a+b+c$ D. abc

gate1998 linear-algebra matrix normal

Answer key 

2.9.10 Matrix: GATE CSE 2001 | Question: 1.1



Consider the following statements:

- S1: The sum of two singular $n \times n$ matrices may be non-singular
- S2: The sum of two $n \times n$ non-singular matrices may be singular

Which one of the following statements is correct?

- A. $S1$ and $S2$ both are true B. $S1$ is true, $S2$ is false
C. $S1$ is false, $S2$ is true D. $S1$ and $S2$ both are false

gatecse-2001 linear-algebra normal matrix

Answer key 

2.9.11 Matrix: GATE CSE 2004 | Question: 26



The number of different $n \times n$ symmetric matrices with each element being either 0 or 1 is: (Note: power $(2, X)$ is same as 2^X)

- A. power $(2, n)$ B. power $(2, n^2)$
C. power $\left(2, \frac{(n^2+n)}{2}\right)$ D. power $\left(2, \frac{(n^2-n)}{2}\right)$

gatecse-2004 linear-algebra normal matrix

Answer key 

2.9.12 Matrix: GATE CSE 2004 | Question: 27



Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then B^{-1} is

- A. $D^{-1}C^{-1}A^{-1}$ B. CDA
C. ADC D. Does not necessarily exist

gatecse-2004 linear-algebra normal matrix

Answer key 

2.9.13 Matrix: GATE CSE 2004 | Question: 76



In an $M \times N$ matrix all non-zero entries are covered in a rows and b columns. Then the maximum number of non-zero entries, such that no two are on the same row or column, is

- A. $\leq a+b$ B. $\leq \max(a, b)$
C. $\leq \min(M-a, N-b)$ D. $\leq \min(a, b)$

gatecse-2004 linear-algebra normal matrix

Answer key 

2.9.14 Matrix: GATE CSE 2006 | Question: 23



F is an $n \times n$ real matrix. b is an $n \times 1$ real vector. Suppose there are two $n \times 1$ vectors, u and v such that, $u \neq v$ and $Fu = b, Fv = b$. Which one of the following statements is false?

- A. Determinant of F is zero.
- B. There are an infinite number of solutions to $Fx = b$
- C. There is an $x \neq 0$ such that $Fx = 0$
- D. F must have two identical rows

gatecse-2006 linear-algebra normal matrix

Answer key 

2.9.15 Matrix: GATE CSE 2015 Set 1 | Question: 18



In the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is _____.

gatecse-2015-set1 linear-algebra matrix numerical-answers

Answer key 

2.9.16 Matrix: GATE CSE 2015 Set 2 | Question: 27



Perform the following operations on the matrix $\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$

- i. Add the third row to the second row
- ii. Subtract the third column from the first column.

The determinant of the resultant matrix is _____.

gatecse-2015-set2 linear-algebra matrix easy numerical-answers

Answer key 

2.9.17 Matrix: GATE CSE 2022 | Question: 10



Consider the following two statements with respect to the matrices $A_{m \times n}, B_{n \times m}, C_{n \times n}$ and $D_{n \times n}$.

Statement 1 : $tr(AB) = tr(BA)$

Statement 2 : $tr(CD) = tr(DC)$

where $tr()$ represents the trace of a matrix. Which one of the following holds?

- A. Statement 1 is correct and Statement 2 is wrong.
- B. Statement 1 is wrong and Statement 2 is correct.
- C. Both Statement 1 and Statement 2 are correct.
- D. Both Statement 1 and Statement 2 are wrong.

gatecse-2022 linear-algebra matrix 1-mark

Answer key 

2.9.18 Matrix: GATE IT 2004 | Question: 36

If matrix $X = \begin{bmatrix} a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$ and $X^2 - X + I = O$ (I is the identity matrix and O is the zero matrix), then the inverse of X is

A. $\begin{bmatrix} 1-a & -1 \\ a^2 & a \end{bmatrix}$

B. $\begin{bmatrix} 1-a & -1 \\ a^2 - a + 1 & a \end{bmatrix}$

C. $\begin{bmatrix} -a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$

D. $\begin{bmatrix} a^2 - a + 1 & a \\ 1 & 1 - a \end{bmatrix}$

gateit-2004 linear-algebra matrix normal

Answer key

2.9.19 Matrix: GATE IT 2008 | Question: 29

If M is a square matrix with a zero determinant, which of the following assertion (s) is (are) correct?

S1: Each row of M can be represented as a linear combination of the other rows

S2: Each column of M can be represented as a linear combination of the other columns

S3: $MX = 0$ has a nontrivial solution

S4: M has an inverse

A. S3 and S2

B. S1 and S4

C. S1 and S3

D. S1, S2 and S3

gateit-2008 linear-algebra normal matrix

Answer key

2.10**Matrix Algebra (7)****2.10.1 Matrix Algebra: GATE CH 2021 | Question: 26**

Let A be a square matrix of size $n \times n$ ($n > 1$). The elements of $A = \{a_{ij}\}$ are given by

$$a_{ij} = \begin{cases} i \times j, & \text{if } i \geq j \\ 0, & \text{if } i < j \end{cases}$$

The determinant of A is

A. 0

B. 1

C. $n!$

D. $(n!)^2$

gatech-2021 linear-algebra matrix-algebra determinant

2.10.2 System of Equations: GATE ECE 2011 | Question: 35

The system of equations

$$x + y + z = 6$$

$$x + 4y + 6z = 20$$

$$x + 4y + \lambda z = \mu$$

has **NO** solution for values of λ and μ given by

A. $\lambda = 6, \mu = 20$

B. $\lambda = 6, \mu \neq 20$

C. $\lambda \neq 6, \mu = 20$

D. $\lambda \neq 6, \mu \neq 20$

gate2011-ec linear-algebra system-of-equations

Answer key

2.10.3 Matrix Algebra: GATE Civil 2021 Set 2 | Question: 4

If A is a square matrix then orthogonality property mandates

- A. $AA^T = I$ B. $AA^T = 0$ C. $AA^T = A^{-1}$ D. $AA^T = A^2$

gatecivil-2021-set2 linear-algebra matrices matrix-algebra

2.10.4 Matrix Algebra: GATE Mechanical 2014 Set 1 | Question: 4

The matrix form of the linear syatem $\frac{dx}{dt} = 3x - 5y$ and $\frac{dy}{dt} = 4x + 8y$ is

- A. $\frac{d}{dt} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix}$
 B. $\frac{d}{dt} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & -5 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix}$
 C. $\frac{d}{dt} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & -5 \\ 3 & 8 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix}$
 D. $\frac{d}{dt} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & 8 \\ 3 & -5 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix}$

gateme-2014-set1 linear-algebra matrices matrix-algebra

2.10.5 Matrix Algebra: GATE Mechanical 2014 Set 2 | Question: 18

If there are m sources and n destinations in a transportation matrix, the total number of basic variables in a basic feasible solution is

- A. $m + n$ B. $m + n + 1$ C. $m + n - 1$ D. m

gateme-2014-set2 linear-algebra matrix-algebra

2.10.6 Matrix Algebra: GATE Mechanical 2014 Set 4 | Question: 1

Which one of the following equations is a correct identity for arbitrary 3×3 real matrices P , Q and R ?

- A. $P(Q + R) = PQ + RP$ B. $(P - Q)^2 = P^2 - 2PQ - Q^2$
 C. $\det(P + Q) = \det P + \det Q$ D. $(P + Q)^2 = P^2 + PQ + QP + Q^2$

gateme-2014-set4 linear-algebra matrices matrix-algebra

Answer key

2.10.7 Matrix Algebra: GATE Mechanical 2021 Set 2 | Question: 27

Let the superscript T represent the transpose operation. Consider the function $f(x) = \frac{1}{2}x^T Qx - r^T x$, where x and r are $n \times 1$ vectors and Q is a symmetric $n \times n$ matrix. The stationary point of $f(x)$ is

- A. $Q^T r$
 B. $Q^{-1} r$
 C. $\frac{r}{r^T r}$
 D. r

2.11

Numerical Methods (2)

2.11.1 Numerical Methods: GATE IT 2006 | Question: 76



- $x + y/2 = 9$
- $3x + y = 10$

The value of the Frobenius norm for the above system of equations is

- A. 0.5 B. 0.75 C. 1.5 D. 2.0

gateit-2006 linear-algebra normal numerical-methods non-gatecse

Answer key

2.11.2 Numerical Methods: GATE IT 2006 | Question: 77



- $x + y/2 = 9$
- $3x + y = 10$

What can be said about the Gauss-Siedel iterative method for solving the above set of linear equations?

- A. it will converge B. It will diverse
C. It will neither converge nor diverse D. It is not applicable

gateit-2006 linear-algebra normal numerical-methods non-gatecse

Answer key

2.12

Out of Gatecse Syllabus (1)

2.12.1 Out of Gatecse Syllabus: GATE CSE 1988 | Question: 16ii-iii



If $\|x\|_\infty = 1 < i^{max} < n \max(|x_1|)$ for the vector $\underline{x} = (x_1, x_2 \dots x_n)$ and $\|A\|_\infty = \sup \frac{\|Ax\|_\infty}{\|\underline{x}\|_\infty}$ is

the corresponding matrix norm, calculate $\|A\|_\infty$ for the matrix $A = \begin{bmatrix} 2 & 5 & 9 \\ 4 & 6 & 5 \\ 8 & 2 & 3 \end{bmatrix}$ using a known property

of this norm.

Although this norm is very easy to calculate for any matrix, explain why the condition number is difficult (i.e. expensive) to calculate.

gate1988 descriptive matrix out-of-gatecse-syllabus

2.13

Rank of Matrix (6)

2.13.1 Rank of Matrix: GATE CSE 1994 | Question: 1.9



The rank of matrix $\begin{bmatrix} 0 & 0 & -3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix}$ is:

- A. 0 B. 1 C. 2 D. 3

gate1994 linear-algebra matrix rank-of-matrix easy

Answer key 

2.13.2 Rank of Matrix: GATE CSE 1995 | Question: 1.24



The rank of the following $(n+1) \times (n+1)$ matrix, where a is a real number is

$$\begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix}$$

- A. 1
C. n

- B. 2
D. Depends on the value of a

gate1995 linear-algebra matrix normal rank-of-matrix

Answer key 

2.13.3 Rank of Matrix: GATE CSE 1998 | Question: 2.1



The rank of the matrix given below is:

$$\begin{bmatrix} 1 & 4 & 8 & 7 \\ 0 & 0 & 3 & 0 \\ 4 & 2 & 3 & 1 \\ 3 & 12 & 24 & 21 \end{bmatrix}$$

- A. 3

- B. 1

- C. 2

- D. 4

gate1998 linear-algebra matrix normal rank-of-matrix

Answer key 

2.13.4 Rank of Matrix: GATE CSE 2002 | Question: 1.1



The rank of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is

- A. 4

- B. 2

- C. 1

- D. 0

gatecse-2002 linear-algebra easy rank-of-matrix

Answer key 

2.13.5 Rank of Matrix: GATE CSE 2020 | Question: 27



Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M , respectively. Consider the following statements.

- I. $\text{rank}(AB) = \text{rank}(A)\text{rank}(B)$
- II. $\det(AB) = \det(A)\det(B)$
- III. $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$
- IV. $\det(A+B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

- A. I and II only

- B. I and IV only

- C. II and III only

- D. III and IV only

gatecse-2020 linear-algebra matrix 2-marks rank-of-matrix

Answer key 

2.13.6 Rank of Matrix: GATE CSE 2021 Set 2 | Question: 24



Suppose that P is a 4×5 matrix such that every solution of the equation $Px=0$ is a scalar multiple of $\begin{bmatrix} 2 & 5 & 4 & 3 & 1 \end{bmatrix}^T$. The rank of P is _____

gatecse-2021-set2 numerical-answers linear-algebra matrix rank-of-matrix 1-mark

Answer key

2.14 Second Order Differential Equation (1)

2.14.1 Second Order Differential Equation: GATE Civil 2024 Set 1 | Question: 2



The second-order differential equation in an unknown function $u : u(x, y)$ is defined as

$$\frac{\partial^2 u}{\partial x^2} = 2$$

Assuming $g : g(x)$, $f : f(y)$, and $h : h(y)$, the general solution of the above differential equation is

- A. $u = x^2 + f(y) + g(x)$ B. $u = x^2 + xf(y) + h(y)$
C. $u = x^2 + xf(y) + g(x)$ D. $u = x^2 + f(y) + yg(x)$

gatecivil-2024-set1 second-order-differential-equation

Answer key

2.15 Sequence (1)

2.15.1 Matrices: GATE ECE 2012 | Question: 47



Given that

$$A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix} \text{ and } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ the value of } A^3 \text{ is}$$

- A. $15A + 12I$ B. $19A + 30I$ C. $17A + 15I$ D. $17A + 21I$

gate2012-ec linear-algebra matrices

2.16 State Equations (2)

2.16.1 State Equations: GATE Electrical 2013 | Question: 50



The state variable formulation of a system is given as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad x_1(0) = 0, \quad x_2(0) = 0 \text{ and } y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The system is

- A. controllable but not observable B. not controllable but observable
C. both controllable and observable D. both not controllable and not observable

gate2013-ee linear-algebra matrices state-equations system-of-linear-equations

2.16.2 State Equations: GATE Electrical 2013 | Question: 51



The state variable formulation of a system is given as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad x_1(0) = 0, \quad x_2(0) = 0 \text{ and } y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The response $y(t)$ to a unit step input is

- A. $\frac{1}{2} - \frac{1}{2}e^{-2t}$ B. $1 - \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-t}$ C. $e^{-2t} - e^{-t}$ D. $1 - e^{-t}$

gate2013-ee linear-algebra matrices state-equations system-of-linear-equations

2.17

Subspace (1)

2.17.1 Subspace: GATE DS&AI 2024 | Question: 37



Select all choices that are subspaces of $\mathbb{R}^3$.

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- B. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- C. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0 \right\}$
- D. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

gate-ds-ai-2024 linear-algebra vector-space subspace

Answer key

2.18

System of Equations (19)

2.18.1 System of Equations: GATE CSE 1996 | Question: 1.7



Let $Ax = b$ be a system of linear equations where A is an $m \times n$ matrix and b is a $m \times 1$ column vector and X is an $n \times 1$ column vector of unknowns. Which of the following is false?

- A. The system has a solution if and only if, both A and the augmented matrix $[Ab]$ have the same rank.
- B. If $m < n$ and b is the zero vector, then the system has infinitely many solutions.
- C. If $m = n$ and b is a non-zero vector, then the system has a unique solution.
- D. The system will have only a trivial solution when $m = n$, b is the zero vector and $\text{rank}(A) = n$.

gate1996 linear-algebra system-of-equations normal

Answer key

2.18.2 System of Equations: GATE CSE 1998 | Question: 1.2



Consider the following set of equations

- $x + 2y = 5$
- $4x + 8y = 12$
- $3x + 6y + 3z = 15$

This set

- A. has unique solution
- B. has no solution
- C. has finite number of solutions
- D. has infinite number of solutions

gate1998 linear-algebra system-of-equations easy

Answer key 

2.18.3 System of Equations: GATE CSE 1998 | Question: 9



Derive the expressions for the number of operations required to solve a system of linear equations in n unknowns using the Gaussian Elimination Method. Assume that one operation refers to a multiplication followed by an addition.

gate1998 linear-algebra system-of-equations descriptive

Answer key 

2.18.4 System of Equations: GATE CSE 2003 | Question: 41



Consider the following system of linear equations

$$\begin{pmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha \\ 5 \\ 7 \end{pmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α , does this system of equations have infinitely many solutions?

- A. 0
- B. 1
- C. 2
- D. 3

gatecse-2003 linear-algebra system-of-equations normal

Answer key 

2.18.5 System of Equations: GATE CSE 2004 | Question: 71



How many solutions does the following system of linear equations have?

- $-x + 5y = -1$
- $x - y = 2$
- $x + 3y = 3$

- A. infinitely many
- B. two distinct solutions
- C. unique
- D. none

gatecse-2004 linear-algebra system-of-equations normal

Answer key 

2.18.6 System of Equations: GATE CSE 2005 | Question: 48



Consider the following system of linear equations :

$$2x_1 - x_2 + 3x_3 = 1$$

$$3x_1 + 2x_2 + 5x_3 = 2$$

$$-x_1 + 4x_2 + x_3 = 3$$

The system of equations has

- A. no solution
C. more than one but a finite number of solutions
- B. a unique solution
D. an infinite number of solutions

gatecse-2005 linear-algebra system-of-equations normal

Answer key 

2.18.7 System of Equations: GATE CSE 2008 | Question: 3



The following system of equations

- $x_1 + x_2 + 2x_3 = 1$
- $x_1 + 2x_2 + 3x_3 = 2$
- $x_1 + 4x_2 + \alpha x_3 = 4$

has a unique solution. The only possible value(s) for α is/are

- A. 0
B. either 0 or 1
C. one of 0, 1, or -1
D. any real number

gatecse-2008 easy linear-algebra system-of-equations

Answer key 

2.18.8 System of Equations: GATE CSE 2014 Set 1 | Question: 4



Consider the following system of equations:

- $3x + 2y = 1$
- $4x + 7z = 1$
- $x + y + z = 3$
- $x - 2y + 7z = 0$

The number of solutions for this system is _____

gatecse-2014-set1 linear-algebra system-of-equations numerical-answers normal

Answer key 

2.18.9 System of Equations: GATE CSE 2015 Set 3 | Question: 33



If the following system has non-trivial solution,

- $px + qy + rz = 0$
- $qx + ry + pz = 0$
- $rx + py + qz = 0,$

then which one of the following options is TRUE?

- A. $p - q + r = 0$ or $p = q = -r$
C. $p + q + r = 0$ or $p = q = r$
- B. $p + q - r = 0$ or $p = -q = r$
D. $p - q + r = 0$ or $p = -q = -r$

gatecse-2015-set3 linear-algebra system-of-equations normal

Answer key 

2.18.10 System of Equations: GATE CSE 2016 Set 2 | Question: 04



Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution.
II. If $m > n$, then none of these systems has a solution.
III. If $m = n$, then there exists a system which has a solution.

Which one of the following is **CORRECT**?

- A. I, II and III are true. B. Only II and III are true.
C. Only III is true. D. None of them is true.

gatecse-2016-set2 linear-algebra system-of-equations normal

Answer key 

2.18.11 System of Equations: GATE CSE 2017 Set 1 | Question: 3



Let $c_1, \dots, c_n$ be scalars, not all zero, such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are column vectors in R^n .

Consider the set of linear equations

$$Ax = b$$

where $A = [a_1 \dots a_n]$ and $b = \sum_{i=1}^n a_i$. The set of equations has

- A. a unique solution at $x = J_n$ where J_n denotes a n -dimensional vector of all 1.
B. no solution
C. infinitely many solutions
D. finitely many solutions

gatecse-2017-set1 linear-algebra system-of-equations normal

Answer key 

2.18.12 System of Equations: GATE CSE 2022 | Question: 35



Consider solving the following system of simultaneous equations using LU decomposition.

$$x_1 + x_2 - 2x_3 = 4$$

$$x_1 + 3x_2 - x_3 = 7$$

$$2x_1 + x_2 - 5x_3 = 7$$

where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32} , U_{33} , and x_1 ?

- A. $L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$ B. $L_{32} = 2, U_{33} = 2, x_1 = -1$
C. $L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$ D. $L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

gatecse-2022 linear-algebra matrix system-of-equations 2-marks

Answer key 

2.18.13 System of Equations: GATE CSE 2024 | Set 1 | Question: 39



Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- A. There exist at least $m - n$ linearly independent solutions to this system
B. There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors
C. There exists a non-zero solution in which at least $m - n$ variables are 0
D. There exists a solution in which at least n variables are non-zero

Answer key

2.18.14 System of Equations: GATE Chemical 2016 | Question: 26



A set of simultaneous linear algebraic equations is represented in a matrix form as shown below.

$$\begin{bmatrix} 0 & 0 & 0 & 4 & 13 \\ 2 & 5 & 5 & 2 & 10 \\ 0 & 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 4 & 5 \\ 2 & 3 & 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 46 \\ 161 \\ 61 \\ 30 \\ 81 \end{bmatrix}$$

The value (rounded off to the nearest integer) of x_3 is _____

gate2016 numerical-answers linear-algebra system-of-equations

2.18.15 System of Equations: GATE ECE 2015 Set 1 | Question: 1



Consider a system of linear equations:

$$\begin{aligned} x - 2y + 3z &= -1, \\ x - 3y + 4z &= 1, \text{ and} \\ -2x + 4y - 6z &= k. \end{aligned}$$

The value of k for which the system has infinitely many solutions is _____

gate2015-ec-1 numerical-answers linear-algebra system-of-equations

2.18.16 System of Equations: GATE ECE 2018 | Question: 22



Consider matrix $A = \begin{bmatrix} k & 2k \\ k^2 - k & k^2 \end{bmatrix}$ and vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. The number of distinct real value of k for which the equation $Ax = 0$ has infinitely many solutions is _____.

gate2018-ec numerical-answers linear-algebra system-of-equations

2.18.17 System of Equations: GATE ECE 2020 | Question: 26



Consider the following system of linear equations.

$$x_1 + 2x_2 = b_1; \quad 2x_1 + 4x_2 = b_2; \quad 3x_1 + 7x_2 = b_3; \quad 3x_1 + 9x_2 = b_4$$

Which one of the following conditions ensures that a solution exists for the above system?

- A. $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$ B. $b_3 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$
 C. $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$ D. $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

gate2020-ec linear-algebra system-of-equations

2.18.18 System of Equations: GATE IT 2004 | Question: 6



What values of x , y and z satisfy the following system of linear equations?

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}$$

- A. $x = 6, y = 3, z = 2$ B. $x = 12, y = 3, z = -4$
 C. $x = 6, y = 6, z = -4$ D. $x = 12, y = -3, z = 0$

Answer key **2.18.19 System of Equations: GATE ME 2012 | Question: 47**

$$x + 2y + z = 4$$

$$2x + y + 2z = 5$$

$$x - y + z = 1$$

The system of algebraic equations given above has

- A. a unique solution of $x = 1, y = 1$ and $z = 1$ B. only the two solutions of $(x = 1, y = 1, z = 1)$ and $(x = 2, y = 1, z = 0)$
- C. infinite number of solutions D. no feasible solution

gateme-2012 linear-algebra system-of-equations

2.19**Transition Matrix (1)****2.19.1 Transition Matrix: GATE Electrical 2014 Set 2 | Question: 18**

The state transition matrix for the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$
 is

A.

$$\begin{bmatrix} e^t & 0 \\ e^t & e^t \end{bmatrix}$$

B.

$$\begin{bmatrix} e^t & 0 \\ t^2 e^t & e^t \end{bmatrix}$$

C.

$$\begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$$

D.

$$\begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}$$

gate2014-ee-2 linear-algebra matrices transition-matrix

2.20**Vector Analysis (1)****2.20.1 Vector Analysis: GATE ECE 2024 | Question: 20**

Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors

$$\{[2 \quad -3 \quad \alpha], [3 \quad -1 \quad 3], [1 \quad -5 \quad 7]\}$$

does not form a basis of $\mathbb{R}^3$ is _____.

gateece-2024 numerical-answers linear-algebra vector-analysis

Answer key **2.21****Vector Space (5)****2.21.1 Vector Space: GATE CSE 1993 | Question: 02.3**

If the linear velocity $\vec{V}$ is given by

$$\vec{V} = x^2 y \hat{i} + xyz \hat{j} - yz^2 \hat{k}$$

The angular velocity $\vec{\omega}$ at the point $(1, 1, -1)$ is _____

Answer key **2.21.2 Vector Space: GATE CSE 1995 | Question: 2.13**

A unit vector perpendicular to both the vectors $a = 2i - 3j + k$ and $b = i + j - 2k$ is:

- A. $\frac{1}{\sqrt{3}}(i + j + k)$ B. $\frac{1}{3}(i + j - k)$ C. $\frac{1}{3}(i - j - k)$ D. $\frac{1}{\sqrt{3}}(i + j - k)$

gate1995 linear-algebra normal vector-space

Answer key **2.21.3 Vector Space: GATE CSE 2007 | Question: 27**

Consider the set of (column) vectors defined by

$$X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0, \text{ where } x^T = [x_1, x_2, x_3]^T\}$$

.Which of the following is TRUE?

- A. $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X .
 B. $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X .
 C. X is not a subspace of R^3 .
 D. None of the above

gatecse-2007 linear-algebra normal vector-space

Answer key **2.21.4 Vector Space: GATE CSE 2014 Set 3 | Question: 5**

If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V , then the smallest possible dimension of $V_1 \cap V_2$ is _____.

gatecse-2014-set3 linear-algebra vector-space normal numerical-answers

Answer key **2.21.5 Vector Space: GATE CSE 2017 Set 1 | Question: 30**

Let u and v be two vectors in $\mathbf{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v ?

- A. 2 B. $\frac{1}{2}$ C. 1 D. $\frac{-1}{2}$

gatecse-2017-set1 linear-algebra normal vector-space

Answer key **Answer Keys**

2.1.1	A	2.2.1	B	2.2.2	A	2.2.3	A	2.2.4	0
2.2.5	D	2.2.6	B	2.2.7	A;B	2.2.8	0	2.2.9	D
2.2.10	A	2.3.1	B;D	2.3.2	N/A	2.3.3	B	2.3.4	C
2.3.5	A	2.3.6	D	2.3.7	A	2.3.8	D	2.3.9	0

2.3.10	6	2.3.11	A	2.3.12	D	2.3.13	6	2.3.14	B
2.3.15	15	2.3.16	0.125	2.3.17	B	2.3.18	2	2.3.19	5
2.3.20	3	2.3.21	D	2.3.22	12	2.3.23	3 : 3	2.3.24	A;C;D
2.3.25	2	2.3.26	B	2.3.27	B	2.3.28	B	2.3.29	A
2.3.30	B	2.4.1	D	2.4.2	B	2.4.3	7	2.4.4	13.9:14.1
2.4.5	14:14	2.4.6	B	2.4.7	A	2.4.8	A	2.4.9	Q-Q
2.4.10	A	2.4.11	0.99 : 1.01	2.4.12	B	2.4.13	16.5 : 17.5	2.4.14	0.95 : 1.05
2.4.15	C	2.4.16	C	2.4.17	3 : 3	2.4.18	10.0 : 10.0	2.4.19	B
2.4.20	D	2.4.21	2.9 : 3.1	2.4.22	A	2.4.23	B	2.4.24	A
2.4.25	3.0 : 03.0	2.4.26	A	2.4.27	D	2.4.28	A	2.4.29	A;D
2.4.30	B	2.4.31	A	2.4.32	D	2.4.33	C	2.4.34	B
2.4.35	B	2.4.36	C	2.4.37	D	2.4.38	D	2.4.39	B
2.4.40	D	2.5.1	A	2.6.1	C	2.7.1	B	2.8.1	8 : 8
2.8.2	3 : 3	2.8.3	B	2.8.4	1:1	2.8.5	C	2.8.6	16
2.8.7	B	2.8.8	C	2.8.9	B	2.8.10	D	2.8.11	0.99 : 1.01
2.8.12	199 : 201	2.8.13	B	2.8.14	48.9 : 49.1	2.8.15	B	2.8.16	D
2.8.17	D	2.8.18	C	2.8.19	C	2.8.20	230 : 240	2.8.21	2.9 : 3.1
2.8.22	D	2.8.23	C	2.8.24	4 : 4	2.8.25	D	2.8.26	D
2.8.27	C	2.8.28	B	2.8.29	5.5 : 5.5	2.8.30	0.9 : 1.1	2.8.31	C
2.8.32	1 : 1	2.8.33	A	2.8.34	4 : 4	2.8.35	4 : 4	2.8.36	A
2.9.1	C	2.9.2	N/A	2.9.3	N/A	2.9.4	D	2.9.5	N/A
2.9.6	N/A	2.9.7	A	2.9.8	C	2.9.9	B	2.9.10	A
2.9.11	C	2.9.12	B	2.9.13	D	2.9.14	D	2.9.15	5
2.9.16	0	2.9.17	C	2.9.18	B	2.9.19	D	2.10.1	D
2.10.2	D	2.10.3	A	2.10.4	A	2.10.5	C	2.10.6	D
2.10.7	B	2.11.1	D	2.11.2	A	2.12.1	N/A	2.13.1	C
2.13.2	A	2.13.3	A	2.13.4	C	2.13.5	C	2.13.6	4 : 4
2.14.1	B	2.15.1	B	2.16.1	A	2.16.2	A	2.17.1	A;C
2.18.1	C	2.18.2	B	2.18.3	N/A	2.18.4	B	2.18.5	C
2.18.6	B	2.18.7	X	2.18.8	1	2.18.9	C	2.18.10	C
2.18.11	C	2.18.12	D	2.18.13	A	2.18.14	14.9:15.1	2.18.15	2
2.18.16	2 : 2	2.18.17	A	2.18.18	C	2.18.19	C	2.19.1	C
2.20.1	5:5	2.21.1	N/A	2.21.2	A	2.21.3	A	2.21.4	2
2.21.5	A								

3.1

Binomial Distribution (6)

3.1.1 Binomial Distribution: GATE CSE 2002 | Question: 2.16



Four fair coins are tossed simultaneously. The probability that at least one head and one tail turn up is

- A. $\frac{1}{16}$ B. $\frac{1}{8}$ C. $\frac{7}{8}$ D. $\frac{15}{16}$

gatecse-2002 probability easy binomial-distribution

Answer key

3.1.2 Binomial Distribution: GATE CSE 2005 | Question: 52



A random bit string of length n is constructed by tossing a fair coin n times and setting a bit to 0 or 1 depending on outcomes head and tail, respectively. The probability that two such randomly generated strings are not identical is:

- A. $\frac{1}{2^n}$ B. $1 - \frac{1}{n}$ C. $\frac{1}{n!}$ D. $1 - \frac{1}{2^n}$

gatecse-2005 probability binomial-distribution easy

Answer key

3.1.3 Binomial Distribution: GATE CSE 2006 | Question: 21



For each element in a set of size $2n$, an unbiased coin is tossed. The $2n$ coin tosses are independent. An element is chosen if the corresponding coin toss was a head. The probability that exactly n elements are chosen is

- A. $\frac{{}^{2n}C_n}{4^n}$ B. $\frac{{}^{2n}C_n}{2^n}$ C. $\frac{1}{{}^{2n}C_n}$ D. $\frac{1}{2}$

gatecse-2006 probability binomial-distribution normal

Answer key

3.1.4 Binomial Distribution: GATE IT 2005 | Question: 32



An unbiased coin is tossed repeatedly until the outcome of two successive tosses is the same. Assuming that the trials are independent, the expected number of tosses is

- A. 3 B. 4 C. 5 D. 6

gateit-2005 probability binomial-distribution expectation normal

Answer key

3.1.5 Binomial Distribution: GATE IT 2006 | Question: 22



When a coin is tossed, the probability of getting a Head is $p, 0 < p < 1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the expected value of N is

- A. $\frac{1}{p}$ B. $\frac{1}{(1-p)}$ C. $\frac{1}{p^2}$ D. $\frac{1}{(1-p^2)}$

gateit-2006 probability binomial-distribution expectation normal

Answer key

3.1.6 Binomial Distribution: GATE IT 2007 | Question: 1



Suppose there are two coins. The first coin gives heads with probability $\frac{5}{8}$ when tossed, while the second coin gives heads with probability $\frac{1}{4}$. One of the two coins is picked up at random with equal probability and tossed. What is the probability of obtaining heads ?

- A. $\left(\frac{7}{8}\right)$ B. $\left(\frac{1}{2}\right)$ C. $\left(\frac{7}{16}\right)$ D. $\left(\frac{5}{32}\right)$

gateit-2007 probability normal binomial-distribution

Answer key 

3.2

Conditional Probability (15)

3.2.1 Conditional Probability: GATE CSE 1994 | Question: 1.4, ISRO2017-2



Let A and B be any two arbitrary events, then, which one of the following is TRUE?

- A. $P(A \cap B) = P(A)P(B)$ B. $P(A \cup B) = P(A) + P(B)$
C. $P(A | B) = P(A \cap B)P(B)$ D. $P(A \cup B) \leq P(A) + P(B)$

gate1994 probability conditional-probability normal isro2017

Answer key 

3.2.2 Conditional Probability: GATE CSE 1994 | Question: 2.6



The probability of an event B is P_1 . The probability that events A and B occur together is P_2 while the probability that A and $\bar{B}$ occur together is P_3 . The probability of the event A in terms of P_1, P_2 and P_3 is

gate1994 probability normal conditional-probability fill-in-the-blanks

Answer key 

3.2.3 Conditional Probability: GATE CSE 2003 | Question: 3



Let $P(E)$ denote the probability of the event E . Given $P(A) = 1, P(B) = \frac{1}{2}$, the values of $P(A | B)$ and $P(B | A)$ respectively are

- A. $\left(\frac{1}{4}\right), \left(\frac{1}{2}\right)$ B. $\left(\frac{1}{2}\right), \left(\frac{1}{4}\right)$ C. $\left(\frac{1}{2}\right), 1$ D. $1, \left(\frac{1}{2}\right)$

gatecse-2003 probability easy conditional-probability

Answer key 

3.2.4 Conditional Probability: GATE CSE 2005 | Question: 51



Box P has 2 red balls and 3 blue balls and box Q has 3 red balls and 1 blue ball. A ball is selected as follows: (i) select a box (ii) choose a ball from the selected box such that each ball in the box is equally likely to be chosen. The probabilities of selecting boxes P and Q are $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Given that a ball selected in the above process is a red ball, the probability that it came from the box P is:

- A. $\frac{4}{19}$ B. $\frac{5}{19}$ C. $\frac{2}{9}$ D. $\frac{19}{30}$

gatecse-2005 probability conditional-probability normal

Answer key 

3.2.5 Conditional Probability: GATE CSE 2008 | Question: 27



Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a

day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

- A. 0.24 B. 0.36 C. 0.4 D. 0.6

gatecse-2008 probability normal conditional-probability

Answer key 

3.2.6 Conditional Probability: GATE CSE 2009 | Question: 21



An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same. If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3?

- A. 0.453 B. 0.468 C. 0.485 D. 0.492

gatecse-2009 probability normal conditional-probability

Answer key 

3.2.7 Conditional Probability: GATE CSE 2011 | Question: 3



If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

- A. $\left(\frac{1}{3}\right)$ B. $\left(\frac{1}{4}\right)$ C. $\left(\frac{1}{2}\right)$ D. $\left(\frac{2}{3}\right)$

gatecse-2011 probability easy conditional-probability

Answer key 

3.2.8 Conditional Probability: GATE CSE 2012 | Question: 33



Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2, or 3, the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6?

- A. $\frac{10}{21}$ B. $\frac{5}{12}$ C. $\frac{2}{3}$ D. $\frac{1}{6}$

gatecse-2012 probability conditional-probability normal

Answer key 

3.2.9 Conditional Probability: GATE CSE 2016 Set 2 | Question: 05



Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____.

gatecse-2016-set2 probability conditional-probability normal numerical-answers

Answer key 

3.2.10 Conditional Probability: GATE CSE 2017 Set 2 | Question: 26



P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for this job is

A. $\left(\frac{4}{5}\right)$

B. $\left(\frac{5}{6}\right)$

C. $\left(\frac{7}{8}\right)$

D. $\left(\frac{11}{12}\right)$

gatecse-2017-set2 probability conditional-probability

Answer key **3.2.11 Conditional Probability: GATE CSE 2018 | Question: 44**

Consider Guwahati, (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D | H_G) = 0.40$. Similarly, the next two entries are $P(M_D | H_G) = 0.48$ and $P(L_D | H_G) = 0.12$. Similarly for the other rows.

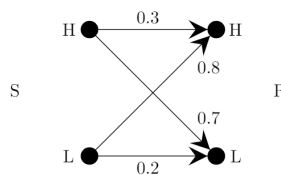
If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

gatecse-2018 probability conditional-probability numerical-answers 2-marks

Answer key **3.2.12 Conditional Probability: GATE CSE 2021 Set 1 | Question: 54**

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).

In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H , is 0.7.



If the received signal is H , the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability conditional-probability numerical-answers 2-marks

Answer key **3.2.13 Conditional Probability: GATE CSE 2024 | Set 1 | Question: 53**

A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____.

gatecse2024-set1 numerical-answers probability conditional-probability

Answer key 

3.2.14 Conditional Probability: GATE IT 2006 | Question: 1



In a certain town, the probability that it will rain in the afternoon is known to be 0.6. Moreover, meteorological data indicates that if the temperature at noon is less than or equal to 25°C , the probability that it will rain in the afternoon is 0.4. The temperature at noon is equally likely to be above 25°C , or at/below 25°C . What is the probability that it will rain in the afternoon on a day when the temperature at noon is above 25°C ?

- A. 0.4 B. 0.6 C. 0.8 D. 0.9

gateit-2006 probability normal conditional-probability

Answer key 

3.2.15 Conditional Probability: GATE Suitability Test | Test 1 | Question: 8



The chance of rainfall in the afternoon is twice that in the morning and there is no chance of rain happening at both morning and afternoon. If it rained for 3 days, what is the probability that there was rainfall in the morning on all three days? (rounded to two decimal places)

gate-suitability-test-1 conditional-probability numerical-answers

3.3 Continuous Distribution (1)

3.3.1 Continuous Distribution: GATE CSE 2016 Set 1 | Question: 04



A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____.

gatecse-2016-set1 probability normal numerical-answers continuous-distribution

Answer key 

3.4 Expectation (11)

3.4.1 Expectation: GATE CSE 1999 | Question: 1.1



Suppose that the expectation of a random variable X is 5. Which of the following statements is true?

- A. There is a sample point at which X has the value 5.
B. There is a sample point at which X has value greater than 5.
C. There is a sample point at which X has a value greater than equal to 5.
D. None of the above.

gate1999 probability expectation easy

Answer key 

3.4.2 Expectation: GATE CSE 2004 | Question: 74



An examination paper has 150 multiple choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 marks. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

- A. 0 B. 2550 C. 7525 D. 9375

gatecse-2004 probability expectation normal

Answer key 

3.4.3 Expectation: GATE CSE 2006 | Question: 18



We are given a set $X = \{X_1, \dots, X_n\}$ where $X_i = 2^i$. A sample $S \subseteq X$ is drawn by selecting each X_i independently with probability $P_i = \frac{1}{2}$. The expected value of the smallest number in sample S is:

- A. $\left(\frac{1}{n}\right)$ B. 2 C. $\sqrt{n}$ D. n

gatecse-2006 probability expectation normal

Answer key

3.4.4 Expectation: GATE CSE 2011 | Question: 18



If the difference between the expectation of the square of a random variable $(E[X^2])$ and the square of the expectation of the random variable $(E[X])^2$ is denoted by R , then

- A. $R=0$ B. $R<0$ C. $R \geq 0$ D. $R>0$

gatecse-2011 probability random-variable expectation normal

Answer key

3.4.5 Expectation: GATE CSE 2013 | Question: 24



Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is $\frac{1}{2}$. What is the expected number of unordered cycles of length three?

- A. $\frac{1}{8}$ B. 1 C. 7 D. 8

gatecse-2013 probability expectation normal

Answer key

3.4.6 Expectation: GATE CSE 2014 Set 2 | Question: 2



Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is _____. (The answer should be rounded to one decimal place.)

gatecse-2014-set2 probability expectation numerical-answers easy

Answer key

3.4.7 Expectation: GATE CSE 2017 Set 2 | Question: 31



For any discrete random variable X , with probability mass function $P(X=j)=p_j, p_j \geq 0, j \in \{0, \dots, N\}$, and $\sum_{j=0}^N p_j = 1$, define the polynomial function $g_x(z) = \sum_{j=0}^N p_j z^j$. For a certain discrete random variable Y , there exists a scalar $\beta \in [0, 1]$ such that $g_y(z) = (1 - \beta + \beta z)^N$. The expectation of Y is

- A. $N\beta(1-\beta)$ B. $N\beta$
C. $N(1-\beta)$ D. Not expressible in terms of N and β alone

gatecse-2017-set2 probability random-variable difficult expectation

Answer key

3.4.8 Expectation: GATE CSE 2021 Set 1 | Question: 35



Consider the two statements.

- S_1 : There exist random variables X and Y such that $(\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))])^2 > \text{Var}[X]\text{Var}[Y]$
- S_2 : For all random variables X and Y , $\text{Cov}[X, Y] = \mathbb{E}[|X - \mathbb{E}[X]| |Y - \mathbb{E}[Y]|]$

Which one of the following choices is correct?

- A. Both S_1 and S_2 are true
B. S_1 is true, but S_2 is false
C. S_1 is false, but S_2 is true
D. Both S_1 and S_2 are false

gatecse-2021-set1 probability random-variable difficult 2-marks expectation

Answer key 

3.4.9 Expectation: GATE CSE 2021 Set 2 | Question: 29



In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

question	probability of answering correctly	marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

- A. First QuesA and then QuesB. Expected marks 14.
B. First QuesB and then QuesA. Expected marks 14.
C. First QuesB and then QuesA. Expected marks 22.
D. First QuesA and then QuesB. Expected marks 16.

gatecse-2021-set2 probability expectation 2-marks

Answer key 

3.4.10 Expectation: GATE DS&AI 2024 | Question: 26



A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently.

What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?

- A. 2 B. 4 C. 6 D. 8

gate-ds-ai-2024 probability expectation

Answer key 

3.4.11 Expectation: GATE Suitability Test | Test 1 | Question: 3



A coin is such that after every toss the probability of same side coming again increases by 50% from the initial value. If initially the probability of head and tail are the same, what is the expected number of tosses until we get a head?

gate-suitability-test-1 expectation probability numerical-answers

Answer key 

3.5.1 Exponential Distribution: GATE CSE 2021 Set 1 | Question: 18



The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability random-variable numerical-answers 1-mark exponential-distribution

Answer key

3.5.2 Exponential Distribution: GATE DS&AI 2024 | Question: 47



Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ indicate the expectation and variance of X , respectively, the value of λ is _____ (rounded off to one decimal place).

gate-ds-ai-2024 numerical-answers probability random-variable exponential-distribution

Answer key

3.5.3 Exponential Distribution: GATE IT 2004 | Question: 33



Let X and Y be two exponentially distributed and independent random variables with mean α and β , respectively. If $Z = \min(X, Y)$, then the mean of Z is given by

- A. $\left(\frac{1}{\alpha + \beta} \right)$ B. $\min(\alpha, \beta)$
C. $\left(\frac{\alpha\beta}{\alpha + \beta} \right)$ D. $\alpha + \beta$

gateit-2004 probability exponential-distribution random-variable normal

Answer key

3.6 Independent Events (5)

3.6.1 Independent Events: GATE CSE 1994 | Question: 2.8



Let A, B , and C be independent events which occur with probabilities 0.8, 0.5, and 0.3 respectively. The probability of occurrence of at least one of the event is _____

gate1994 probability normal numerical-answers independent-events

Answer key

3.6.2 Independent Events: GATE CSE 1999 | Question: 2.1



Consider two events E_1 and E_2 such that probability of E_1 , $P_r[E_1] = \frac{1}{2}$, probability of E_2 , $P_r[E_2] = \frac{1}{3}$, and probability of E_1 and E_2 , $P_r[E_1 \text{ and } E_2] = \frac{1}{5}$. Which of the following statements is/are true?

- A. $P_r[E_1 \text{ or } E_2]$ is $\frac{2}{3}$ B. Events E_1 and E_2 are independent
C. Events E_1 and E_2 are not independent D. $P_r[E_1 | E_2] = \frac{4}{5}$

gate1999 probability normal independent-events

Answer key

3.6.3 Independent Events: GATE CSE 2000 | Question: 2.2



E_1 and E_2 are events in a probability space satisfying the following constraints:

- $Pr(E_1) = Pr(E_2)$
- $Pr(E_1 \cup E_2) = 1$
- E_1 and E_2 are independent

The value of $Pr(E_1)$, the probability of the event E_1 , is

- A. 0 B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. 1

gatecse-2000 probability easy independent-events

Answer key

3.6.4 Independent Events: GATE CSE 2023 | Question: 43



Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

- A. A and B are independent. B. A and C are independent.
C. B and C are independent. D. $\text{Prob}(B | C) = \text{Prob}(B)$

gatecse-2023 probability independent-events multiple-selects 2-marks

Answer key

3.6.5 Independent Events: GATE DS&AI 2024 | Question: 2



Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails.

What is the probability of the event $T \cap S$?

- A. 0 B. 0.5 C. 0.25 D. 1

gate-ds-ai-2024 probability independent-events

Answer key

3.7 Normal Distribution (2)

3.7.1 Normal Distribution: GATE CSE 2008 | Question: 29



Let X be a random variable following normal distribution with mean $+1$ and variance 4 . Let Y be another normal variable with mean -1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

- A. 3 B. 2 C. $\sqrt{2}$ D. 1

gatecse-2008 random-variable normal-distribution probability normal

Answer key

3.7.2 Normal Distribution: GATE CSE 2017 Set 1 | Question: 19



Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

gatecse-2017-set1 probability numerical-answers normal-distribution

Answer key

3.8 Poisson Distribution (4)

3.8.1 Poisson Distribution: GATE CSE 1989 | Question: 4-viii



$P_n(t)$ is the probability of n events occurring during a time interval t . How will you express $P_0(t+h)$ in terms of $P_0(h)$, if $P_0(t)$ has stationary independent increments? (Note: $P_t(t)$ is the probability density function).

gate1989 descriptive probability poisson-distribution

Answer key

3.8.2 Poisson Distribution: GATE CSE 2013 | Question: 2



Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

- A. $\frac{8}{(2e^3)}$ B. $\frac{9}{(2e^3)}$ C. $\frac{17}{(2e^3)}$ D. $\frac{26}{(2e^3)}$

gatecse-2013 probability poisson-distribution normal

Answer key

3.8.3 Poisson Distribution: GATE CSE 2017 Set 2 | Question: 48



If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(x+2)^2]$ equals ____.

gatecse-2017-set2 expectation poisson-distribution numerical-answers probability

Answer key

3.8.4 Poisson Distribution: GATE IT 2007 | Question: 57



In a multi-user operating system on an average, 20 requests are made to use a particular resource per hour. The arrival of requests follows a Poisson distribution. The probability that either one, three or five requests are made in 45 minutes is given by :

- A. $6.9 \times 10^6 \times e^{-20}$ B. $1.02 \times 10^6 \times e^{-20}$
C. $6.9 \times 10^3 \times e^{-20}$ D. $1.02 \times 10^3 \times e^{-20}$

gateit-2007 probability poisson-distribution normal

Answer key

3.9 Probability (27)

3.9.1 Probability: GATE CSE 1995 | Question: 1.18



The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is:

- A. $\frac{16}{25}$ B. $\left(\frac{9}{10}\right)^3$ C. $\frac{27}{75}$ D. $\frac{18}{25}$

gate1995 probability normal

Answer key

3.9.2 Probability: GATE CSE 1995 | Question: 2.14



A bag contains 10 white balls and 15 black balls. Two balls are drawn in succession. The probability that one of them is black and the other is white is:

- A. $\frac{2}{3}$ B. $\frac{4}{5}$ C. $\frac{1}{2}$ D. $\frac{1}{3}$

Answer key **3.9.3 Probability: GATE CSE 1996 | Question: 1.5**

Two dice are thrown simultaneously. The probability that at least one of them will have 6 facing up is

- A. $\frac{1}{36}$ B. $\frac{1}{3}$ C. $\frac{25}{36}$ D. $\frac{11}{36}$

gate1996 probability easy

Answer key **3.9.4 Probability: GATE CSE 1996 | Question: 2.7**

The probability that top and bottom cards of a randomly shuffled deck are both aces is

- A. $\frac{4}{52} \times \frac{4}{52}$ B. $\frac{4}{52} \times \frac{3}{52}$
C. $\frac{4}{52} \times \frac{3}{51}$ D. $\frac{4}{52} \times \frac{4}{51}$

gate1996 probability easy

Answer key **3.9.5 Probability: GATE CSE 1997 | Question: 1.1**

The probability that it will rain today is 0.5. The probability that it will rain tomorrow is 0.6. The probability that it will rain either today or tomorrow is 0.7. What is the probability that it will rain today and tomorrow?

- A. 0.3 B. 0.25 C. 0.35 D. 0.4

gate1997 probability easy

Answer key **3.9.6 Probability: GATE CSE 1998 | Question: 1.1**

A die is rolled three times. The probability that exactly one odd number turns up among the three outcomes is

- A. $\frac{1}{6}$ B. $\frac{3}{8}$ C. $\frac{1}{8}$ D. $\frac{1}{2}$

gate1998 probability easy

Answer key **3.9.7 Probability: GATE CSE 2001 | Question: 2.4**

Seven (distinct) car accidents occurred in a week. What is the probability that they all occurred on the same day?

- A. $\frac{1}{7^7}$ B. $\frac{1}{7^6}$ C. $\frac{1}{2^7}$ D. $\frac{7}{2^7}$

gatecse-2001 probability normal

Answer key **3.9.8 Probability: GATE CSE 2004 | Question: 25**

If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

A. $\frac{3}{8}$

B. $\frac{1}{2}$

C. $\frac{5}{8}$

D. $\frac{3}{4}$

gatecse-2004 probability easy

Answer key **3.9.9 Probability: GATE CSE 2010 | Question: 26**

Consider a company that assembles computers. The probability of a faulty assembly of any computer is p . The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q . What is the probability of a computer being declared faulty?

A. $pq + (1-p)(1-q)$

B. $(1-q)p$

C. $(1-p)q$

D. pq

gatecse-2010 probability easy

Answer key **3.9.10 Probability: GATE CSE 2010 | Question: 27**

What is the probability that divisor of 10^{99} is a multiple of 10^{96} ?

A. $\left(\frac{1}{625}\right)$

B. $\left(\frac{4}{625}\right)$

C. $\left(\frac{12}{625}\right)$

D. $\left(\frac{16}{625}\right)$

gatecse-2010 probability normal

Answer key **3.9.11 Probability: GATE CSE 2011 | Question: 34**

A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

A. $\left(\frac{1}{5}\right)$

B. $\left(\frac{4}{25}\right)$

C. $\left(\frac{1}{4}\right)$

D. $\left(\frac{2}{5}\right)$

gatecse-2011 probability normal

Answer key **3.9.12 Probability: GATE CSE 2014 Set 1 | Question: 48**

Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $\frac{X}{1296}$. The value of X is _____

gatecse-2014-set1 probability numerical-answers normal

Answer key **3.9.13 Probability: GATE CSE 2014 Set 2 | Question: 1**

The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p . Then $100p =$ _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key 

3.9.14 Probability: GATE CSE 2014 Set 2 | Question: 48



The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key

3.9.15 Probability: GATE CSE 2014 Set 3 | Question: 48



Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)P(B)$ is _____.

gatecse-2014-set3 probability numerical-answers normal

Answer key

3.9.16 Probability: GATE CSE 2016 Set 1 | Question: 29



Consider the following experiment.

Step 1. Flip a fair coin twice.

Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places)

gatecse-2016-set1 probability normal numerical-answers

Answer key

3.9.17 Probability: GATE CSE 2018 | Question: 15



Two people, P and Q , decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q . Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____

gatecse-2018 probability normal numerical-answers 1-mark

Answer key

3.9.18 Probability: GATE CSE 2021 Set 2 | Question: 33



A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

A.

$$\frac{r}{r+b}$$

B.

$$\frac{r}{r+b+3}$$

C.

$$\frac{r+3}{r+b+3}$$

D. $\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

gatecse-2021-set2 probability normal 2-marks

Answer key 

3.9.19 Probability: GATE CSE 2024 | Set 1 | Question: 17



Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

- A. The two events A and B are independent
- B. $P(A \cup B) = 0.7$
- C. $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- D. $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B , respectively

gatecse2024-set1 multiple-selects probability

Answer key 

3.9.20 Probability: GATE CSE 2024 | Set 2 | Question: 8



When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (*i. e.*, 1, 2, 3, 4, 5, and 6) is

- A. $\frac{1}{324}$
- B. $\frac{5}{324}$
- C. $\frac{7}{324}$
- D. $\frac{11}{324}$

gatecse2024-set2 probability

Answer key 

3.9.21 Probability: GATE DS&AI 2024 | Question: 48



Consider two events T and S . Let $\bar{T}$ denote the complement of the event T . The probability associated with different events are given as follows:

$$P(T) = 0.6, \quad P(S | T) = 0.3, \quad P(S | \bar{T}) = 0.6$$

Then, $P(\bar{T} | S)$ is _____ (rounded off to two decimal places).

gate-ds-ai-2024 numerical-answers probability

Answer key 

3.9.22 Probability: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 7



A fair coin is flipped twice and it is known that at least one tail is observed. The probability of getting two tails is

- A. $\frac{1}{2}$
- B. $\frac{1}{3}$
- C. $\frac{2}{3}$
- D. $\frac{1}{4}$

gateda-sample-paper-2024 engineering-mathematics probability

Answer key 

3.9.23 Probability: GATE IT 2004 | Question: 1

In a population of N families, 50% of the families have three children, 30% of the families have two children and the remaining families have one child. What is the probability that a randomly picked child belongs to a family with two children?

- A. $\left(\frac{3}{23}\right)$ B. $\left(\frac{6}{23}\right)$ C. $\left(\frac{3}{10}\right)$ D. $\left(\frac{3}{5}\right)$

gateit-2004 probability normal

Answer key

3.9.24 Probability: GATE IT 2005 | Question: 1

A bag contains 10 blue marbles, 20 green marbles and 30 red marbles. A marble is drawn from the bag, its colour recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour is

- A. $\left(\frac{1}{36}\right)$ B. $\left(\frac{1}{6}\right)$ C. $\left(\frac{1}{4}\right)$ D. $\left(\frac{1}{3}\right)$

gateit-2005 probability normal

Answer key

3.9.25 Probability: GATE IT 2008 | Question: 2

A sample space has two events A and B such that probabilities $P(A \cap B) = \frac{1}{2}$, $P(A') = \frac{1}{3}$, $P(B') = \frac{1}{3}$. What is $P(A \cup B)$?

- A. $\left(\frac{11}{12}\right)$ B. $\left(\frac{10}{12}\right)$ C. $\left(\frac{9}{12}\right)$ D. $\left(\frac{8}{12}\right)$

gateit-2008 probability easy

Answer key

3.9.26 Probability: GATE IT 2008 | Question: 23

What is the probability that in a randomly chosen group of r people at least three people have the same birthday?

- A. $1 - \frac{365 \cdot 364 \dots (365 - r + 1)}{365^r}$
 B. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} + {}^r C_1 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r+2}}$
 C. $1 - \frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} - {}^r C_2 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r-2}}$
 D. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r}$

gateit-2008 probability normal

Answer key

3.9.27 Probability: GATE Suitability Test | Test 1 | Question: 5

In a group of 11 people what is the probability that exactly two people have birthday on same day ?

gate-suitability-test-1 probability

Answer key

Probability Density Function (1)

3.10.1 Probability Density Function: GATE CSE 2003 | Question: 60, ISRO2007-45



A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

- A. $f_1(t) + f_2(t)$
B. $\int_0^t f_1(x)f_2(x)dx$
C. $\int_0^t f_1(x)f_2(t-x)dx$
D. $\max\{f_1(t), f_2(t)\}$

gatecse-2003 probability normal isro2007 probability-density-function

Answer key

3.11

Queuing Theory (1)

3.11.1 Queuing Theory: GATE CSE 1989 | Question: 3-vii



Which of the following statements are FALSE?

- A. For poisson distribution, the mean is twice the variance.
- B. In queuing theory, if arrivals occur according to poisson distribution, then the inter-arrival time is exponentially distributed.
- C. The distribution of waiting time is independent of the service discipline used in selecting the waiting customers for service.
- D. If the time between successive arrivals is exponential, then the time between the occurrences of every third arrival is also exponential.

gate1989 normal probability poisson-distribution queuing-theory out-of-gatecse-syllabus

Answer key

3.12

Random Variable (5)

3.12.1 Random Variable: GATE CSE 2005 | Question: 12, ISRO2009-64



Let $f(x)$ be the continuous probability density function of a random variable x , the probability that $a < x < b$ is :

- A. $\int_a^b f(b-a)$
 B. $\int_a^b f(b) - f(a)$
 C. $\int_a^b f(x) dx$
 D. $\int_a^b xf(x) dx$

gatecse-2005 probability random-variable easy isro2009

Answer key

3.12.2 Random Variable: GATE CSE 2011 | Question: 33



Consider a finite sequence of random values $X = [x_1, x_2, \dots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X . Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence.

Which one of the following statements is **INCORRECT**?

- A. Index position of mode of X in X is the same as the index position of mode of Y in Y
 B. Index position of median of X in X is the same as the index position of median of Y in Y
 C. $\mu_y = a\mu_x + b$
 D. $\sigma_y = a\sigma_x + b$

Answer key **3.12.3 Random Variable: GATE CSE 2012 | Question: 21**

Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

- A. 0 and 0.5 B. 0 and 1 C. 0.5 and 1 D. 0.25 and 0.75

gatecse-2012 probability random-variable easy

Answer key **3.12.4 Random Variable: GATE CSE 2015 Set 3 | Question: 37**

Suppose X_i for $i = 1, 2, 3$ are independent and identically distributed random variables whose probability mass functions are $Pr[X_i = 0] = Pr[X_i = 1] = \frac{1}{2}$ for $i = 1, 2, 3$. Define another random variable $Y = X_1 X_2 \oplus X_3$, where $\oplus$ denotes XOR. Then $Pr[Y = 0 | X_3 = 0] = \underline{\hspace{2cm}}$.

gatecse-2015-set3 probability random-variable normal numerical-answers

Answer key **3.12.5 Random Variable: GATE DS&AI 2024 | Question: 1**

Consider the following statements:

- The mean and variance of a Poisson random variable are equal.
- For a standard normal random variable, the mean is zero and the variance is one.

Which **ONE** of the following options is **correct**?

- A. Both (i) and (ii) are true B. (i) is true and (ii) is false
C. (ii) is true and (i) is false D. Both (i) and (ii) are false

gate-ds-ai-2024 probability random-variable

Answer key **3.13 Statistics (3)****3.13.1 Statistics: GATE CSE 2021 Set 2 | Question: 22**

For a given biased coin, the probability that the outcome of a toss is a head is 0.4 . This coin is tossed $1,000$ times. Let X denote the random variable whose value is the number of times that head appeared in these $1,000$ tosses. The standard deviation of X (rounded to 2 decimal place) is $\underline{\hspace{2cm}}$

gatecse-2021-set2 numerical-answers probability random-variable 1-mark statistics

Answer key **3.13.2 Statistics: GATE CSE 2024 | Set 2 | Question: 34**

Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

- A. $\bar{z} = \bar{x}\bar{y}$ B. $\bar{z} \leq \bar{x}\bar{y}$
C. $\bar{z} \geq \bar{x}\bar{y}$ D. $\bar{z} \leq \bar{x}$

gatecse2024-set2 probability random-variable statistics

Answer key 

3.13.3 Statistics: GATE DS&AI 2024 | Question: 17



Let the minimum, maximum, mean and standard deviation values for the attribute income of data scientists be ₹46000, ₹ 170000, ₹ 96000, and ₹ 21000, respectively. The z -score normalized income value of ₹ 106000 is closest to which ONE of the following options?

- A. 0.217 B. 0.476 C. 0.623 D. 2.304

gate-ds-ai-2024 statistics probability non-gatecse

Answer key 

3.14

Uniform Distribution (9)

3.14.1 Uniform Distribution: GATE CSE 1998 | Question: 3a



Two friends agree to meet at a park with the following conditions. Each will reach the park between 4:00 pm and 5:00 pm and will see if the other has already arrived. If not, they will wait for 10 minutes or the end of the hour whichever is earlier and leave. What is the probability that the two will not meet?

gate1998 probability normal numerical-answers uniform-distribution

Answer key 

3.14.2 Uniform Distribution: GATE CSE 2004 | Question: 78



Two n bit binary strings, S_1 and S_2 are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

- A. $\frac{{}^nC_d}{2^n}$ B. $\frac{{}^nC_d}{2^d}$ C. $\frac{d}{2^n}$ D. $\frac{1}{2^d}$

gatecse-2004 probability normal uniform-distribution

Answer key 

3.14.3 Uniform Distribution: GATE CSE 2004 | Question: 80



A point is randomly selected with uniform probability in the $X - Y$ plane within the rectangle with corners at $(0,0)$, $(1,0)$, $(1,2)$ and $(0,2)$. If p is the length of the position vector of the point, the expected value of p^2 is

- A. $\left(\frac{2}{3}\right)$ B. 1 C. $\left(\frac{4}{3}\right)$ D. $\left(\frac{5}{3}\right)$

gatecse-2004 probability uniform-distribution expectation normal

Answer key 

3.14.4 Uniform Distribution: GATE CSE 2007 | Question: 24



Suppose we uniformly and randomly select a permutation from the $20!$ permutations of $1, 2, 3, \dots, 20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

- A. $\left(\frac{1}{2}\right)$ B. $\left(\frac{1}{10}\right)$ C. $\left(\frac{9!}{20!}\right)$ D. None of these

gatecse-2007 probability easy uniform-distribution

Answer key 

3.14.5 Uniform Distribution: GATE CSE 2014 Set 1 | Question: 2



Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is _____.

gatecse-2014-set1 probability uniform-distribution expectation numerical-answers normal

Answer key

3.14.6 Uniform Distribution: GATE CSE 2019 | Question: 47



Suppose Y is distributed uniformly in the open interval $(1, 6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____

gatecse-2019 numerical-answers engineering-mathematics probability uniform-distribution 2-marks

Answer key

3.14.7 Uniform Distribution: GATE CSE 2020 | Question: 45



For $n > 2$, let $a \in \{0, 1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0, 1\}^n$. Then, the probability that $\sum_{i=1}^n a_i x_i$ is an odd number is _____

gatecse-2020 numerical-answers probability uniform-distribution 2-marks

Answer key

3.14.8 Uniform Distribution: GATE CSE 2024 | Set 1 | Question: 4



Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

- A. The events X and Y are mutually exclusive
- B. The events X and Y are independent
- C. Either event X or Y must occur
- D. Event X is more likely than event Y

gatecse2024-set1 probability uniform-distribution

Answer key

3.14.9 Uniform Distribution: GATE DS&AI 2024 | Question: 46



Let X be a random variable uniformly distributed in the interval $[1, 3]$ and Y be a random variable uniformly distributed in the interval $[2, 4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).

gate-ds-ai-2024 numerical-answers random-variable probability uniform-distribution

Answer key

Answer Keys

3.1.1	C	3.1.2	D	3.1.3	A	3.1.4	A	3.1.5	A
3.1.6	C	3.2.1	D	3.2.2	N/A	3.2.3	D	3.2.4	A
3.2.5	C	3.2.6	B	3.2.7	A	3.2.8	B	3.2.9	0.55
3.2.10	A	3.2.11	0.60 : 0.61	3.2.12	0.04 : 0.04	3.2.13	0.375	3.2.14	C
3.2.15	0.035:0.04	3.3.1	0.5	3.4.1	C	3.4.2	D	3.4.3	D
3.4.4	C	3.4.5	C	3.4.6	3.8 : 3.9	3.4.7	B	3.4.8	D

3.4.9	D
3.5.3	C
3.6.5	A
3.8.3	54
3.9.4	C
3.9.9	A
3.9.14	0.259 : 0.261
3.9.19	B;C
3.9.24	B
3.11.1	A;C
3.12.5	A
3.14.2	A
3.14.7	0.5

3.4.10	C
3.6.1	0.93
3.7.1	A
3.8.4	B
3.9.5	D
3.9.10	A
3.9.15	0.25
3.9.20	B
3.9.25	B
3.12.1	C
3.13.1	15.00 : 16.00
3.14.3	D
3.14.8	B

3.4.11	3
3.6.2	C
3.7.2	0
3.9.1	D
3.9.6	B
3.9.11	A
3.9.16	0.33 : 0.34
3.9.21	0.25
3.9.26	X
3.12.2	D
3.13.2	D
3.14.4	B
3.14.9	0.125

3.5.1	0.35 : 0.39
3.6.3	D
3.8.1	N/A
3.9.2	C
3.9.7	B
3.9.12	10
3.9.17	0.0230 : 0.0232
3.9.22	B
3.9.27	0.14:0.15
3.12.3	C
3.13.3	B
3.14.5	0.24 : 0.27

3.5.2	0.2
3.6.4	C;D
3.8.2	C
3.9.3	D
3.9.8	A
3.9.13	11.85 : 11.95
3.9.18	A
3.9.23	B
3.10.1	C
3.12.4	0.75
3.14.1	0.69:0.70
3.14.6	0.8